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CATS INSIGHTS

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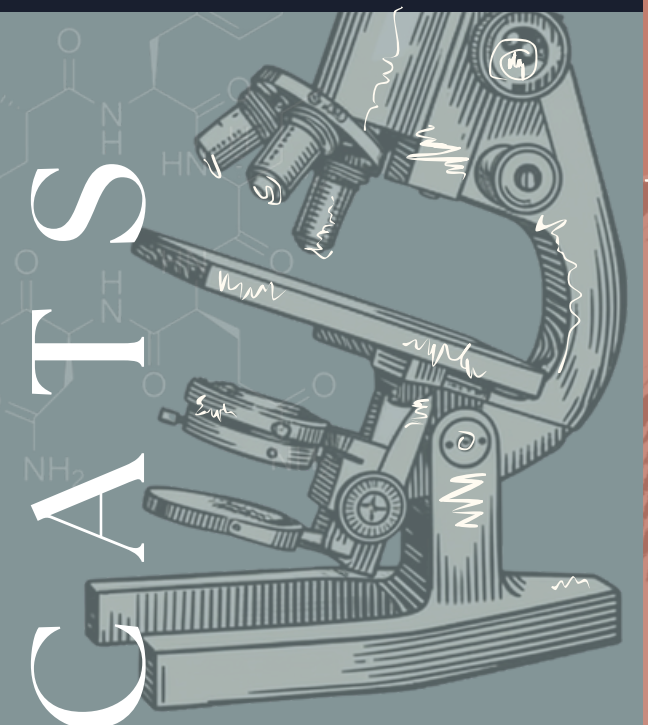

Ocular immune privilege is a highly regulated state in which standard immune responses are modulated to preserve vision as a primary function of the eye and protect intraocular structures from inflammation-induced damage. It is maintained via blood–ocular barriers, immunosuppressive microenvironments and systemic immune tolerance. Despite such autonomy, the protection is not absolute. Ocular trauma or surgery may lead to the disruption of immune privilege

π

s history starts out at around 1900 to 1860 BC where Ancient Babylonians commonly used the number 3 in place of pi as it gave a close enough estimate to the area of a circle, though the value of the area could be skewed once larger radii were involved.

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THE WORTHGATE SCHOOL CANTERBURY

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EDITORIAL

A year ago CATS Insights did not exist.
There was no editorial board, no review process, no house style, and no evidence that a group of students could build any of it. What instead existed was a conviction that students, given the platform and opportunity, could and would produce work worth reading.

I started it for a selfish reason.

I found that I always preferred to read, watch or listen to topics I genuinely found intriguing rather than just what is decided for me from my syllabus.
For me that was John Snow (not that one) and his revolutionary work in Broad Street against cholera which led to an entirely new branch of medical science, epidemiology. I was fascinated for around a week before my interest and curiosity began to wander to numerous other topics such as the South Sea Bubble or the Punic Wars and before I knew it, I had buried myself in a dozen rabbit holes that nobody else knew I was down.

Most things are better shared, and I had nobody to share my passion with. It seemed unlikely I was the only person with that problem, so I built CATS Insights to find my community.

Reminiscing on where we started, it is awe-spiring to see how far we have come and this publication is the materialisation of a year's worth of memories and development culminated.

This edition contains sixteen articles by eleven students. They were written over the course of a year and edited through five stages. Every reference was checked against its source. Every little mistake or slip up in grammar, structure and terminology was scoured for. Every figure had to be checked and cleared for rights or painstakingly made by hand.
A reader will find the immune privilege of the human eye alongside the collapse of a bridge in Kyiv; two thousand years of calculating pi alongside the chemistry of a Kent hop field; superconductivity alongside a question about what an idea actually is. None of it was commissioned to fit a theme. It is what happens when you ask people what they actually find interesting and then hold them to a standard.

We are not leading experts in the numerous fields of study that are included in this publication, yet, a handful of exceptional students took on the Herculean project of researching, writing and producing excellent academic scholarship.

A tremendous amount of time, effort and care had to be dedicated from people who are still in school, have homework, coursework and deadlines which is why I am immensely proud of this publication and everyone who worked on it.

It is my sincerest aspiration that CATS Insights endeavours and thrives in the coming years, surpassing it's precursor to reach new zeniths.

The intent behind this publication is to be the spark that ignites the fire of curosity in the reader, my hope is that it imparts inspiration to those who read it so they look for more.

I hope you find something here worth chasing.

To everyone who wrote for this edition, you committed to something that did not exist yet, and you did the work when it would have been easier not to. Thank you for your perservearance and dedication.

To everyone who supported CATS Insights, the list of acknowledgements at the back is longer than I expected, thank you.

The first edition of anything is a statement of intent. This is ours.

Kyal Sin Ye Yint
Founder and President
CATS Insights
Canterbury, September 2026

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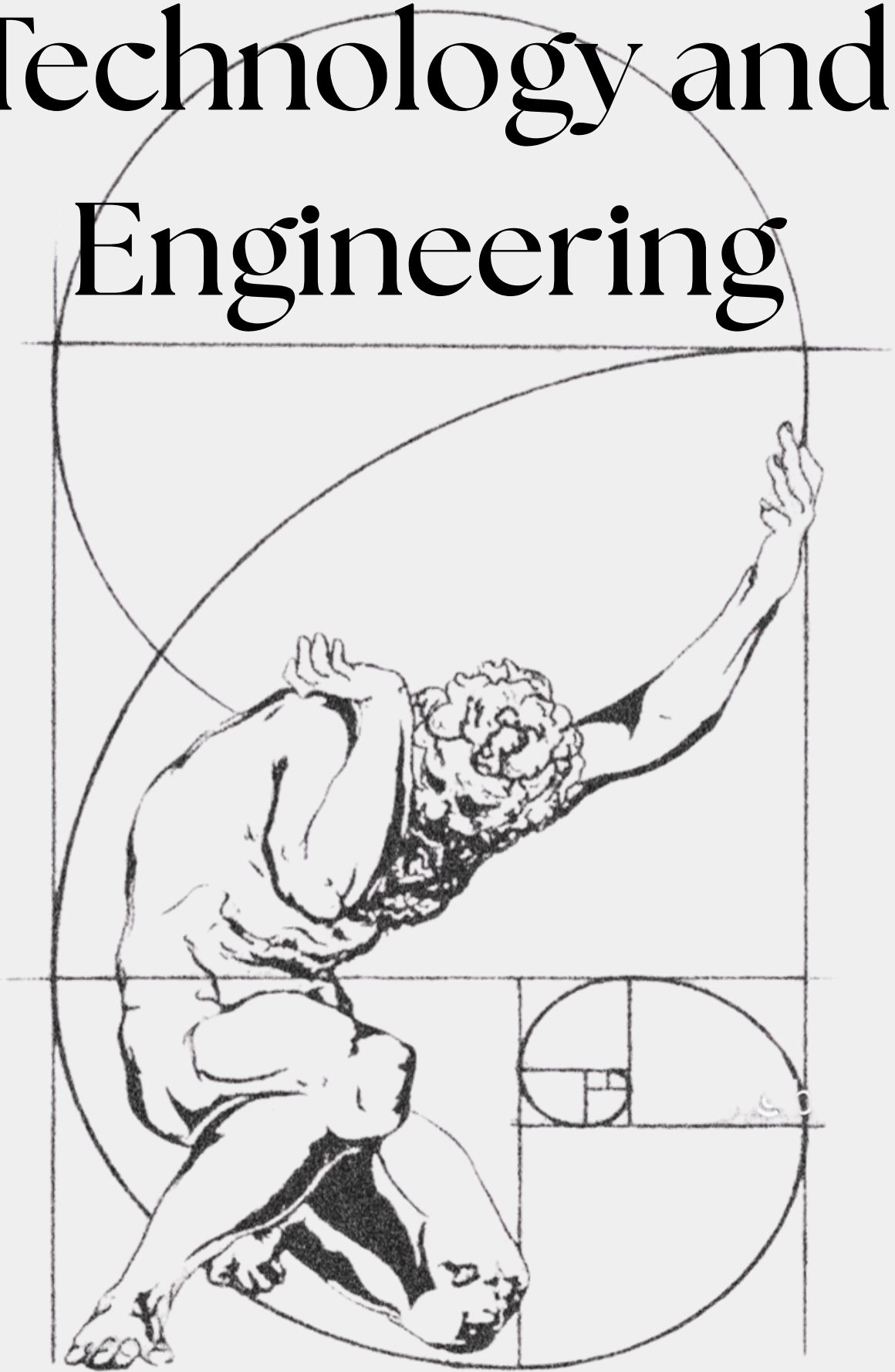

3.14159 and so on: How Pi is Calculated Over the Years

Author: Aung Khant Pyaet

Abstract:

This article covers the majority of the timeline of pi and its development over time, ranging from Ancient Babylonian estimates around 1900 BC to around 1999 in the present day. Most of the approximations follow methods featuring calculus with the remaining few following geometrical methods with various mathematicians from the likes of Euler, Ramanujan, Archimedes, James Gregory, etc.

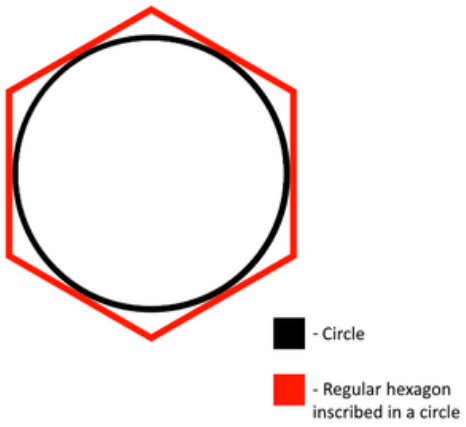

Figure 2.1 - Ancient Babylonian Approximation of Pi
Diagram: Aung Khant Pyaet, CATS Insights, 2026.

I. Introduction

Pi, as most people know, is the ratio between the circle's circumference to the diameter of a circle. This ratio is what allows mathematicians and students alike to find the area of a circle, integrate certain functions, etc. This value has been led up by a wide variety of efforts made by mathematicians of a plethora of backgrounds from as early as 1900 BC to around the modern day with efforts made in approximating this ratio more accurately to this day for seemingly a simple albeit essential numerical value. This article will cover the majority of pi's history aside from a few of the modern methods that are utilised through advanced calculus such as the Jacobian theta function.

II. Early Discovery Through Geometry

2.1 Ancient Babylonian Approximations

Pi's history starts out at around 1900 to 1860 BC where Ancient Babylonians commonly used the number 3 in place of pi as it gave a close enough estimate to the area of a circle, though the value of the area could be skewed once larger radii were involved. [2][3] One of the closer approximations that was used by Ancient Babylonians around the same time was 3.125 which was written on a Babylonian tablet. This estimate was found by inscribing a hexagon inside a circle of diameter d and finding the hexagon's perimeter to find the ratio between the perimeter of the hexagon and the perimeter of the circle as seen in figure 2.1. [3]

III. The First Attempt at Expanding Pi

The algorithm used to find an iteration of the value of pi was first made by Archimedes between 287-212 BC [2]. This system was considered as one of the efficient methods to calculate pi. The way his algorithm worked was by taking a circle with unit diameter and both inscribing and circumscribing a polygon with sides 32n (where $n \in \mathbb{N}$) [1] [9] as seen in Figure 3.1.

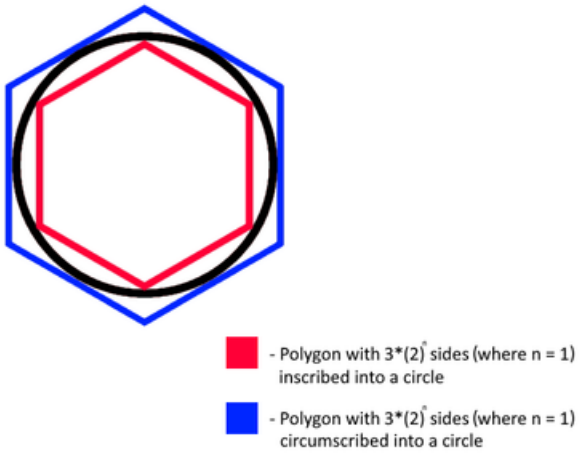

Figure 3.1 - Archimedes's model used to approximate pi
Diagram: Aung Khant Pyaet, CATS Insights, 2026.

With this, 2 series are derived both the perimeters of polygons as:

$$a_1 = 2\sqrt{3} \qquad b_1 = 3$$

$$a_{n+1} = \frac{2a_n b_n}{a_n + b_n} \quad b_{n+1} = \sqrt{a_{n+1} b_n}$$

(where series an represents the perimeter of the circumscribed polygon and series bn represents the perimeter of the inscribed polygon and $n \in \mathbb{N}$. [1] [9] Since the perimeter of b_n is less than the perimeter of the circle and the perimeter of a_n is always greater than the perimeter of the circle as n increases, $b_n < \pi < a_n$. In Archimedes's case, the approximation was calculated to n = 5 which led to the close approximation of an = 3.1415 and bn = 3.1417, which is close to the actual ratio of pi. [1] [9]

IV. Geometrical Approximations Afterwards

4.1 Hiatus in development

There wasn't much development made on increasing the accuracy of the value for the next 1800 years aside from Zu Chongzhi's approach made around 429 to 501 A.D. where he supposedly approached the problem with a similar method that Archimedes used. However, this calculation would end up being lost to time as the book was lost, with little knowledge about how he achieved such a value. The only thing left behind proving his contribution is the value he calculated. [2] This pause was largely attributed towards more of the mathematicians focusing more on fine tuning or using Archimedes's method to produce more accurate values made by mathematicians such as Liu Hui (250 AD) who calculated to n = 6, Zu Chongzhi (500 AD) who calculated to n = 13, Fibonacci (1200) who calculated to n = 5, Al-Kashi (1430) who calculated to an accuracy of 14 decimal places, and Adrian Anthoniszoon(1600) who calculated to an accuracy of 6 decimal places. [3]

4.2 Viete's Method

After this hiatus, a different approach to more accurately find the value of pi was developed by Francois Viete in 1593 which, similarly to Archimedes's method, was based on geometry. [1] His approach featured infinitely nested square roots using the equation

This equation was derived by taking a circle with diameter of 2 and approximating its area by the area of an inscribed square, the area of an inscribed octagon, the area of an inscribed 16 sided polygon, and so on with areas of regular polygons with numbers of sides squared front the last. [1] With this method, Viete's calculation of pi had an accuracy of 9 decimal places.

V. Approximations Through Calculus

5.1 Brouncker's expansion

In 1655, Brouncker expanded pi as a continued fraction

$$\frac{4}{\pi} = 1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{(2n+1)^2}{\ddots}}} \quad \text{(where } n \in \mathbb{N}\text{)}$$

. [5] Increasing the number of fractions inside the continued fraction provides a more accurate value of pi by finding the reciprocal of the sum and multiplying the sum.

5.2 Newton's Approximation

Between 1665 and 1680, Newton and Leibniz had started developing the basis for calculus using work from works from previous mathematicians from the likes of John Wallism, Johannes Kepler, etc. [4] This basis essentially made way for other mathematicians to find another way to achieve a closer value of pi. It was around this time in 1665 where Newton developed the alternative method to finding pi. [1] This alternative method was discovered when Newton realised that the area bounded by a circle with equation

$$\left(x - \frac{1}{2}\right)^2 + y^2 = \frac{1}{4}$$

and the line $x = \frac{1}{4}$ along with the x axis is equal to the difference between the area of a circular sector of 60 degrees with a triangle of the same angle as seen in figure 5.1.

With this, he integrated the function $\int_0^{\frac{1}{4}} \sqrt{x - x^2} dx$ into the form $\int_0^{\frac{1}{4}} \sqrt{x}\sqrt{1-x} dx = \frac{\pi}{24} - \frac{\sqrt{3}}{32}$.

Newton had solved this integral by expanding the function $\sqrt{1-x}$ using the binomial expansion formula. With this expansion, Newton's method gave pi as

$$\frac{3\sqrt{3}}{4} + 2 - \frac{3}{4}\sum_{n=0}^{\infty} \frac{{}^{2n}C_n}{(n+1)(2n+5)16^n} \text{ (where } n \in \mathbb{N}\text{) [1]}$$

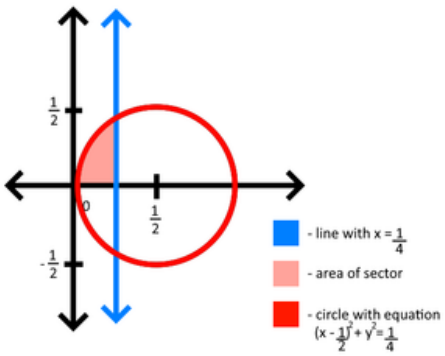

Figure 5.1 - Newton's Discovery
Diagram: Aung Khant Pyaet, CATS Insights, 2026.

The ${}^{2n}C_n$ in the equation can be represented as the fraction $\frac{(2n)!}{n!\,(2n-n)!}$ or $\frac{(2n)!}{n!\,(n)!}$ with each increasing value of n so once summed up, the series $\sum_{n=0}^{\infty} \frac{{}^{2n}C_n}{(n+1)\,(2n+5)16^n}$ can be written as $\frac{1}{5} + \frac{2}{2\times7\times16} + \frac{720}{4\times3\times9\times16^2} + \ldots$ and so on. This series essentially can be used to calculate pi in a similar way to Archimedes's method where the greater the number of terms in the series, the more accurate the value of pi is.

5.3 Expansions through Gregory's function

Similarly, another development was made by James Gregory in 1671 with a different approach in calculus. Gregory's approach utilises the Maclaurin Series of $\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \ldots$. [1] Although he stopped there, other mathematicians could use his function to calculate pi in other ways by integrating both sides of the function which gives the equation of the form $tan^{-1}(x) = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \frac{1}{7}x^7 + \ldots$. In 1674, Leibniz then realised that once x=1 is substituted into the equation, the value of pi could be found such that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots$ where the more terms in the series are included, the more accurate the value of pi similar to the previous other attempts at calculation.

Alternatively, Alternatively, in 1699, Abraham Sharp realised the value $x = \frac{\sqrt{3}}{3}$ into the same equation also grants a similar equation in the form of $\frac{\pi}{6} = \frac{\sqrt{3}}{3}\left(1 - \frac{1}{3\times3} + \frac{1}{5\times3\times3} - \frac{1}{7\times3\times3} + \cdots\right)$ or in the form of

$$\pi = 2\sqrt{3}\sum_{n=0}^{\infty}\frac{(-3)^n}{n+2}$$

It is this equation which allowed him to calculate 71 accurate digits for pi. [1] However similar versions to Sharp's method was made around the fourteenth century by Madhava of Sangamagramma who calculated up to 11 decimal places and another around the sixteenth century by Nilakantha Somayaji with it being found in Sanskrit texts made by him. [1] [6] In 1706, John Machin derived his own formula of $\frac{\pi}{4} = 4tan^{-1}\left(\frac{1}{5}\right) - tan^{-1}\left(\frac{1}{239}\right)$ which allowed him to calculate 100 digits of pi. [7]

Around 1755, Euler proved Brouncker's expression from 1655 and remodelled his expression as the fraction

$$a_1 + a_1a_2 + a_1a_2a_3 + \ldots = \frac{a_1}{1 - \frac{a_{n+1}}{1+a_{n+2}}\ldots} \text{ (where } n \in \mathbb{N}\text{)}$$

by rewriting an iteration of Gregory's function as

$$tan^{-1}(x) = x + x\left(\frac{-x^2}{3}\right) + x\left(\frac{-x^2}{3}\right)\left(-\frac{3x^2}{5}\right) + x\left(-\frac{x^2}{3}\right)\left(-\frac{3x^2}{5}\right)\left(-\frac{5x^2}{7}\right) + \ldots$$

where substituting x = 1 gives the Brouncker's expression.

5.4 Euler's Expansion

Around 1736, Euler managed to prove that $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \ldots = \frac{\pi^2}{6}$. [1] This was possible by taking the series

$$\frac{sin\,(x)}{x} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \ldots$$

from the MaClaurin series of sin(x), an alternative method to defining the function of sin(x) so as to find proofs for various other equations, and dividing the values in the series by x as well as another equation setting sin(x) as equal to 0 while x isn't equal to 0. Since when $sin\,(x) = 0, x = n\pi$ (where $n \in \mathbb{N}$). This allowed Euler to to rewrite the equation of $\frac{sin\,(x)}{x}$ as a polynomial

$$\left(1 - \frac{x^2}{\pi^2}\right)\left(1 - \frac{x^2}{4\pi^2}\right)\left(1 - \frac{x^2}{9\pi^2}\right)\ldots \text{[1]}$$

Since this series is equal to $1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \ldots$, setting all of the x^2 coefficients equal by adding all of the values of x^2 from the polynomial such that

$$\frac{x^2}{\pi^2} + \frac{x^2}{4\pi^2} + \frac{x^2}{9\pi^2} + \frac{x^2}{16\pi^2} + \ldots = \frac{x^2}{3!}$$

Multiplying both sides by π^2 and dividing both sides of the equation by the value x^2 grants the equation given at the start of the paragraph. There were also more similar equations made by Euler such as

$$\sum_{n=1}^{\infty}\frac{1}{(2n+1)^2} = \frac{\pi^2}{8} \qquad \sum_{n=1}^{\infty}\frac{1}{n^4} = \frac{\pi^4}{90}$$

$$\sum_{n=1}^{\infty}\frac{1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}}{n^3} = \frac{\pi^4}{72}$$

which were all proven in a similar manner to the previous equation. [1]

5.5 Ramanujan's Vision

It was around 1914 when Ramanujan discovered a new iteration of pi, which was done by creating the elliptic integral

$$\int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1-k^2 sin^2(\Phi)}} d\Phi = 1 + \left(\frac{1}{2}\right)^2 k^2 + \left(\frac{1\times 3}{2\times 4}\right)^2 k^4 + \dots$$ [1]

The constant value k in the expression represents the elliptical modulus where $k \in [0,1]$. Ramanujan then writes the functions $r\frac{L'}{L} = \frac{K'}{K}$ or $Q^r = q$ where $q = e^{-\pi\frac{K'}{K}}$ and $Q = e^{-\pi\frac{L'}{L}}$. In these, K, K', L, and L' denote elliptical integrals of modulus k, k', l, and l' such that $K = \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1-k^2 sin^2(\Phi)}} d\Phi$,

$K' = \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1-(k')^2 sin^2(\Phi)}} d\Phi, L = \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1-l^2 sin^2(\Phi)}} d\Phi$

$L' = \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1-(l')^2 sin^2(\Phi)}} d\Phi$. [1]

When $k = l'$ and $k' = l$, $q = e^{-\pi\sqrt{r}}$ with the variable r coming from the previous equation where $r\frac{L'}{L} = \frac{K'}{K}$. With this, he was able to derive a wide variety of formulas to derive pi such as the equations :

$$\sum_{n=0}^{\infty} \frac{(2n)^3!}{(n!)^6} \times \frac{1}{2^{8n}}(6n+1) = \frac{4}{\pi}$$

$$\frac{\sqrt{8}}{9801} \sum_{n=0}^{\infty} \frac{(4n)!}{(n!)^4} \times \frac{(1103+26390n)}{(396)^{4n}} = \frac{1}{\pi}$$

which can then be flipped to approximate the value of pi. [9]

5.6 Comtet's Approximation

Another approach using calculus was developed by mathematician Comtet in 1974 which is similar to Newton's as it utilises the binomial expansion as well. With the identity

$$2\int_0^{\frac{\pi}{3}} \theta log^2\left(2sin\left(\frac{\theta}{2}\right)\right) d\theta = \sum_{n=1}^{\infty} \frac{1}{n^4 \times {}^{2n}C_n} \text{ (where } n \in \mathbb{N})$$

rearranging the values after integrating the function give the following equation

$$\pi^4 = \frac{3240}{17} \sum_{n=1}^{\infty} \frac{1}{n^4 \times {}^{2n}C_n}$$ [1]

This equation, similar to the previous attempts on calculating pi, increases its accuracy through increasing the value of n.

5.6 Gauss-Legendere Algorithm

One of the algorithms combining other previous systems was the Gauss-Legendre algorithm which was discovered in 1975 by Richard Brent and Eugene Salamin. [8] This calculation was done by setting values

$$a_0 = 1, \quad b_0 = \frac{\sqrt{2}}{2}, \; p_0 = 1, \; t_0 = \frac{1}{4}$$

with the functions

$$a_{n+1} = \frac{a_n + b_n}{2}, \quad b_{n+1} = \sqrt{a_n b_n}, \quad p_{n+1} = 2p_n$$

$$t_{n+1} = t_n - p_n(a_{n+1} - a_n)^2$$ [8]

The value of pi can then be iterated as

$$\pi \approx \frac{(a_{n+1} + b_{n+1})^2}{4t_{n+1}}$$

which can then be iterated several times to slowly calculate the accurate value of pi. Although its method requires heavy computing power as several calculations are required to be able to produce the most accurate result. [8]

VI. Conclusion

While it seems absurd to be spending so much time and resources to be able to reach an accurate value of pi (especially since 39 places of pi already suffice in calculating the circumference of a circle surrounding the known universe with the margin of error of a hydrogen atom [9]), the race to calculating alternative methods to accurately calculate pi has produced a wide array of methods that is used commonly.

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Superconductivity: Materials and Underlying Mechanisms

Author: Oleksandr Boyko

Abstract:

This article explains the phenomenon of superconductivity as well as fundamental properties of superconductors. It examines the quantum mechanical mechanisms that enable electrical current to flow within the superconductor without resistance, as well as the expulsion of magnetic fields, by a superconducting material, known as the Meissner effect. In addition the article outlines the history of invention and improvement of superconducting materials, and their real world applications.

I. Introduction

Metals are widely used as electrical conductors due to their ability to allow the flow of electric current with relatively low resistance. The conductivity of the material is inversely proportional to its resistivity. However, there are certain materials that allow current to flow without energy loss, resulting in effectively zero electrical resistance. These materials are known as superconductors.

II. What are superconductors?

Superconductors are materials whose electrical resistance drops to zero when they are cooled below their transition temperature (the specific, often extremely low, temperature at which a material transitions from a normal state to a zero-resistance superconducting state)[1]. There are a number of metallic elements (aluminium, lead, mercury) that exhibit superconductivity if cooled sufficiently, although their transition temperature is extremely low (<10 K). This makes them unsuitable for reliable and practical use due to the logistical and monetary issues that arise from the necessity of obtaining such low temperatures.

As a result of extensive research in the field of superconductivity, some materials with a much higher transition temperature were discovered. Notably, ceramic compounds such as yttrium barium copper oxide (YBCO) and bismuth strontium calcium copper oxide (BSCCO) exhibit superconductivity at temperatures above 93 K and 108 K respectively[2]. These so-called high-temperature superconductors have significantly increased the feasibility of practical applications, particularly in areas where liquid nitrogen cooling can be used.

II. Mechanism of Superconductivity

2.1 Zero resistance below their transition temperature

In conventional conductors, electrical resistance decreases gradually as temperature is reduced, primarily due to reduced lattice vibrations and fewer collisions between the passing electrons and atomic nuclei. This implies that the resistivity of materials decreases gradually with the decrease in temperature[3]. However, this classic behaviour cannot account for superconductivity, where resistance drops abruptly to zero at a transition temperature.

While electrons behave as particles, they undergo scattering from lattice imperfections and collisions with atoms, which means that for non-superconductors the resistivity will never reach zero. However, once a superconductor reaches its transition temperature, the phenomenon known as Cooper Pair formation occurs. When the electron travels through the lattice of positively charged ions, it induces a slight distortion of the lattice structure, creating a region of increased positive charge density. Under sufficiently low temperatures and appropriate lattice structure, this area of positive charge attracts another electron and, as a result, a pair of electrons is formed, which is called a Cooper Pair[4] (Figure 1.1).

Under normal conditions, the electron is a subatomic particle classified as a fermion, which means it has a half-integral spin (a quantum mechanical property of particles where their intrinsic angular momentum is a half-integer multiple of the reduced Planck constant), obeying the Pauli Exclusion Principle (no two electrons can have the same four electronic quantum numbers, meaning they cannot exist at the same place at the same time), but once 2 electrons form a pair their spins align in the opposite direction to one another, resulting in the net spin of 0.

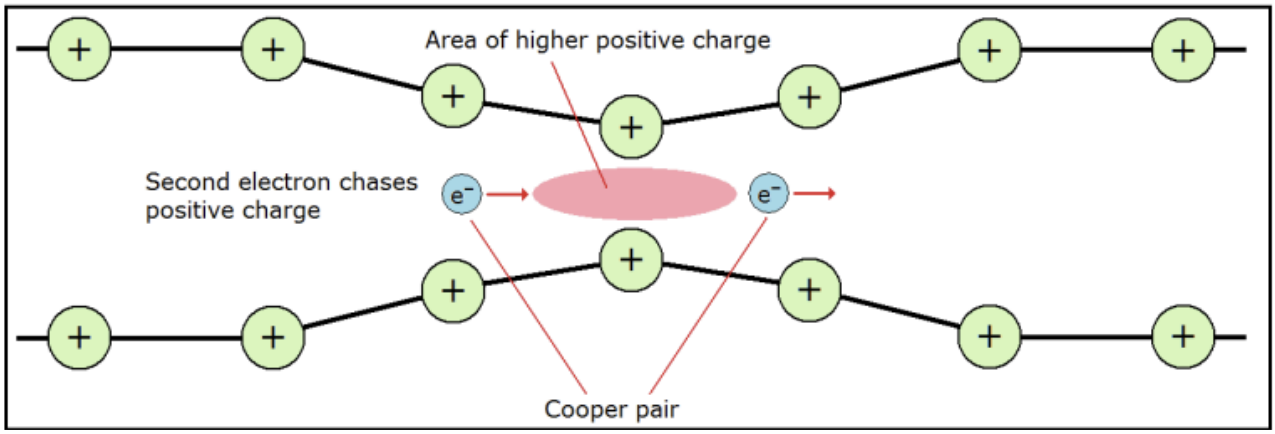

Figure 1.1 The simplified model of formation of Cooper Pairs
Source: PHYS 498 ART Spring 2018 by Karmela Padavić

This means that 2 electrons in a Cooper Pair behave like a composite boson rather than a fermion. Bosons are not subject to the Pauli Exclusion Principle, allowing them to occupy the same quantum state (similarly to how photons can occupy the same quantum space in a laser), forming a wave described by a single macroscopic wavefunction[4].

This coherent quantum state allows electrons to move through the material without scattering, eliminating electrical resistance. Unlike in normal conductors, where electrons behave as independent particles, superconducting electrons act collectively, preventing energy dissipation. As a result, the wavefunction they abide by is robust to the scattering from the atoms or imperfections in the substance, effectively creating a current with 0 resistance.

This explains why in conventional conductors resistivity gradually decreases with the decrease of the temperature, whereas in superconductors it drops abruptly to zero at the transition temperature (Figure 1.2)

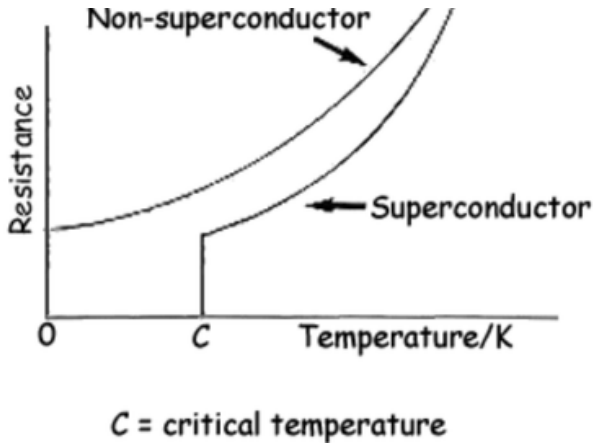

Figure 1.2 The relationship between the resistance and temperature for the superconductors and non-superconductors.
Source: cyberphysics

2.2 No interior magnetic field

Another key property of all superconductors is the complete expulsion of external magnetic fields. This property arises as a result of a unique superconducting state, rather than simply the absence of electrical resistance. The cause of this phenomenon can be explained by The Meissner effect.

As stated above superconductors do not have resistance thus once there is a current in the circuit made entirely of a superconducting material it theoretically will never diminish unless the external force acts upon it. Once the superconductor is introduced to a magnetic field, persistent surface currents are induced within the material - eddy currents (a localized electric current induced in a conductor by a varying magnetic field) (Figure 2.1). Since such currents are not affected by resistance within the superconductor their magnitude is directly proportional to the strength of the magnetic field to which they are exposed. However, those currents also produce their own magnetic field. This field adjusts in response to the applied magnetic field, opposing and cancelling it within the interior of the superconductor. This means that the magnetic field produced by the current in the superconductor is equal in magnitude and mirrored (opposite) in direction to the external magnetic field that acts upon the superconductor, effectively counteracting it.

Moreover, unlike a conventional conductor, a superconductor will expel a magnetic field even if that field is already present when the material is cooled below its critical temperature.

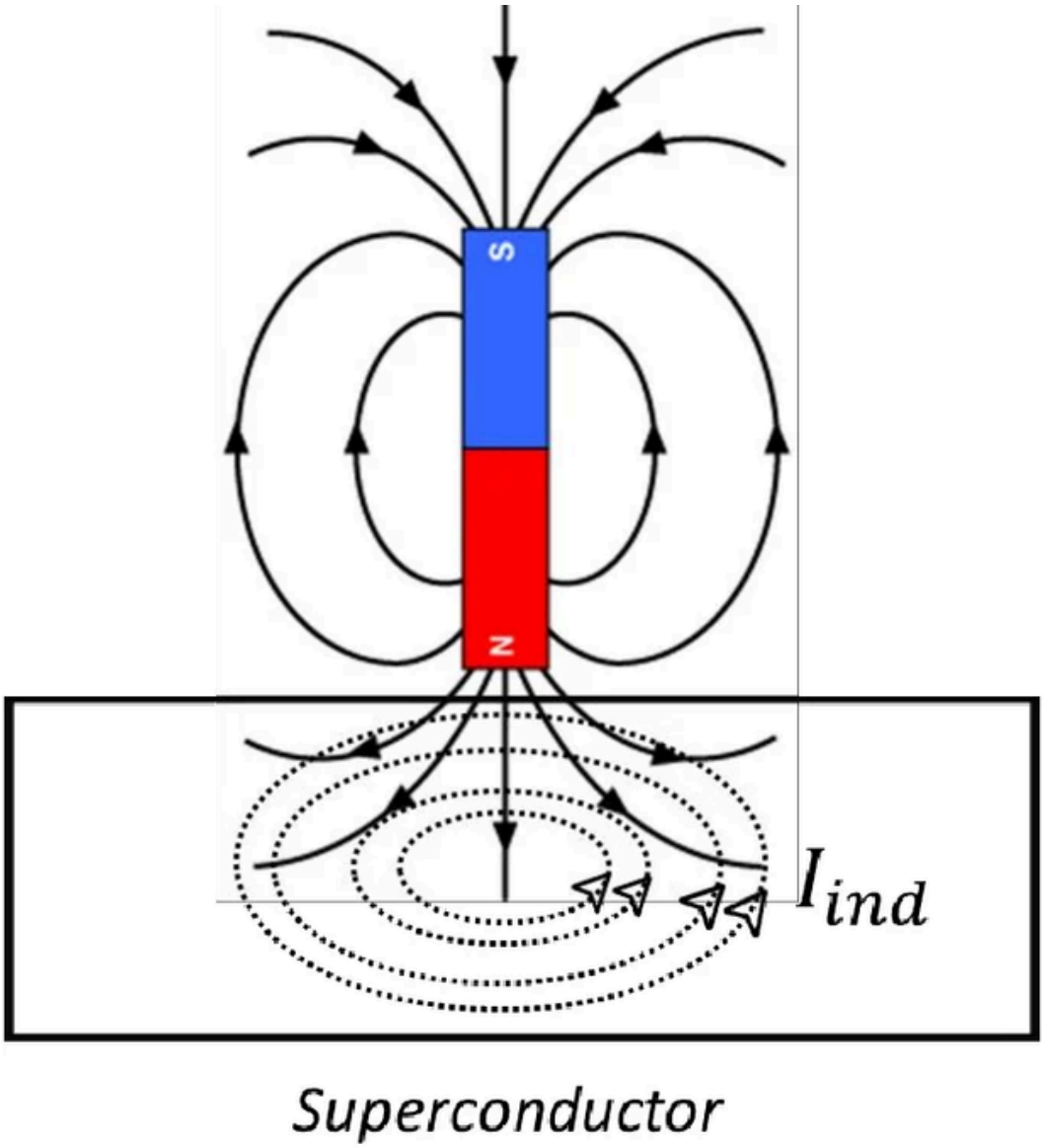

Figure 2.1 Diagram, showing the creation of eddy currents induced by the external magnetic field
Source: Researchgate 2022 by Nabin Raut

This demonstrates that the effect is not simply due to electromagnetic induction, but is instead a fundamental property of the superconducting state.

III. History of superconductors

The discovery of superconductors was accidental. In 1911 the Dutch physicist Heike Kamerlingh Onnes in his laboratory noticed mercury's zero resistance at temperatures lower than 4.2 K. It has led to the creation of an entirely new field within physics called superconductivity. Since then the physics behind superconductors is rigorously studied leading to the discovery of new materials and methods of their synthesis.

Their previously unseen properties have puzzled physicists all around the world, and many explanations of the phenomena were proposed with little to no success. Until in 1935, the theory proposed by Fritz and Heinz London (known as the London theory) was the first success in the field, which was followed by another theory put forward by Vitaly Ginzburg and Lev Landau in the 1950s.

Though the proposed theory was satisfactory it was not perfect and soon a more complete explanation emerged with the work of John Bardeen, Leon Cooper and John Robert Schrieffer. It was nicknamed the BCS theory in their honour. This theory successfully explained superconductivity in conventional materials through the formation of Cooper pairs and remains fundamental to the field.

Later another breakthrough in physics occurred - the discovery of the cuprate high temperature superconductors (HTS) in 1986, which gave a new momentum to the field and intensified the research towards the development of superconductors that can operate at even higher, potentially room, temperatures, which continues to this day.

IV. Real life applications of superconductors

Superconductors already have a range of practical applications, as cooling materials to the boiling points of liquid helium or hydrogen is technically feasible. However, conditions of their use are quite demanding in the face of high costs and complicated logistics used to maintain sufficient temperatures.

Superconducting materials have been explored for use in high-speed connections between components inside high-level computer chips, where reduced electrical resistance can significantly improve performance. In medicine, superconducting coils enabled the generation of extremely strong magnetic fields in devices such as magnetic resonance imaging (MRI) scanners, which are used to produce detailed images of soft tissues within the human body, improving treatments that involve precise surgeries like brain tumor removal.

In countries like Japan and Germany, superconducting magnets are used in experimental magnetic levitation (maglev) trains, allowing them to hover above the tracks and therefore minimise friction, improving efficiency and speed. Similarly, superconducting magnets play a crucial role in particle accelerators, where they are used to guide and focus particle beams for the study of fundamental atomic and subatomic structures.

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Regulations of invisible Traffic Jams

Author: Abdulaziz Maksumov

I. Introduction

Today people hardly imagine life without electronic devices such as mobile phones and computers. One of their important uses is to make distant communication possible. Signals sent by the devices to a certain address travel through the air in high-frequency wave form and through wires as electrical signals. In highly dense settlements hundreds of thousands of users use their devices simultaneously to exchange messages with each other within the small volume of air relative to the number of the users, hence there will arise invisible traffic jams in the air. Modern technology enables billions of devices to share the same space to transmit data simultaneously without having a large signal interference. The aim of the article is to give readers a general view of what the wireless communication system is like.

II. Cellphones

Analogue signals, the data pieces in different forms (e.g., electrical signal, audio, etc.), are converted into digits of 0s and 1s [3]. A *transmitter* (the apparatus that generates radio waves, modulates their amplitude or frequency, and transmits them by means of an antenna [1]) of the phone turns the digits to an electromagnetic wave of a certain frequency [2]. The wave travels to the nearest *base station*. The base station transmits the data to the *Mobile Telephone Switching Office* (MTSO) in the same form. Each call provider company owns one MTSO.

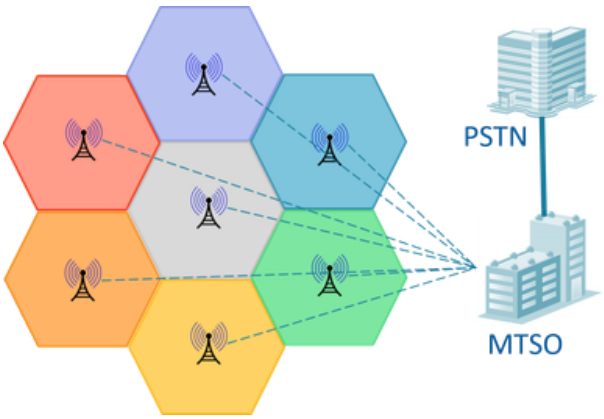


Figure 2: The cells, the MTSO, and the Phone Switched,
Source: Sciencebydegrees.com, 2018

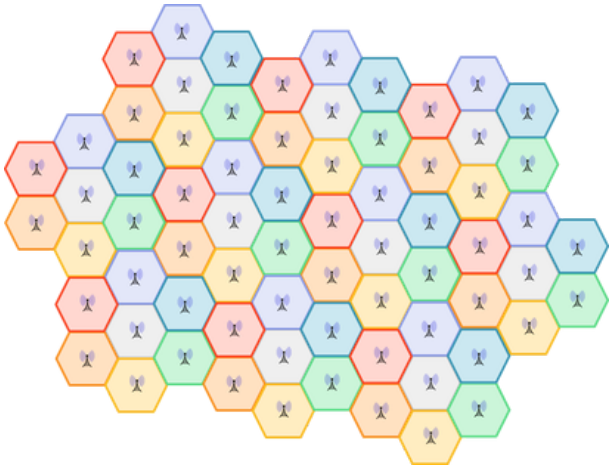


Figure 1: The cells of base stations
Source: Sciencebydegrees.com, 2018

The MTSO then sends the data to the *base station closest* to the receiver's cellphone. Finally, the data reaches the receiver's cell phone from the base station.

Each base station in a certain region makes its own cell. The handset measures the neighbouring cell signal strength, triggering a network-controlled handover at a defined threshold.

The example above illustrates the communication system in a single city. When the communication range goes to the intercity or international level, the data relay would continue forward after the MTSO, and physical wires would be included in there [9].

III. Regulations

It's known that waves of the same frequency interfere. When they interfere the amplitude of the net wave changes continuously causing wrong receiving of the digital data by the devices. This may lead to inaccurate and highly inefficient communication in everyday life.

On the other hand, there are many frequency bands, each of which contains a unique trade-off between coverage range and data capacity. Using the wave frequency of cellular networks to send data to space stations will not give a result just as desired.

To handle this issue, regulatory bodies assign specific frequency bands to specific technologies at international and national levels.

At the international level, the International Telecommunication Union, an agency run by the UN, controls global radio regulations.

The ITU allocates frequency bands to different radio services through a structured international process. The World Radiocommunication Conference (WRC) is held every 3-4 years. National delegations negotiate changes to the Table of Frequency Allocations [7].

The ITU divides the world into three regions: Region 1: Europe, Africa, the Middle East, and the CIS states. Region 2: the Americas. Region 3: Asia-Pacific, Australia. Different regions may have different allocations for the same frequency band.

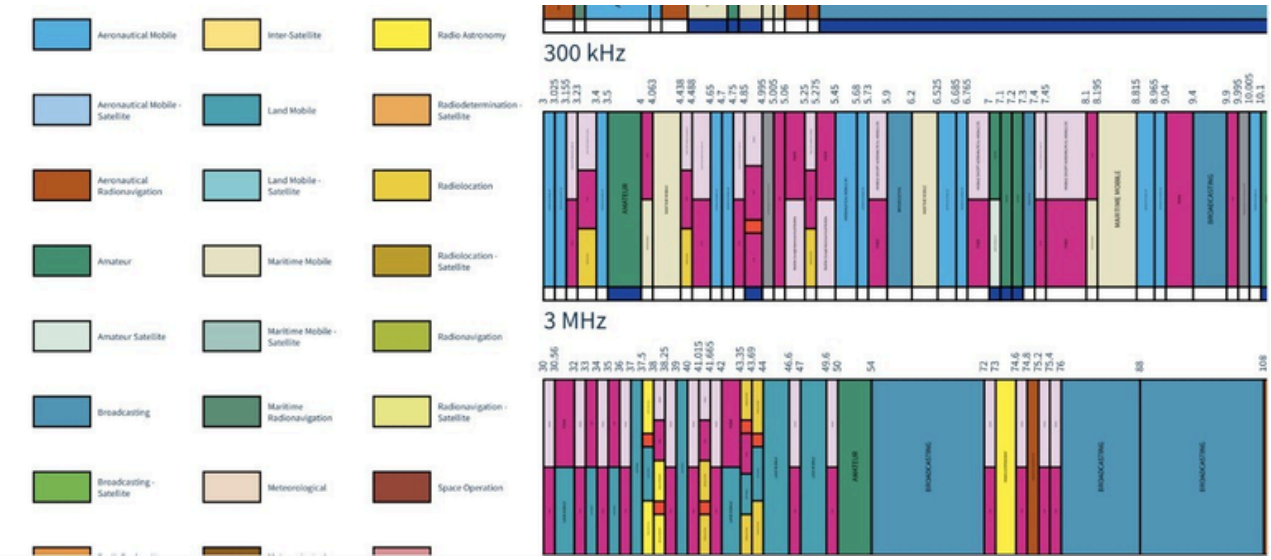


Figure 3: Example of allocated frequency bands to specific areas
Source: U.S. Department of Commerce, National Telecommunications and Information Administration (NTIA), United States Frequency Allocations, September 2025. US government work, public domain.

Example: 902-928 MHz: Region 2 (Americas): allocated to ISM (Industrial, Scientific, and Medical) and used for unlicensed devices (LoRa, Part 15). Region 1 (Europe): allocated to GSM-R (railway mobile) and not available for unlicensed use. All other countries use certain frequency ranges for certain technologies in correspondence with ITU [8].

IV: Multiple Access Techniques

A single cell might have hundreds of active users. In this case, the frequency band assigned to that specific cell may not be enough to offer a unique route for each user at the same time. Splitting radio waves into separate channels for every user of millions wastes available network resources. because each frequency band is considered a regionally limited resource.

Multiple access (MA) techniques allow users to use the same frequency channel by managing allocation of resources such as time, frequency, or code to different users to ensure that they do not interfere with each other.

The fundamental types of multiple access techniques are as follows:

- *Frequency-division multiple access (FDMA)*
- *Time-division multiple access (TDMA)*
- *Code-division multiple access (CDMA)*

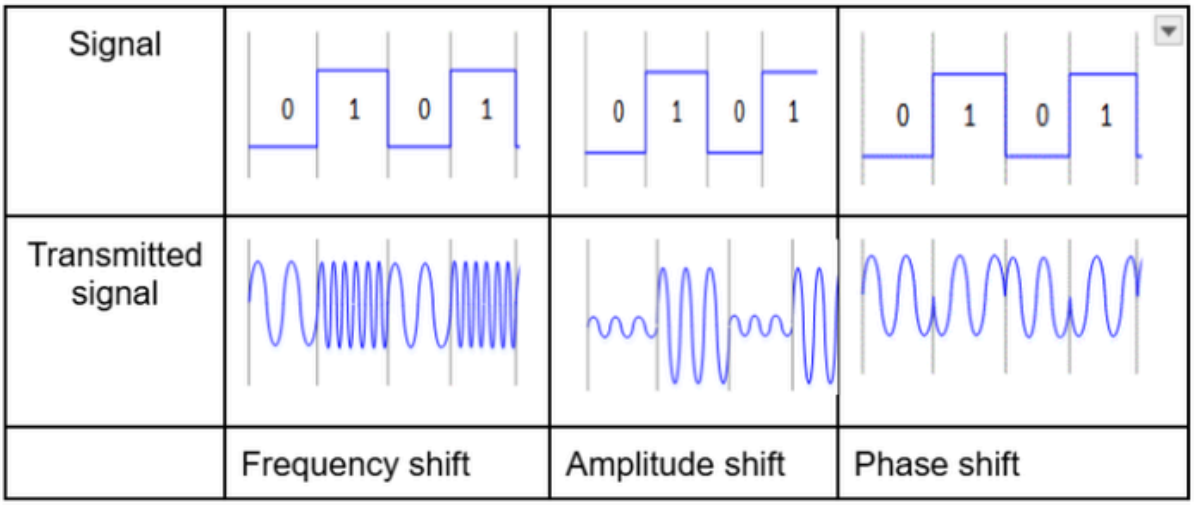
FDMA is the process of dividing one channel or bandwidth into multiple individual bands, each for use by a single user. Each individual band or channel is wide enough to accommodate the signal spectra of the transmissions to be propagated. FDMA allows continuous transmission; it is good for real-time communication [6].

TDMA divides transmission of the same frequency into frames, and each frame is further divided into time slots (e.g., a 10 ms time frame taken on a specific wave frequency is divided for 5 users chronologically, leaving each user with a time slot of 2 ms). Each user transmits during their allocated time slot, and only one user transmits at a time. This technique works well with slow voice data signals and is useful for compressed video and other high-speed data [10].

In CDMA, each user is assigned a unique spreading code which is used to modulate their signal. Before going to CDMA's working principle, a *digital modulation* should be reviewed.

Data in different forms (e.g., voice, text, picture, etc.) is converted to digital data signals of 1s and 0s. The signal first travels through wire inside devices in the form of alternating current with current corresponding to high and low potential differences representing 1 and 0, respectively.

This signal then is converted into a wave by transmitters. To represent 1s and 0s, one of three physical properties, amplitude, frequency or phase, of the wave is changed; usually high means 1 and low means 0 [4].



Source: Circuit Design Inc., Digital Modulation

Now returning to CDMA, the spreading code increases the bandwidth of the signal, allowing it to spread over a wide frequency band. The assigned code to the user, consisting of 1s and 0s, is attached to the original data when transmitted. At the receiver end, the signal is demodulated using the same code to extract the original information. For example, using 64 unique codes allows up to 64 users to occupy the same frequency channel at the same time without interfering with each other. CDMA is used in GPS (Global Positioning System) and in 3G mobile networks [6].

Another advanced access technique is *Orthogonal Frequency Division Multiple Access*, or OFDMA, which is used in Long-Term Evolution, LTE, systems to accommodate multiple users in a given bandwidth. OFDMA is a modulation method that divides a channel into multiple narrow orthogonal (it means unrelated, not affected by one another) bands that are spaced so they overlap but do not interfere with each other. Every band is divided into hundreds or even thousands of 15-kHz-wide subcarriers. The data to be transmitted is divided into many lower-speed bit streams and modulated onto the subcarriers. Time slots within each subcarrier are used to package the data to be transmitted. This allows for high data rates and reduces interference [6].

V. Conclusion

Cellphone working principles show how data travels from one device to the other.
Frequency bands for specific areas are allocated by the International Telecommunication Union (ITU). The communication systems use combinations of shifts of the available communication resources, such as time, empty space, and wave physical properties (amplitude, phase, and frequency), to make productive distant communication existent for as many users as possible. TDMA, FDMA, and CDMA are fundamental multiple access techniques, which use those resources.

There are and will be discovered many other MA techniques and technologies. Those who show relatively higher productivity will be widely released for use and remain until being superseded by the better ones.

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A Review of Hybrid Renewable Energy Systems: Modelling and Integration

Author: Gabriel Ruli

Abstract:

Rising global energy demand and mounting concern over the environmental impact of fossil fuel use have driven a sustained shift towards renewable energy sources such as wind, solar photovoltaics, hydropower, and fuel cells. Although these technologies offer a low-carbon alternative to conventional generation, their output depends heavily on weather and other environmental conditions, which limits the ability of any single renewable source to provide a stable, continuous power supply.

This review examines hybrid renewable energy systems (HRES), which combine two or more energy sources — often pairing complementary renewables or supplementing them with a non-renewable backup — to overcome the intermittency limitations of standalone systems.

Drawing on the existing literature, wind–solar, solar–hydro, fuel-cell-based, and solar–diesel configurations are examined alongside the modelling tools, control strategies, and optimisation methods used to design and evaluate them. The review finds that complementary resource pairing, particularly between solar and wind, substantially reduces intermittency, while fuel cells and diesel generators remain widely used as backup sources despite their higher cost or environmental trade-offs. Simulation and optimisation tools such as HOMER, Particle Swarm Optimisation, and Genetic Algorithms are shown to play a central role in sizing HRES components to balance reliability against cost, and real-world case studies confirm that these systems can achieve lower running costs and reduced fossil fuel consumption compared with single-source alternatives.

These findings underline the practical value of hybrid configurations as an adaptable framework for improving the reliability, cost-effectiveness, and environmental performance of renewable power generation across a wide range of geographical and economic contexts.

I. Introduction

Energy demand is rising. Whether driven by growth in the world's population or by industrial development, the need for greater energy supply grows with every passing moment. The moral imperative to reduce the rapid depletion of natural resources and the environmental damage caused by the use of gas and crude oil directs attention towards renewable energy. These sources, while addressing environmental damage and carbon footprints, pose a major challenge: unreliability [1]. Sources such as solar or wind suffer from significant reliability issues due to meteorological conditions, while others, such as biomass or geothermal, face reliability issues that depend on location. The issue of unreliability can be addressed with Hybrid Renewable Energy Systems (HRES) [2]. By combining two or more energy sources, the weaknesses of individual systems can be compensated for, increasing overall reliability.

The literature also shows a gradual shift in focus, moving away from systems that pair renewable sources with non-renewable backup generation towards configurations built entirely from renewable sources. This shift reflects the broader push towards full decarbonisation, and it is a theme this review returns to in later sections when considering the long-term direction of HRES design.

One of the most promising renewable energy sources is photovoltaic (PV)/solar generators. These systems convert sunlight directly into electricity using the photovoltaic effect, in which photons strike a semiconductor material (usually silicon) and release electrons from its atoms. These electrons are then pushed in one direction, due to a built-in electric field, generating an electric current. This technology is an excellent alternative to traditional fossil fuels or crude oil due to its clean nature and scalability. However, these systems do have their limitations. Their output relies solely on sunlight availability (solar irradiation), which varies with time of day, weather conditions, and seasonal changes.

Due to this disadvantage, PV systems can be less reliable as standalone power sources. Paired with other energy sources, PV systems prove to be not only viable but also an influential contributor to clean, environmentally friendly energy generation [3], as illustrated in Fig. 1.

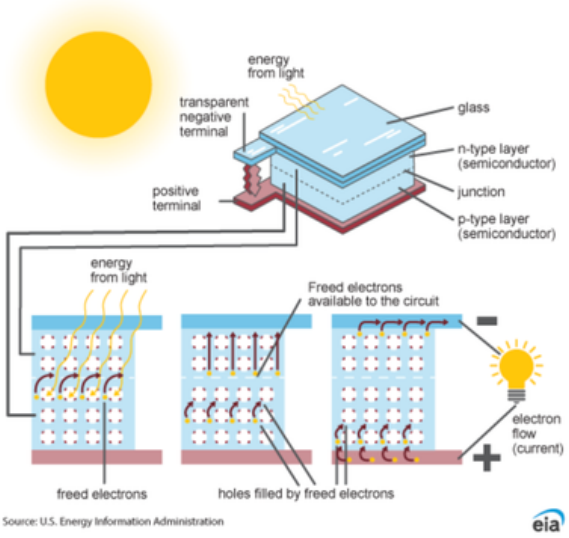

Figure 1: Inside a photovoltaic cell
Source: U.S. Energy Information Administration (US government work, public domain).

Another widely used renewable energy source is wind generation. Wind turbines convert the kinetic energy of moving air into mechanical power, which is then converted to electricity by a generator. When wind passes the turbines, it creates lift, causing the blades to rotate. This rotation powers the generator, producing electricity without the need for combustion or harmful emissions. Wind power is widely considered a clean alternative to energy generation and can be implemented in large-scale wind farms as well as smaller setups, whether grid-connected or not. Wind turbines, however, are not without their own challenges. Their energy output is heavily dependent on wind speed and wind direction, which can be influenced by changes in weather conditions, seasonal changes, and geographical location. These issues may hinder the reliability of this energy source when used alone. However, combined with other renewable or conventional energy sources, wind generation becomes a highly effective contributor to sustainable power production [4], [5], as shown in Fig. 2.

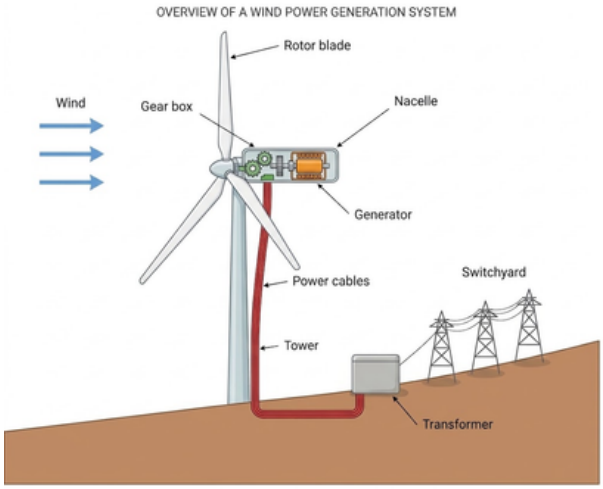

Figure 2. Overview of a Wind Power Generation System diagram, rendered using generative AI [26].

Hydropower is one of the oldest and most established forms of renewable energy, using the natural flow of water to produce electricity. In a typical system, flowing water drives a turbine, which rotates and moves a generator to create electricity. Because the water cycle is continuous and replenishes itself with rainfall and evaporation, hydropower can provide a stable, large-scale energy supply with minimal loss and minimal emissions. Unlike solar and wind, which suffer from intermittency due to the nature of their power production, hydroelectric power plants are often capable of providing a consistent power output. This energy source relies mainly on the availability of water resources, meaning that geographical location and seasonal patterns play a significant role in performance. Additionally, large hydropower plants can raise environmental concerns including habitat disruption and impacts on ecosystems. Use of these systems in hybrid renewable energy systems can enhance efficiency, as well as energy security [6], [7], as shown in Fig. 3.

Fuel cells are another highly efficient and sustainable energy technology. These systems use an electrochemical reaction, through which they generate electricity. Most commonly, this reaction happens between hydrogen and oxygen, without the need for combustion. Inside this cell, hydrogen molecules are split into protons and electrons at the anode. These protons then pass through an electrolyte membrane, forcing the electrons to travel through an external circuit, thus creating an electric current.

At the cathode, protons and electrons recombine with oxygen to form water, making water the by-product of this process. Fuel cells are a very attractive alternative to fossil fuels, offering clean energy generation with high efficiency and minimal emissions. However, their widespread use faces challenges, including high production costs and durability issues. Despite these limitations, the integration of fuel cells into hybrid energy systems holds great potential as a clean energy source and as an output regulator, given their nature [8], as shown in Fig. 4.

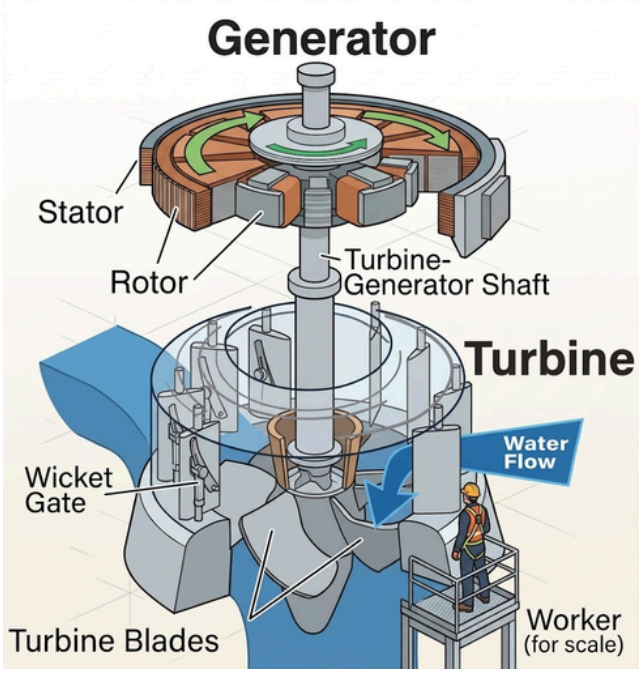

Figure 3: Schematic diagram of a hydroelectric generator and turbine, rendered via generative AI [24].

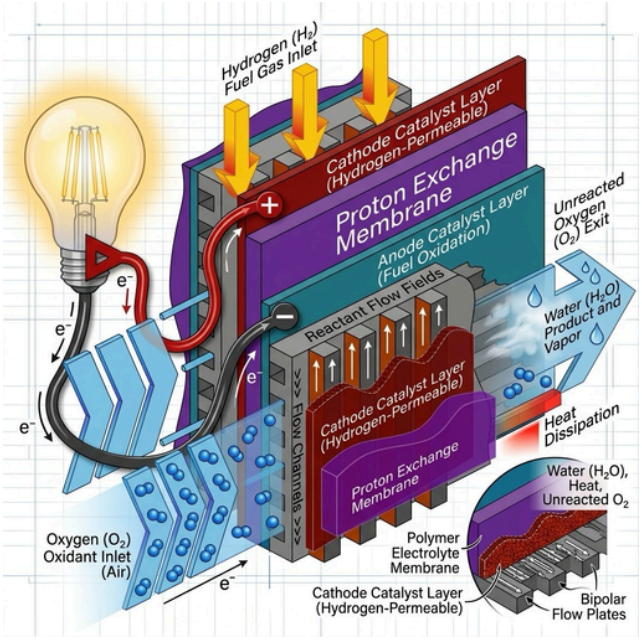

Figure 4: Schematic illustration of a proton exchange membrane fuel cell, rendered via generative AI [25]

II. What is a HRES?

Hybrid renewable energy systems (HRES) are integrated power generation systems that combine two or more energy sources, such as solar, wind, hydro, biomass, fuel cells, and even conventional non-renewable sources such as diesel or natural gas, into a single interconnected system. The purpose of these systems is to solve the above-mentioned issues and increase efficiency as well as reliability. For example, solar power is abundant during the daytime but unavailable at night. At the same time, wind generation can be stronger during evenings or seasonal changes that dampen solar irradiation, allowing one source to balance and fill in the shortcomings of the other. These systems usually incorporate batteries or hydrogen systems to ensure a continuous energy output and address intermittency. By addressing the intermittency issues of standalone renewable systems, these setups help minimise energy shortages and reduce dependence on fossil fuels. They can be implemented in both grid-connected and off-grid applications, making them suitable for both urban and rural areas. Overall, hybrid renewable energy systems present a practical solution to achieving cleaner, more resilient, and more efficient energy production, while aligning with global sustainability goals. For these reasons, an analysis of different hybrid renewable energy systems is presented [9].

III. Literature Review, Modelling, Utilisation, and Analysis of Hybrid Renewable Energy Systems

An integration scheme has been proposed for a wind and solar hybrid energy system that uses both the rotor and grid converters of a Doubly Fed Induction Generator (DFIG) to transfer power from the photovoltaic array to the electrical grid [10]. In this configuration, the DFIG typically used in wind turbines serves as a central interface for managing and optimising the flow of energy from different sources. By channelling photovoltaic-generated electricity through the existing DFIG converters, the system effectively minimises the need for additional hardware, reduces conversion losses, and simplifies overall design.

This proposed method increases efficiency by reducing unnecessary power losses and helps maintain stable power generation, providing a cost-effective way of integrating a photovoltaic generator into the grid.

Bae and Kwasinski [11] present a simulation of a 30 kW hybrid microgrid comprising wind and solar systems, along with a separate energy storage system. The wind power is modelled using a permanent magnet synchronous generator, which produces electricity without needing extra parts like gearboxes. The solar panels are modelled to respond to changes in sunlight. Both sources, along with the load, are tested under real-world conditions where wind, sun and demand continuously change. To manage this, the system uses a special DC-DC converter that can take in power from multiple sources and connect them to one common DC bus. A smart controller (MPPT) was implemented and simulated for each source to ensure efficient use of the system's power output. The simulation shows that this system stays stable and efficient even when conditions vary.

Because the converter combines the system's different energy outputs in a single stage rather than running them through separate stages, the system reduces energy losses and costs. Overall, the results suggest that this system is reliable and affordable as an option for microgrids, especially those with constantly shifting power supply and demand.

Erdinc and Uzunoglu [9] discuss various modelling and sizing methods for hybrid renewable energy (HRE) systems. The combination of different hybrid renewable energy sources with backup units is stated to offer a more economical, environmentally friendly, and reliable power supply compared to single renewable source systems. The key challenge discussed is optimally sizing the components to meet load requirements while still keeping costs low. Different sizing approaches are proposed, such as using software like HOMER to optimally size and simulate HRE systems, or using Genetic Algorithms to account for meteorological changes and fluctuations, and the ways they impact the system. HOMER works by simulating different combinations of power sources, storage, and loads, then comparing them to find the most cost-effective and reliable hybrid energy system design, as shown in Fig. 5.

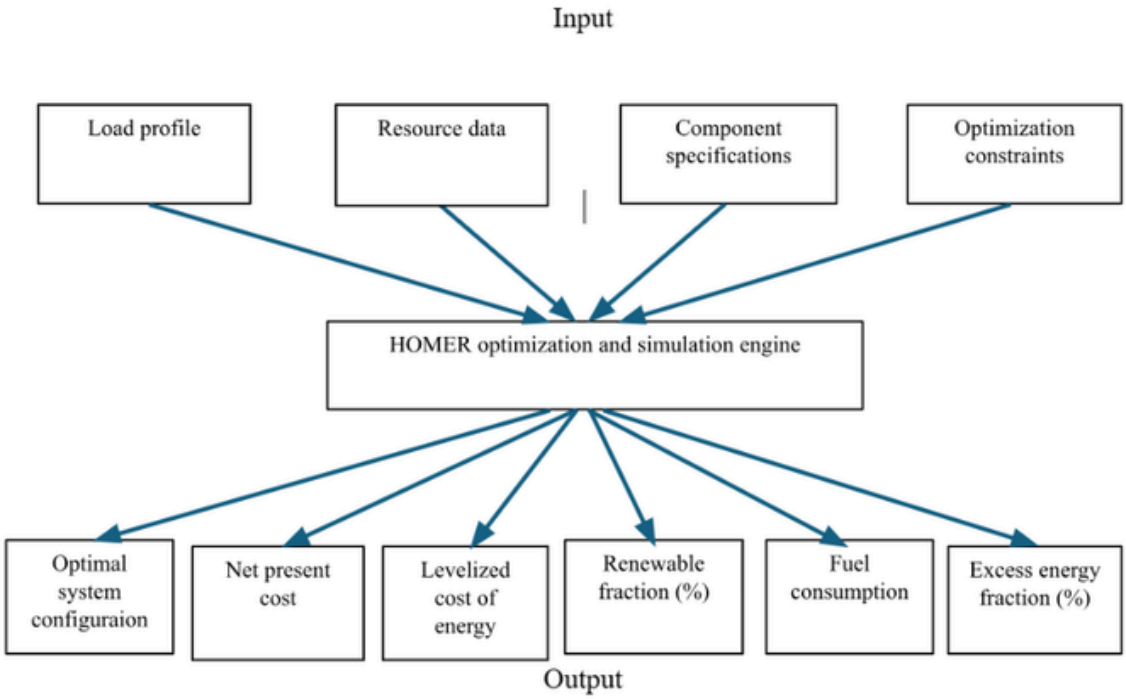

Figure 5. Generalised HOMER methodology for sizing and comparing renewable energy systems
Source: Gabriel Ruli - Original diagram, redrawn from the general methodology described in [9]

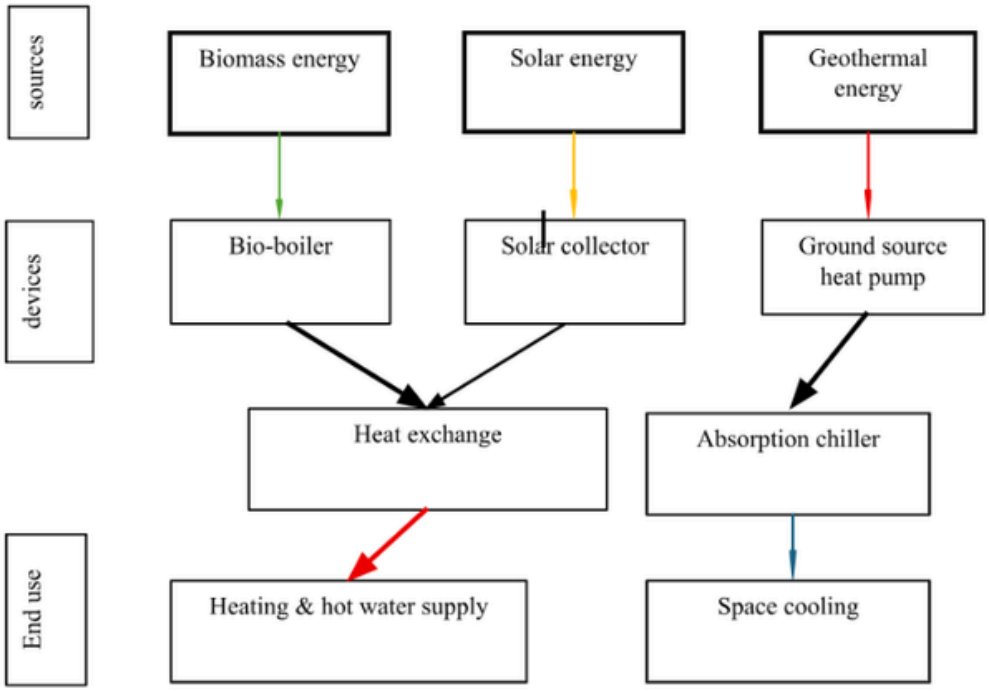

Figure 6: Example use of a solar/biomass/geothermal hybrid system for space heating, hot water, and cooling (Original diagram, redrawn from the system described in [15])

Currently, a significant portion of total energy consumption in most countries goes to space heating and cooling, as well as hot water usage [12]. To reduce the use of fossil fuels as energy generators as well as their impact on the environment, the use of widely available renewable energy systems, such as solar, should be considered for the specific use of space heating and hot water usage. However, the issue of solar irradiance fluctuation has been raised, and the combination of solar energy generators with other renewable energy systems, such as biomass for regions rich in both resources or geothermal for regions with geothermal activity as well as solar irradiance, has been proposed [13], [14].

PV modules produce electricity when solar irradiation is available, storing the produced electricity in a series of batteries, while the biomass subsystem, consisting of a biomass gasifier and a gas net, is coupled to the main alternating current to provide a continuous power supply, as shown in Fig. 6. The system was evaluated using HOMER, and results indicate that it outperforms alternatives with a levelised cost of energy (LCOE) of €0.33/kWh, versus €0.42/kWh for a solar and diesel system [15].

Karami et al. [18] designed and proposed a hybrid energy system consisting of photovoltaic panels, a fuel cell, a battery, and a supercapacitor. Each source plays its own role: the PV panels provide renewable and sustainable power, the fuel cell ensures backup power when sunlight is low, and the battery and supercapacitor help balance changes in demand. To tie it all together, a buck converter — a simple circuit that steps the voltage down while keeping efficiency high [16] — was used, along with a type-III compensator, a control method that keeps the output voltage stable even when the load changes [17]. The PV part of the system also uses MPPT to ensure it always operates at its most efficient state.

Tests, simulations, and prototypes showed that this setup keeps voltage steady, shares the load smoothly between sources, and does so with relatively low cost and simple hardware.

Distributed generation is an increasingly popular concept in modern power systems. The purpose of such systems is to produce electricity at or near the point of consumption rather than relying solely on large, centralised power plants. This approach utilises both renewable and non-renewable sources, as well as storage technologies, to meet demand. On the non-renewable side, distributed generation systems usually consist of small combustion turbines or gas turbines, which can quickly provide power but still have issues of emissions and fuel dependency.

On the other side, renewable distributed generation uses sources such as photovoltaic panels, wind turbines, or small hydropower systems. These offer clean, sustainable energy while facing intermittency issues. To address this variability, storage solutions such as batteries or flywheels are often integrated into these distributed systems, ensuring a continuous energy output. With the combination of these elements, distributed generation systems enhance energy security and lower transmission losses due to their proximity to demand, all while contributing towards more sustainable energy generation [19], as shown in Fig. 7.

Ye et al. [20] presented a microgrid architecture combining wind turbine generators, photovoltaic panels, and hydro generators, and built a detailed simulation model using EMTP/ATP software (software used to test components or machines under real-world scenarios, such as simulating changes in voltage and/or current that occur in power systems when something disturbs the system) to test how all parts interact. In their model, each power source is represented with its own electronic interface (for example, converters or inverters) and control systems so that the full dynamic behaviour (how voltages, currents, and power change over time) can be captured. The control strategies implemented used active power/frequency droop control to manage how much real power each source contributes and to stabilise frequency.

To verify performance, simulations were run under normal conditions as well as under power disturbance scenarios to see how this microgrid would respond. The results show that these models behave in a realistic way and are feasible, and do not show obvious instability or failure, even though the paper does not emphasise strict performance metrics like efficiency or cost in detail.

Ahmad et al. [21] present an experimental evaluation of a small-scale hybrid renewable system composed of a photovoltaic array and a wind turbine installed at the Lebanese University in Tripoli. The system, consisting of three PV panels and a 1 kW wind turbine connected to a battery bank and an inverter, was monitored from 2011 to 2015. The data was collected and analysed to determine the energy contribution of each source individually, as well as the combined performance of the hybrid setup. Results show clear seasonal variations in output, with PV generation dominating during summer months while wind energy is more significant in other periods of the year. Over the five-year measurement period, annual production was estimated at 163.78 kWh for PV and 130.89 kWh for wind.

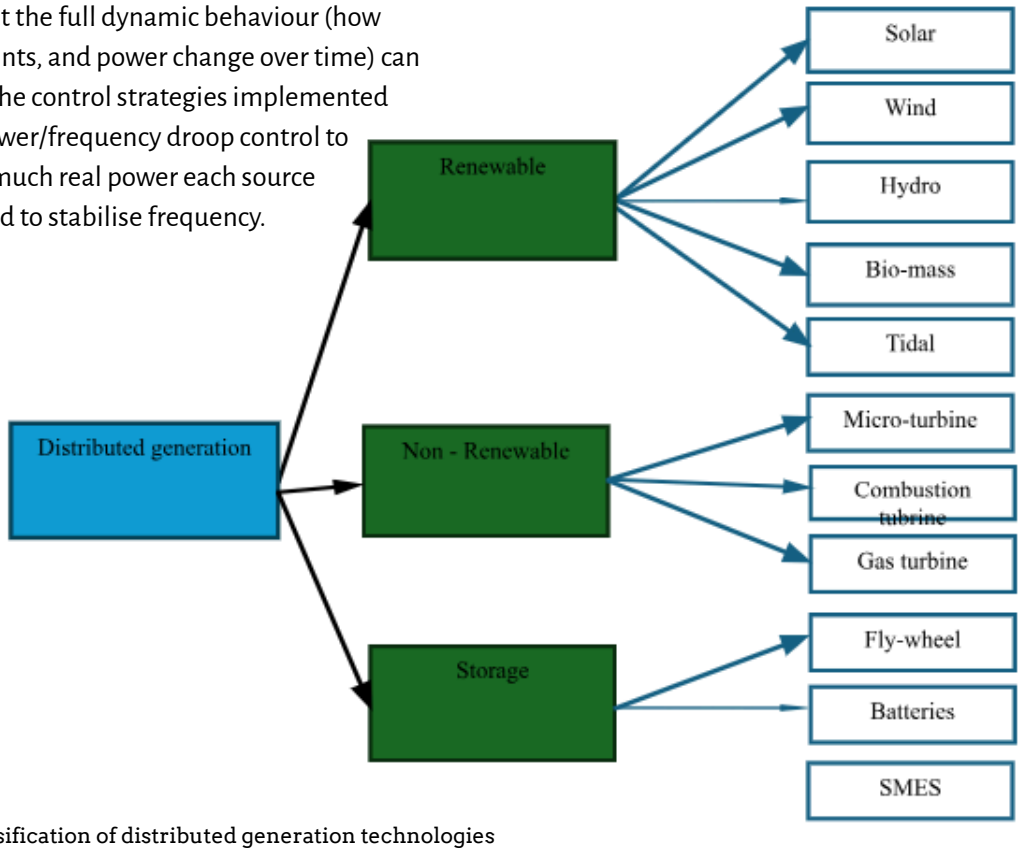

Figure 7: Classification of distributed generation technologies into renewable, non-renewable and storage sources
Source: Gabriel Ruli - Original diagram, redrawn from the system described in [15]

Calculating the data over a 20-year lifetime, the PV system alone produced approximately 16,378 kWh compared to 2,617.8 kWh for wind under a 1 kW baseline. When including economic factors such as installation and maintenance costs, the PV system was determined to have the lowest cost per kilowatt-hour at $0.47, followed by the hybrid system at $0.51, while wind alone reached $3.21. Based on these findings, the authors conclude that PV is the most efficient and profitable option for the studied site, with hybrid systems also showing promise.

Wind and solar energy are widely used renewable energy sources, but face challenges due to intermittency caused by changes in weather or seasonal variations. Ahmed [22] proposes the integration of fuel cells to address these limitations, providing more reliable, continuous power generation. In this system, photovoltaic modules and wind turbines act as the primary sources, connected to DC-DC converters equipped with maximum power point tracking (MPPT) to maximise power output. This combined output is then fed into a common DC bus and linked through an inverter, while the fuel cell subsystem is activated to supply the system when the renewable sources cannot meet demand. Simulation results demonstrate that this system, and in particular the use of fuel cells, effectively suppresses fluctuations caused by wind speed or solar irradiance.

Photovoltaic (PV) paired with diesel hybrid systems are among the simplest and most widely applied hybrid renewable energy systems, especially in rural areas with limited grid access. These systems consist of PV modules that generate electricity during daylight, while a diesel generator provides power during low-irradiance periods, where the PV system is underperforming. Yamegueu et al. [23] investigated a PV-diesel system without an energy storage system, under constant load, to assess its reliability and efficiency in off-grid applications. Their study demonstrated that the system performs efficiently under high solar radiation and heavier load periods, with the diesel generator sized to meet peak demand and maintain a stable energy output. A key advantage of this system is its reduced fossil fuel consumption compared to standalone diesel generators.

This not only lowers operational costs but also facilitates the transition to more renewable-energy-focused power generation, while meeting demand and lowering emissions. The absence of an energy storage system, however, has its own drawbacks, including a heavier reliance on diesel during prolonged periods of low solar irradiance, which also raises costs and environmental impact.

IV. Discussion

The findings from the reviewed literature on hybrid renewable energy systems (HRES) highlight several themes: reliability improvement, cost effectiveness, and sustainability. While standalone renewable systems such as solar, wind or hydro each present environmental advantages over fossil fuel-based generation, their individual limitations, primarily intermittency and location dependence, make them less reliable for a continuous energy supply. Hybrid systems, by combining multiple energy sources with complementary characteristics, have emerged as a practical pathway to achieving both efficiency and reliability while advancing global sustainability goals.

One key point is the role of complementary resource behaviour in reducing intermittency. For example, the integration of solar and wind has shown strong promise, as both peak at different times. While solar irradiance is more abundant during daytime, allowing peak PV system output, wind generation can increase during evenings, nights and seasons when solar irradiance is low, allowing these two systems to almost perfectly complement each other. The Lebanese case study [21] confirmed that this hybrid system can mitigate seasonal variability and balance the output between the two sources. This balance, however, does not entirely eliminate intermittency, which underlines the importance of incorporating storage technologies and units such as fuel cells or diesel generators.

Fuel cells have been repeatedly emphasised as a reliable backup option. Their rapid response in delivering a stable voltage makes them effective at smoothing out fluctuations from variable renewable sources. Simulations demonstrated that systems such as a wind-solar-fuel cell combination could suppress unpredictability while maintaining stable power output.

Even though the results showed a high upfront cost for fuel cells and hydrogen storage systems, their contribution to these systems and the reliability they provide make them an attractive option, especially for grid-connected systems or situations where an uninterrupted supply is essential.

At the same time, simpler systems like solar and diesel remain widespread and practical in rural electrification due to their technical simplicity and cost feasibility in areas where biomass or wind are limited. These systems have been shown to reduce diesel consumption compared to standalone fossil-fuel generation, thereby lowering costs and emissions. Their reliance on non-renewable fuel sources, however, raises questions about their long-term sustainability — a tension that is one of the drivers behind the broader shift towards fully renewable configurations noted in the introduction.

Another point raised by the literature is the challenge of optimal sizing and system design. Improper sizing can erode the efficiency and benefits of HRES, leading to energy shortages or unnecessary investment in oversized components. Various approaches, ranging from simulation software such as HOMER to advanced algorithms such as Particle Swarm Optimisation (PSO) and Genetic Algorithms, have been proposed to account for load profiles, meteorological variability, and cost estimation. These methods are important for ensuring that hybrid systems deliver both technical reliability and economic viability.

Environmental and economic trade-offs are another point of discussion. While systems like solar and biomass hybrids can achieve a lower levelised cost of energy (LCOE) compared to solar and diesel systems, the availability of biomass resources is region-specific, limiting their universal application. Hydropower, though reliable, raises ecological concerns such as habitat disruption. Thus, the selection of hybrid configurations must be context-specific, considering not only technical and economic factors but also social and environmental impacts.

V. Conclusion

The transition towards renewable energy is no longer a choice but a necessity in order to reduce greenhouse gas emissions, slow global warming, and preserve natural resources for future generations. While individual renewable systems such as solar, wind, or hydro offer significant environmental advantages, their dependence on meteorological and geographic factors limits their reliability when used alone. The literature reviewed shows that hybrid renewable energy systems offer an effective solution by combining the complementary strengths of multiple resources into a single integrated setup.

Studies have shown that pairing resources such as solar and wind can reduce intermittency. At the same time, the use of fuel cells or diesel units can ensure a continuous supply even during periods of low renewable output. Simulation tools and optimisation algorithms have also helped improve system design, reduce energy losses, and ensure cost effectiveness. In addition, real-world experiments of such systems confirm the practicality of these setups and their ability to deliver reliable power over extended periods.

However, hybrid systems are not without limitations. The absence or high cost of storage technologies can increase reliance on non-renewable backup sources, while economic feasibility remains highly dependent on local resource availability and regional conditions. Environmental and social considerations also remain important when evaluating specific resources such as hydropower or biomass.

Overall, the reviewed studies suggest that hybrid renewable energy systems are not a one-size-fits-all solution but a flexible framework adaptable to diverse geographical, economic, and social contexts. For developed regions with strong grid infrastructure, the focus lies on optimising renewable energy systems with advanced control and storage integration. In contrast, developing regions and remote areas may continue to rely on transitional systems such as solar-diesel, gradually moving towards cleaner alternatives as technology becomes more affordable.

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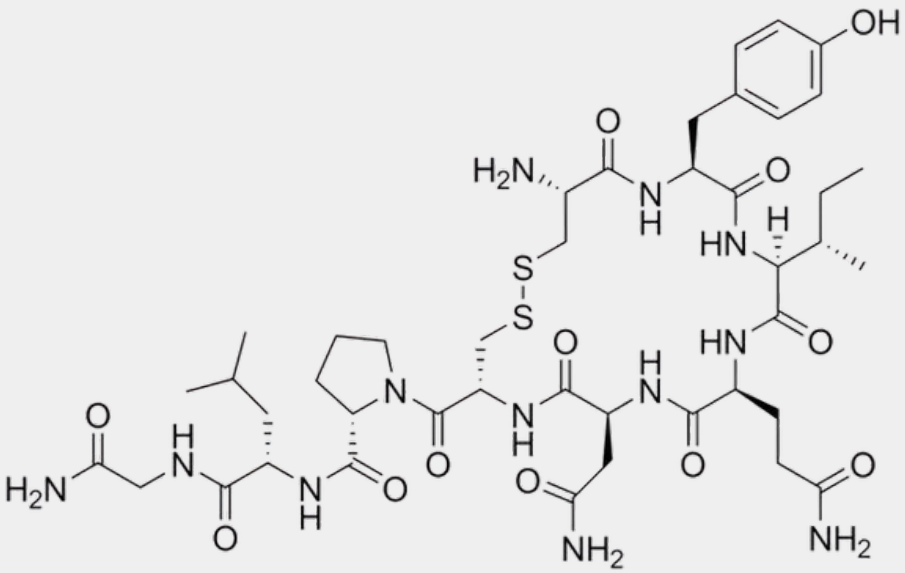

Ocular Immune Privilege: Mechanisms and Pathophysiological Consequences

Author: Anna Sazonova

Abstract:

Ocular immune privilege is a highly regulated state in which standard immune responses are modulated to preserve vision as a primary function of the eye and protect intraocular structures from inflammation-induced damage. It is maintained via blood-ocular barriers, immunosuppressive microenvironments and systemic immune tolerance. Despite such autonomy, the protection is not absolute. Ocular trauma or surgery may lead to the disruption of immune privilege, which then can contribute to sympathetic ophthalmia, a bilateral granulomatous inflammatory disorder. This provides a valuable model for understanding major consequences of tissue-specific immune regulation breakdown with further implications for the field of transplantology. In this review, cellular and molecular mechanisms underlying ocular immune privilege and standard immune response are examined, with particular emphasis on normal physiology followed by a pathophysiological perspective.

I. Introduction

The immune system is indispensable for our survival. It maintains the integrity of the host's cells, whilst actively defending our organism against foreign antigens or pathogens that may invade. Although rapid elimination of 'non-self' is a substantial part of the immune response as a process, a delicate balance between immune activation and regulation is required to maximise its efficiency. An insufficient immune response would prove unsuccessful against the infection, whereas excessive or misdirected immune activity may result in inflammation-induced tissue damage and autoimmunity. This becomes particularly prominent in specialised organs that, if exposed to a standard immune response, may lose their primary function, with consequences of an inflammatory injury extending beyond the initial scale of response itself.

The eye is a naturally occurring example of such. Great reliance on highly specialised structures responsible for vision creates a strong physiological requirement for mechanisms that can limit potentially destructive immune activity without compromising immune defence. Immune privilege introduces a concept of selectivity and, therefore, addresses an important matter as to how an organ with low regenerative ability may retain its capacity for an immune response whilst protected from immune-mediated damage. Understanding the mechanisms underlying this balance, as well as the consequences of its disruption, provides a broader insight into the regulation of immune responses in the body and informs the development of strategies to mitigate immune-mediated rejection in transplantation and promote immune tolerance.

II. Immune System & Response

The immune system is a complex group of defence responses found in humans and other advanced vertebrates that helps repel disease-causing organisms called pathogens1. Normally, its duty would include not only allocating a response to an invading pathogen, but also distinguishing self from non-self. To do that, both *innate* and *adaptive* mechanisms would be used.

2.1 Innate Immune System

First comes what is commonly referred to as the *innate immune system*. Its components include physical and chemical barriers, cellular defences (neutrophils, macrophages, dendritic cells, natural killer cells or NK cells) and proteins that allow immediate, non-specific protection. Additionally, *soluble mediators* such as *cytokines* and *chemokines*, which facilitate cell signalling and immune cell guidance respectively, are of major significance in innate immune response efficiency2.

Innate immune cells detect pathogen-associated molecular patterns (PAMPs) by using pattern recognition receptors (PRPs). The activation of these triggers the production of *anti-inflammatory cytokines*[3], increases *phagocytosis* (cell engulfment) and promotes further immune cell recruitment to the site of infection[4].

2.2 Adaptive Immune System

In contrast, the adaptive immune response involves high specificity in pathogen recognition using gene recombination processes and, therefore, allows for immunological memory to be built[5]. T- and B-lymphocytes, generated namely in the thymus and bone marrow, are predominant in this mechanism. While B lymphocytes are mostly responsible for toxin and pathogen removal, T-lymphocytes coordinate the immune response and engage in homeostasis maintenance, meaning these cells ensure no malignancy is present. However, due to an increased specialisation of T-cells, few of them can directly bind to an antigen unless it has been processed by a dendritic cell. Hence, antigenic peptides, a short form of a precedent antigen, are presented in a complex with a plasma membrane-bound major histocompatibility complex (MHC) class I or class II molecule (MHC-I or MHC-II)[6]. This leads to an accumulation of T-lymphocytes in regional lymphatic nodes that serve as hubs for the body's immune response launch.

CD4+ T-lymphocytes, otherwise known as Th (T helper) cells, regulate the immune response onward, secreting cytokines, whilst CD8+ cytotoxic T-cells recognise pathogen-infected or mutated somatic cells and eliminate them. Concurrently, B-lymphocytes, triggered by cytokines released, proliferate and differentiate into plasma cells that liberate antibodies specific to the antigen bound[7].

2.3 Immune Tolerance

Besides responding rapidly to an invading pathogen, our organism uses an immunological tolerance system. It is a state of unresponsiveness to self-antigens, which helps prevent the immune system from attacking healthy tissue. To put it another way, self-reactive (able to trigger an autoimmune response) T- and B-lymphocytes are destroyed at the maturation sites, suppressed by regulatory T-cells (Treg), or subjected to apoptosis (cell death). Failure to eliminate such cells may lead to the onset of autoimmune disea

III. Ocular Immune Privilege

Immune privilege was considered an evolutionary adaptation to protect irreplaceable tissues and organs from an uncontrolled inflammatory response. In 1948, Medawar defined immune-privileged organs as sites where foreign tissue grafts could survive for extended, often unlimited periods of time, whereas similar grafts placed at regular sites would be rapidly rejected[8]. Except for eyes, other immune-privileged structures remain: brain, hair follicles, cartilages, liver, adrenal cortex, pregnant uterus, placenta, ovary, testis, prostate, tumours and hamster cheek pouch[9]. Despite a variety of tissues belonging to this category, ocular immune privilege is thought to have been studied the most.

3.1 Anatomy & Function as precursors to Immune Privilege

A clear image on the retina is formed through transmission and refraction of light. Transparency and precise arrangement of internal structures, including the cornea and lens, facilitate our vision as the primary function of the eye site.
The existence of ocular immune privilege is necessitated by a limited ability for regeneration of highly specialised compartments. Vision could simply be obliterated and permanently impaired as a result of a normal inflammatory response: swelling from fluid leakage, influx of immune cells, increased vascular permeability and vasodilation would obstruct light paths and, consequently, may lead to irreversible corneal opacity (clouded vision), scar formation, retinal oedema or photoreceptor degeneration[10].

3.2 The Mechanisms of Ocular Immune Privilege

Ocular immune privilege is mostly provided by blood-ocular barriers, namely blood-retinal and blood-aqueous ones, which restrict the free passage of immune cells and large protein molecules into the intraocular environment from the bloodstream[11]. Moreover, it also lacks lymphatic vessels, which are common paths for detecting non-self as discussed in the Immune Response block. The blood-aqueous barrier is formed from non-pigmented ciliary epithelium, located in the ciliary body, and specialised blood vessels of the iris that include non-fenestrated (non-porous) capillaries.

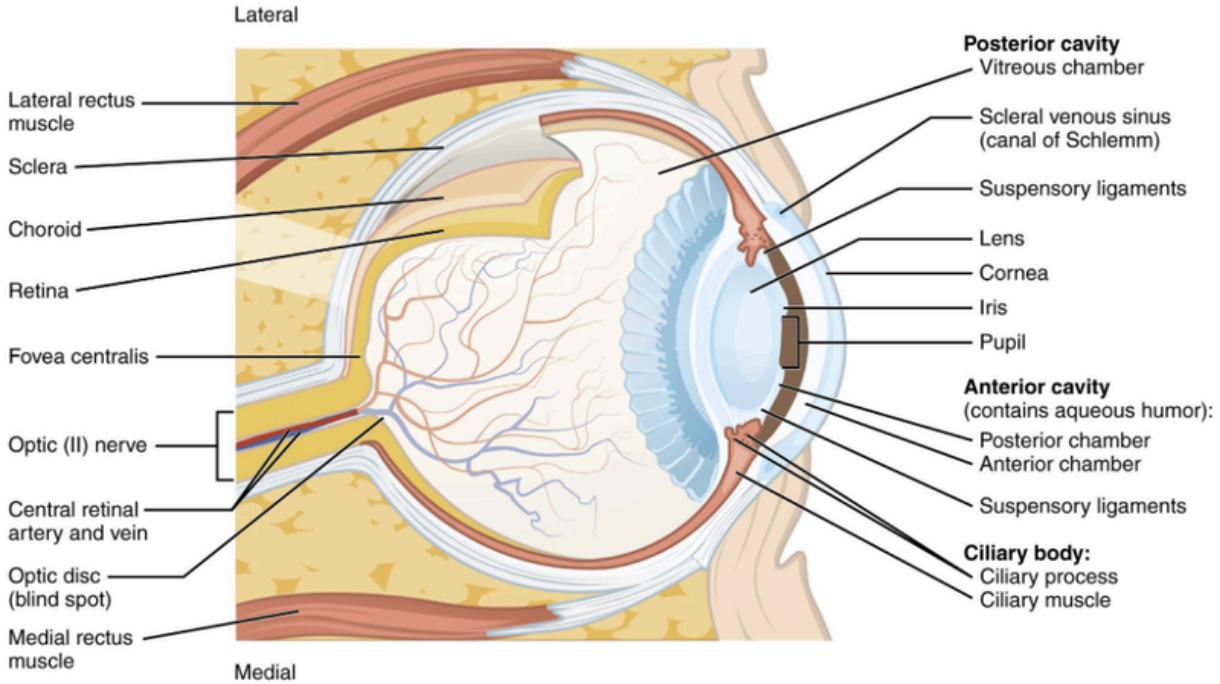

Fig. 1. Anatomy of an eye. Image: OpenStax, Anatomy & Physiology, via University of Hawai'i OER Pressbooks. CC BY 4.0.
Source: https://pressbooks-dev.oer.hawaii.edu/anatomyandphysiology/chapter/sensory-perception/

Tight junctions with the endothelial (lining the vessel internally) cells prevent leakage into the anterior chamber of the eye. Overall, these anatomical structures limit the passage of leukocytes and antibodies into the eye's interior[12].

Nevertheless, ocular immune privilege extends beyond anatomical isolation. An immunosuppressive environment is actively cultivated within the site. Aqueous humour is rich in concentrations of transforming growth factor-beta (TGF-β) and alpha-melanocyte-stimulating hormone (α-MSH), which both contribute to the inhibition of T-cell activation, promoting an anti-inflammatory state11. Vasoactive intestinal peptide (VIP), calcitonin gene-related peptide (CGRP), and interleukin-10 suppress the activation of macrophages, dendritic cells and T-lymphocytes, whilst promoting the differentiation of Treg cells[13],[14],[15]. Consequently, antigen-presenting cells acquire a tolerogenic phenotype and induce not the inflammatory response, but immunological tolerance[16].

An important component of immune privilege is the expression of molecules that induce apoptosis in activated immune cells.

Corneal, iris, ciliary body and retinal pigment epithelial cells express Fas ligand (FasL/CD95L) that interacts with the Fas receptor on activated T-cells, thereby inducing their programmed cell death. The analogous function is attributed to a programmed death-ligand (PD-1/PD-L1) that inhibits T-cell activation and minimises pro-inflammatory cytokine production[17],[18].

One of the most unique mechanisms is an Anterior Chamber-Associated Immune Deviation (ACAID) – a specialised form of immune tolerance[11]. An antigen exposed to the anterior chamber is captured by antigen-presenting cells to migrate to the spleen, where the production of antigen-specific regulatory T-lymphocytes, rather than the conventional immune response activation, is held[19]. As a result, a systemic delayed-type hypersensitivity is suppressed, thereby lowering the chances of a destructive inflammatory response.

In spite of prominent immune defence mechanisms of the eye, immune privilege only shields the respective site to an extent.

Penetrating injury, ocular trauma, infections or surgical interventions may disrupt the integrity of the blood-ocular barriers, leading to the exposure of previously sequestered antigens to the immune system. Presentation of these may result in the breakdown of peripheral immune tolerance and the initiation of an autoimmune response.

IV. Sympathetic ophthalmia

Sympathetic ophthalmia (SO) is a rare, devastating disease defined as bilateral uveitis following an eye injury. This condition is characterised by the development of granulomatous inflammation with distinctive yellow-white choroidal nodules in both eyes, typically emerging within 12 months post-injury or invasive procedure. The primary site of trauma or surgery is referred to as the 'inciting' or 'exciting' eye, whilst the affected area is called the 'sympathising' eye. Although relatively rare, it is considered to be one of the major models of immune privilege destruction, as its pathogenesis is tightly connected to the breakdown of immune tolerance[20].

Under normal physiological conditions, antigens from the uveal tract, retinal pigment epithelium, and photoreceptors are hidden from the immune system by the blood-ocular barriers and immunosuppressive microenvironment. With penetrating injury, these antigens gain access to the local conjunctival lymphatics, where uptake by antigen-presenting cells becomes possible. Then, primarily CD4+ T-cells are activated, which induces a subsequent rise in CD8+ cytotoxic cells, regulating an autoimmune pro-inflammatory response as it progresses. Even though no specific autoantigen has been determined to be responsible for the onset of the disease, several are thought to contribute most actively: interphotoreceptor retinoid-binding protein (IRBP), S-antigen (arrestin), recoverin, rhodopsin, melanocyte-associated tyrosinase and retinal pigment epithelium-associated antigens[20].

Symptoms may occur within the first few weeks, yet cases developing a disease years after ocular injury have also been reported. Historically, more young individuals had been diagnosed with sympathetic ophthalmia due to high volumes of accidental injuries.

However, an increased occurrence amongst adults has been witnessed recently, potentially linked to greater rates of ocular surgeries[20]. Patients typically present with lowered visual acuity, photophobia, pain, epiphora, floaters, and redness. The severity of vision loss can range widely, with some presentations including *Dalen-Fuchs nodules*, yellow-white spots most commonly observed near the equator or peripheral regions of the eye[21].

They comprise lymphocytes and epithelioid cells between the retinal pigment epithelium and Bruch's membrane.

V. Treatment

Initially, the surgical removal of the 'inciting' eye within 14 days of trauma was viewed as a helpful intervention preventing the onset of sympathetic ophthalmia. Nonetheless, the practical application of this method has proved unsuccessful, with the disease occurring regardless. Furthermore, patients statistically had better vision in the 'inciting' eye, meaning its removal would not only fail to mitigate the issue but also not improve the visual acuity. Interestingly, many cases where the traumatic globe was not removed have indicated no development of sympathetic ophthalmia. Yet, the decision of when the radical surgical intervention is needed is of great concern and should not be considered prior to alternatives such as enucleation or evisceration, which preserve extraocular structures and are generally less invasive[20]. A complete loss of an eye may lead to detrimental psychological impacts, including dysmorphia, increased anxiety or depressive symptoms – hence, any procedures should be carefully weighed against ethical considerations and effects on the patient's self-image.

Additionally, high-dose systemic corticosteroid therapy is used as a treatment. Given the substantial risks of relapse and the chronic nature of the disease, most patients require immunosuppressive therapy that may include immunomodulating drugs such as methotrexate, azathioprine, mycophenolate mofetil, cyclosporine, chlorambucil, cyclophosphamide, infliximab and adalimumab. Early diagnosis significantly improves health outcomes and successfully preserves functional vision for the majority[20], [22].

Sympathetic ophthalmia therefore provides compelling evidence for the significance of the ocular immune privilege role in maintaining the functionality and integrity of the eye. Disruption of both anatomical and chemical barriers results in the activation of the autoimmune response.

The research in this field has largely advanced the current understanding of mechanisms underlying sympathetic ophthalmia as well as sophisticated the general concept of immune-privileged sites, contributing to the development of modern immunosuppressive therapeutic strategies.

VI. Conclusion

Ocular immune privilege represents a multilayered system of protection encompassing physical blood-ocular barriers, the production of anti-inflammatory cells, and peripheral immune tolerance. Together, these mechanisms minimise the risks of permanent loss of function and protect irreplaceable tissues from damage. In contrast to the old concepts, modern understanding of this process underpins the dynamic nature involving active interactions of immune cells and the specialised intraocular environment as opposed to passive isolation of eye tissues.

Disruption of the ocular immune privilege following penetrating injury, surgical interventions or infections results in the exposure of sequestered antigens to the immune system, leading to the development of autoimmune disease. Sympathetic ophthalmia, as a primary example of this process, demonstrates the close relationship between immune tolerance breakdown and cell-mediated autoimmune response.

Further investigation into ocular immune privilege is of fundamental and clinical importance. Understanding deep molecular processes of local immune regulation may allow for alternative, more effective ways of treating or preventing autoimmune diseases, presenting us with innovative strategies for improving the success of corneal transplantations and tailoring treatments for inflammatory disorders of the eyes.

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Antibiotic Resistance: A Global Health Crisis in Slow Motion

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Abstract:

Antimicrobial resistance (AMR) occurs when microorganisms that cause infection no longer respond to antimicrobial medicines. This resistance develops naturally over time as pathogens evolve through genetic changes. Antibiotics have been among the greatest advances in modern medicine, greatly reducing illness and death from infectious diseases. However, antibiotic resistance is now a major challenge, leading to increased illness and mortality and posing a serious global health risk. The World Health Organisation (WHO) considers AMR one of the top public health and development threats worldwide.

This article examines the scientific aetiology of antibiotic resistance, providing a comprehensive overview of its development and its implications as a preeminent global health crisis.

I. Introduction

Antibiotics have been a foundation of modern medicine, acting as vital antibacterial agents that treat infections by controlling bacterial growth and reproduction. This medical revolution has saved millions of lives by competently addressing previously life-threatening bacterial infections.

However, the misuse and excessive use of antibiotics, combined with incomplete treatment and inadequate hygiene, have precipitated a worldwide health crisis.

Antimicrobial resistance (AMR) is the process by which microorganisms that cause infections become resistant to antimicrobial drugs. AMR is a natural process that develops over time through genetic changes in pathogens.

As a specific form of antimicrobial resistance (AMR), antibiotic resistance occurs when antibiotic use creates selective pressure, favouring bacteria with genetic variations that enable them to survive exposure to antibacterial agents. Consequently, antibiotics become ineffective, producing prolonged infections, increased therapeutic complexity, and a substantial risk of fatality.

Furthermore, the marked increase in AMR among common bacterial pathogens now threatens treatment efficacy and jeopardises successful outcomes for critically ill patients. Significantly, the World Health Organisation has named antibiotic resistance as one of the three most significant public health threats of the 21st century.

Often described as a '*silent pandemic*', antimicrobial resistance progresses gradually and frequently goes unnoticed. It is directly responsible for 1.3 million deaths annually and contributes to an additional 5 million deaths each year (UN 2024). This silent, rapidly growing threat is straining health systems worldwide. By 2050, it is projected that over 39 million people could die from antibiotic-resistant infections (UN 2024). Given the mounting evidence, it is imperative to implement immediate and coordinated interventions across the human, veterinary, and environmental sectors. Only through such comprehensive action can we effectively curb the anticipated rise in preventable deaths.

II. Genetic Basis of Antibiotic Resistance

Bacteria are genetically resilient to environmental threats. Driven by the misuse and overuse of antibiotics, bacteria develop resistance through major genetic strategies. These can be organism-specific or acquired through spontaneous mutation or horizontal gene transfer. Bacteria can also temporarily exhibit adaptive resistance in response to external stimuli.

2.1 Intrinsic Resistance

Intrinsic resistance is the natural resistance of microbial pathogens to certain antibiotics due to structural or biochemical traits. In contrast to acquired resistance, it is an inherited trait that functions as a natural defence mechanism (Nimmana and Nguyen, 2026).

2.2 Adaptive Resistance

Adaptive resistance refers to transient resistance developed in response to external stresses, including efflux pump expression (transport proteins that expel substances, including some antibiotics) or biofilm formation (Nimmana and Nguyen, 2026). Adaptive resistance is a temporary alteration in bacterial gene expression or physiology in response to environmental conditions, rather than a permanent change in genetic material.

2.3 Acquired Resistance

Acquired resistance is a qualitative alteration in the genetic material of a cell that is phenotypically correlated with a measurable decrease in the cell's sensitivity to one or several chemotherapeutic agents (J. Drews). Bacteria develop resistance through two major genetic mechanisms: genetic mutation and horizontal gene transfer (HGT).

2.3.1 Genetic Mutations

Spontaneous point mutations may alter antibiotic target sites, reduce drug binding affinity, or decrease cellular permeability, causing antibiotic resistance.

2.3.2 Horizontal Gene Transfer (HGT)

HGT is the movement of genetic information between organisms; it includes the spread of antibiotic resistance genes among bacteria, fueling pathogen evolution (Alita R., 2015). This process occurs through three primary mechanisms: conjugation, transformation, and transduction (Alita B. 2015).

3. Mechanistic Basis of Antibiotic Resistance

Bacteria have developed refined mechanisms over millions of years to evade antimicrobial molecules. Resistance is often achieved through multiple simultaneous biochemical pathways, allowing the cell to survive the antibiotic's effects. The primary mechanisms include:

3.1 Modification or Destruction of the Antibiotic

This occurs through enzymatic degradation of the molecule or the addition of specific chemical moieties, both of which prevent the drug from engaging its molecular target. One of the primary resistance strategies involves inactivating antibiotics through enzymatic processes.

3.1.1 Enzymatic Modification

Pathogens produce enzymes that chemically modify an antibiotic's structure, thereby neutralising its inhibitory effects. These enzymatic alterations induce steric hindrance, in which the added chemical moiety physically obstructs the drug, preventing it from binding to its target sites, such as the ribosome or the cell wall (J. , C. 2016). Typical types include **acetylation**, **phosphorylation**, and **nucleotidylation**.

3.1.2 Enzymatic Destruction

Microorganisms produce specialised enzymes that degrade antibiotic molecules before they reach their targets, rendering the drugs ineffective. This mechanism, mainly through **enzymatic hydrolysis**, irreversibly cleaves essential chemical bonds, compromising the drug's molecular integrity (J. , C. 2016).

3.2 Decreased Antibiotic Penetration and Efflux

These mechanisms reduce intracellular drug concentrations, allowing bacteria to survive otherwise lethal antibiotic levels.

3.2.1 Decreased Permeability

To be effective, antibiotics must cross the bacterial membranes of bacteria (particularly in Gram-negative species). Bacteria reduce antimicrobial uptake by limiting entry through the outer or cytoplasmic membranes, preventing the drug from reaching intracellular targets (Munita and Arias, 2016).

3.2.2 Efflux Pumps

Efflux pumps are specialised, proteinaceous, active-transport mechanisms located in the cytoplasmic membrane of both prokaryotic (bacterial) and eukaryotic cells (Oxford, 2003). These pumps actively export toxic substances, including multiple classes of antibiotics, from the cytoplasm to the external environment.

3.3 Alterations in Target Sites

Bacteria evade antimicrobial action by altering the molecular target's structure. Pathogens utilise several strategies (Munita and Arias, 2016):

- **Target Site Mutation**: Introducing genetic mutations that alter the primary structure of the target.
- **Enzymatic Modification**: Chemically modifying the target site to reduce antibiotic affinity.
- **Target Bypass or Replacement**: Developing alternative metabolic pathways that circumvent the inhibited target.
- **Target Protection**: Producing specialised proteins that physically shield the binding site from the antibiotic.

By altering these molecular targets, bacteria inhibit drug binding, thereby neutralising the antibiotic's efficacy and ensuring cellular survival.

3.4 Resistance Through Global Cellular Adaptations

Over evolutionary timescales, bacteria have evolved sophisticated adaptive mechanisms to withstand hostile environments, such as those within the human host. Bacteria compete for nutrients, evade antagonistic molecules from rival microorganisms, and withstand host immune defences to establish themselves in specific niches. Thus, bacterial pathogens employ complex strategies that protect essential cellular processes, including cell wall biosynthesis and membrane homeostasis.

IV. Determinants of Increasing Antibiotic Resistance

Antibiotics represent the most critical instruments of modern medicine; as essential prescription drugs, they often constitute life-preserving treatments. However, the non-prescribed use, unnecessary use, incorrect duration, and excessive use of antibiotics have significantly accelerated the development of AMR.

When antibiotics are administered, susceptible bacteria are eliminated, whereas antibiotic-resistant strains continue to proliferate; consequently, the antibiotic loses efficacy against these increasingly resistant populations (Bala K.N.; Andrew D.N.).

The main factors contributing to this phenomenon include inappropriate antibiotic prescribing and misuse, suboptimal treatment durations, excessive agricultural use, and inadequate hygiene and infection control measures (WHO 2023).

Furthermore, the misuse of antibiotics for viral infections such as the common cold, influenza, sore throat, bronchitis, and many sinus and ear infections is a major cause of AMR. Therefore, antibiotics should be used wisely (Tuba D.).

V. Antimicrobial Resistance: A Multidimensional Threat to Global Health and Modern Medicine

The global increase in antibiotic resistance presents a serious threat by reducing the effectiveness of commonly used antibiotics against widespread bacterial infections, and through research, the WHO declared that the antibiotic resistance problem "threatens the achievements of modern medicine. A post-antibiotic era, in which common infections and minor injuries can kill, is a very real possibility for the 21st century." It is also estimated by the WHO that bacterial AMR was directly responsible for 1.27 million global deaths in 2019 and contributed to a further 4.95 million deaths.

As a foundation of modern medicine, antimicrobial medicines are gradually losing their efficacy. The rise of drug-resistant pathogens threatens our ability to treat common infections and perform vital medical procedures, while also harming animal and plant health, reducing farm productivity, and jeopardising food security.

AMR also imposes substantial costs on both health systems and national economies overall (WHO 2023). For instance, it drives the need for more costly, intensive treatments, reduces patients' or their caregivers' productivity through prolonged hospital stays, and, lastly, harms agricultural productivity. AMR affects countries worldwide, regardless of their income level.

Dr Tedros Adhanom Ghebreyesus, the WHO Director-General, stated: "Antimicrobial resistance could unwind 100 years of medical progress, making infections that are easily treatable today a death sentence."

According to the United Nations, drug-resistant pathogens claim almost 1.3 million deaths each year and contribute to the deaths of nearly five million others. A more recent estimate published in The Lancet (2024) suggests that more than 39 million people will die from antibiotic-resistant infections between now and 2050.

VI. Conclusion

This report aimed to examine the genetic and molecular foundations of antibiotic resistance and their implications for global health. The findings show that bacteria's capacity to develop antibiotic resistance is largely determined by genetic adaptation mechanisms. In particular, HGT, mutations, and changes in resistance-gene expression play key roles in the development of resistance.

AMR is not only a clinical problem but also a multidimensional crisis that threatens global public health, requiring urgent attention from leaders across the security, healthcare and economic arenas. It is driven by the indiscriminate and excessive use of antibiotics, underscoring the critical importance of thorough antimicrobial stewardship and reinforced infection control interventions.

Furthermore, developing new antibiotics and investigating alternative treatments can provide sustainable solutions, but effective progress requires multidisciplinary approaches and stronger international co-operation.

In conclusion, antibiotic resistance remains an ever more prevalent and difficult-to-control public health problem. Therefore, both supporting scientific research and raising public awareness are critical to countering this global threat. Ultimately, the preservation of antibiotic efficacy is a shared global responsibility. Without urgent, coordinated, and evidence-based action, the medical community risks entering a "post-antibiotic era" where infections that were once easily treatable become life-threatening again.

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Six Scientists, Seven Drugs: The Pfizer Sandwich Story

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Abstract:

Between 1957 and 2011, a single pharmaceutical research facility in Sandwich, Kent, produced seven major medicines that collectively transformed the treatment of fungal infections, cardiovascular disease, erectile dysfunction, pulmonary hypertension, and HIV. This article traces the scientific and institutional history of Pfizer's European research headquarters, from a team of six scientists to a campus employing over 3,500 and examines the chemistry underpinning two of its most significant discoveries: fluconazole (Diflucan®), the antifungal that has saved millions of immunocompromised patients, and sildenafil (Viagra®), the PDE5 inhibitor whose serendipitous clinical journey remains one of the most cited examples of drug repositioning in pharmaceutical history. The article concludes by considering the site's transformation into Discovery Park, a science and innovation campus that continues Kent's pharmaceutical legacy.

Figure 1. The Pfizer research campus viewed from the banks of the River Stour, Sandwich, Kent. For over five decades, this site on the outskirts of a mediaeval Cinque Port served as the European research headquarters of one of the world's largest pharmaceutical companies, producing seven major medicines including fluconazole and sildenafil. The campus has since been redeveloped as Discovery Park. Image: Nick Smith / Geograph, CC BY-SA 2.0.

I. Introduction: A Quiet Town, a Global Laboratory

Figure 2. The Barbican, a mediaeval gatehouse on the River Stour in Sandwich, Kent. Sandwich's heritage as a Cinque Port town dates to the eleventh century, but it was the establishment of Pfizer's research campus on the town's outskirts in 1954 that transformed this quiet corner of East Kent into an unlikely centre of global pharmaceutical innovation. Image: Nilfanion / Wikimedia Commons, CC BY-SA 3.0.

Sandwich is a mediaeval town in the Dover District of East Kent, better known for its Cinque Ports heritage than for pharmaceutical innovation. Yet for over five decades, a research campus on its outskirts served as the European headquarters of Pfizer, one of the world's largest pharmaceutical companies. The medicines discovered there, spanning antifungals, antihypertensives, and antiretrovirals, have been prescribed to hundreds of millions of patients worldwide.

The story of Pfizer Sandwich is, at its core, a story about what happens when sustained institutional investment meets scientific curiosity. It is also, inescapably, a story about Kent: about the infrastructure, talent, and geography that made a small English town the unlikely birthplace of blockbuster medicines. For students of chemistry, the site offers something rarer still, a case study in how fundamental concepts from the textbook, enzyme inhibition, structure activity relationships, and pharmacokinetics become medicines in the clinic.

II. Origins: From Folkestone to Sandwich (1952–1971)

Pfizer's presence in Kent began in 1952, when the company established a compounding operation in Folkestone to convert base compounds into medicines suitable for the British market.[1]

Two years later, Pfizer acquired an 80-acre site on the outskirts of Sandwich to expand its Kent-based activities.[1] The campus would eventually grow to encompass over 390 acres.[2] An animal-feed plant was established in 1955, followed by an Agricultural Division in 1957, the same year that research activities commenced at the site with a team of just six scientists.[2]

The scale of ambition grew rapidly. In 1964, Pfizer acquired adjacent land known as the 'West Site', and by 1971, Pfizer Central Research had been formally established at Sandwich with a mandate to discover new pharmaceutical, chemical, and agricultural products for Pfizer's worldwide operations.[1] The site would eventually expand to employ over 3,500 people across 2.2 million square feet of research and development space, making it one of the largest pharmaceutical R&D campuses in Europe.[2]

III. Fluconazole: Redesigning the Antifungal Arsenal

3.1 The Clinical Need

In the late 1970s, the treatment of serious fungal infections was severely limited. The two principal agents available, amphotericin B and flucytosine, each carried significant drawbacks. Amphotericin B, while effective, required *intravenous* administration and was associated with severe *nephrotoxicity*. Flucytosine was orally active but possessed a narrow spectrum and was susceptible to resistance.[3] For immunocompromised patients, particularly those with HIV/AIDS, a population that would expand dramatically through the 1980s, the need for a safe, broad-spectrum, orally bioavailable antifungal was acute.

3.2 The Chemistry of Discovery

In 1978, a research programme was initiated at Pfizer's Sandwich laboratories, led by Dr Ken Richardson, with the objective of developing a new antifungal agent suitable for both oral and intravenous use. The team selected *azole derivatives* as their starting point, owing to their established tolerability and selective mechanism of action: inhibition of fungal sterol C-14 demethylation.[4,3]

The target enzyme, *lanosterol 14a-demethylase*, is a cytochrome P450 enzyme (CYP51) that catalyses an essential step in the biosynthesis of *ergosterol*, the principal sterol in fungal cell membranes, functionally analogous to cholesterol in mammalian cells. By inhibiting this enzyme, azole antifungals disrupt ergosterol production, compromising membrane integrity and ultimately causing fungal cell death.

Initial investigations focused on imidazole derivatives, but the team soon redirected their attention to *triazole analogues*. The rationale was pharmacokinetic: triazoles demonstrated reduced susceptibility to metabolic breakdown compared to imidazoles, potentially enabling better oral bioavailability and a longer half-life.[4] The researchers further emphasised polar derivatives, hypothesising that increased polarity would facilitate higher blood levels and further reduce metabolic degradation.

This strategy yielded a series of novel 1,3-bis-triazole-2-arylpropan-2-ol derivatives with promising activity across multiple models of fungal infection. Among these, the 2,4-difluorophenyl analogue stood out: it was the only derivative that was water-soluble, enabling formulation for both oral and intravenous administration, a critical advantage that none of the competing analogues offered.

3.3 Clinical Impact

Fluconazole was launched commercially as Diflucan® in 1988. In the United States, the FDA granted it fast-track treatment, approving it for use in just nine months, a reflection of the urgency created by the HIV/AIDS epidemic.[5] The drug proved effective against a wide range of fungal infections in immunocompromised patients, including transplant recipients, burn survivors, and individuals undergoing chemotherapy.

Pfizer UK received the Queen's Award for Technological Achievement in 1991 in recognition of the discovery.[1] Dr Richardson himself was appointed an Officer of the Order of the British Empire (OBE) in 1997 and was inducted into the United States National Inventors Hall of Fame in 2008, one of only two Pfizer scientists to receive this honour.[5]

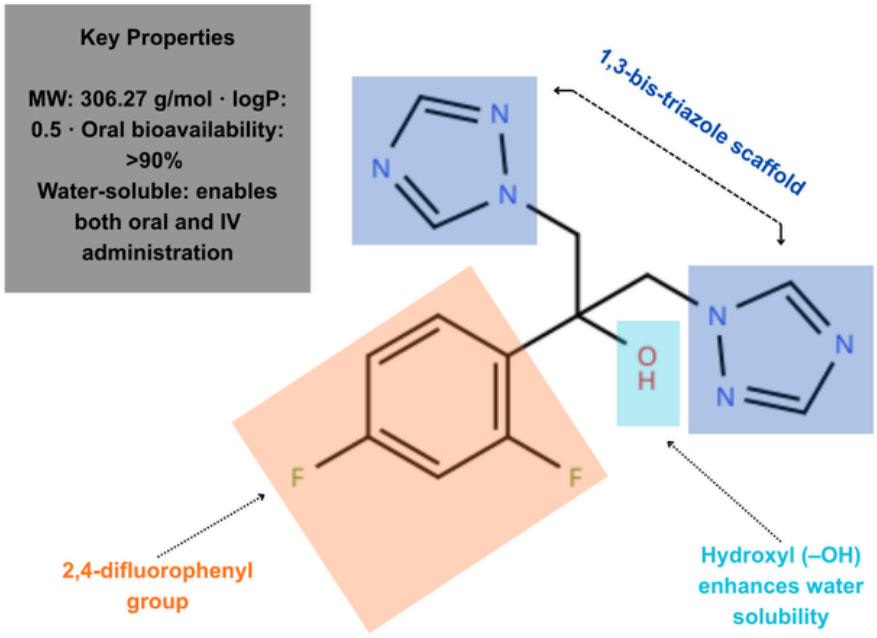

Figure 3. K.S. Ye Yint, CATS Insights, 2026

Figure 3. Structure of fluconazole, a 1,3-bis-triazole antifungal discovered at Pfizer Sandwich. The two triazole rings (blue) provide metabolic stability and selective inhibition of fungal lanosterol 14α-demethylase (CYP51). The 2,4-difluorophenyl group (orange) and central hydroxyl group confer the water solubility that distinguished fluconazole from all competing analogues, enabling formulation for both oral and intravenous administration. Diagram: K.S. Ye Yint, CATS Insights, 2026.

Richardson's personal story carries its own significance. He left school at sixteen to work as a laboratory assistant, earning his chemistry degree through evening classes at Trent Polytechnic before completing a doctorate at the University of Nottingham.[5] His trajectory from evening-class student to National Inventors Hall of Fame inductee illustrates a principle worth noting: access to scientific careers need not depend on conventional academic pathways nor any other limitations.

IV. Sildenafil: Serendipity and the Art of Repositioning

4.1 The Cardiovascular Hypothesis

By the mid-1980s, the cardiovascular research programme at Sandwich was generating significant results. The site had already contributed to the development of doxazosin (Cardura®) and amlodipine (Norvasc®/Istin®), antihypertensive agents that would become widely prescribed.

The scientific rationale was grounded in the nitric oxide (NO)/cGMP signalling pathway. Nitric oxide, released by endothelial cells, activates soluble guanylate cyclase (sGC), which converts guanosine triphosphate (GTP) to cGMP.[7] Elevated cGMP induces smooth muscle relaxation and vasodilation. The duration of this vasodilatory signal is regulated by phosphodiesterase type 5 (PDE5), the enzyme responsible for hydrolysing cGMP back to its inactive form, GMP.[7] The Pfizer team hypothesised that selective inhibition of PDE5 would prolong the vasodilatory effect of endogenous NO, offering a potential treatment for angina.

It is worth placing this work in a broader scientific context. The discovery of nitric oxide as a signalling molecule in the cardiovascular system was recognised with the 1998 Nobel Prize in Physiology or Medicine, awarded to Robert Furchgott, Louis Ignarro, and Ferid Murad.[8] Sildenafil's mechanism of action is directly built upon their foundational research.

4.2 Synthesis and Preclinical Profile

In 1989, a team of medicinal chemists working within a programme led by Simon Campbell synthesised a novel pyrazolopyrimidinone compound designated UK-92,480.[6] The discovery team included Nicholas Terrett, Andrew Bell, David Brown, and Peter Ellis, whose foundational work was published in 1996.[9]

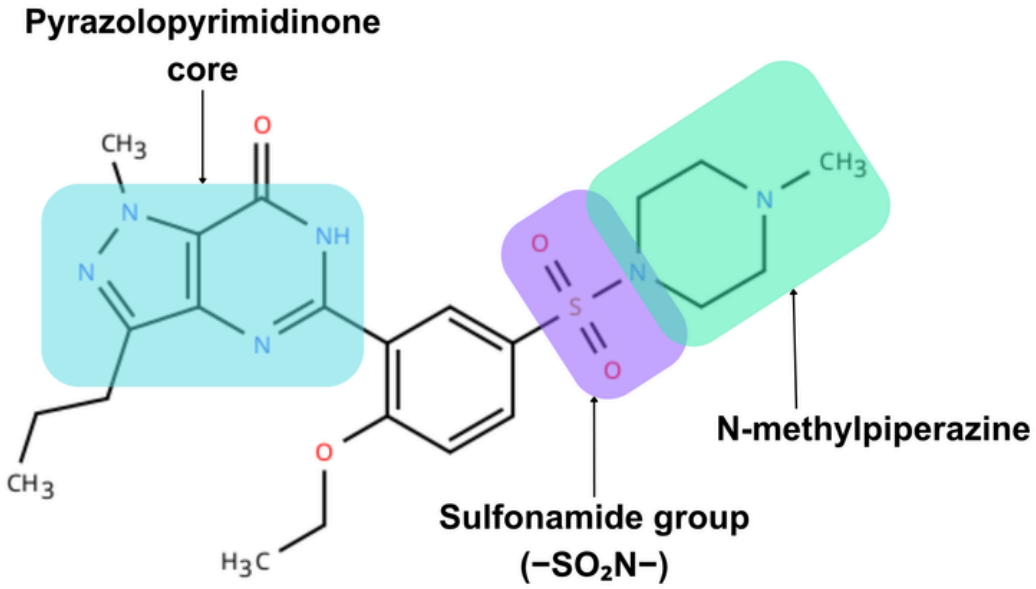

Figure 5. K.S. Ye Yint, CATS Insights, 2026

Figure 5. Structure of sildenafil (UK-92,480), a pyrazolopyrimidinone PDE5 inhibitor discovered at Pfizer Sandwich. The fused pyrazolopyrimidinone core (blue) binds selectively to the active site of PDE5. The compound was originally synthesised as a potential treatment for angina but was repositioned for erectile dysfunction after unexpected clinical observations during Phase I trials. Sildenafil was approved by the FDA on 27 March 1998 as the first oral treatment for erectile dysfunction. Diagram: K.S. Ye Yint, CATS Insights, 2026.

Peter Dunn and Albert Wood subsequently developed the scaleable manufacturing process that enabled commercial production of the compound.[10] UK-92,480 demonstrated potent and selective inhibition of PDE5, with a half-maximal inhibitory concentration (IC_{50}) of 3.5 nM against PDE5 derived from human corpus cavernosum, approximately tenfold more selective than its inhibition of PDE6 and over 700-fold more selective than for PDE3.[9] Preclinical studies confirmed vasodilatory effects and inhibition of platelet aggregation, and UK-92,480 was selected for clinical development.

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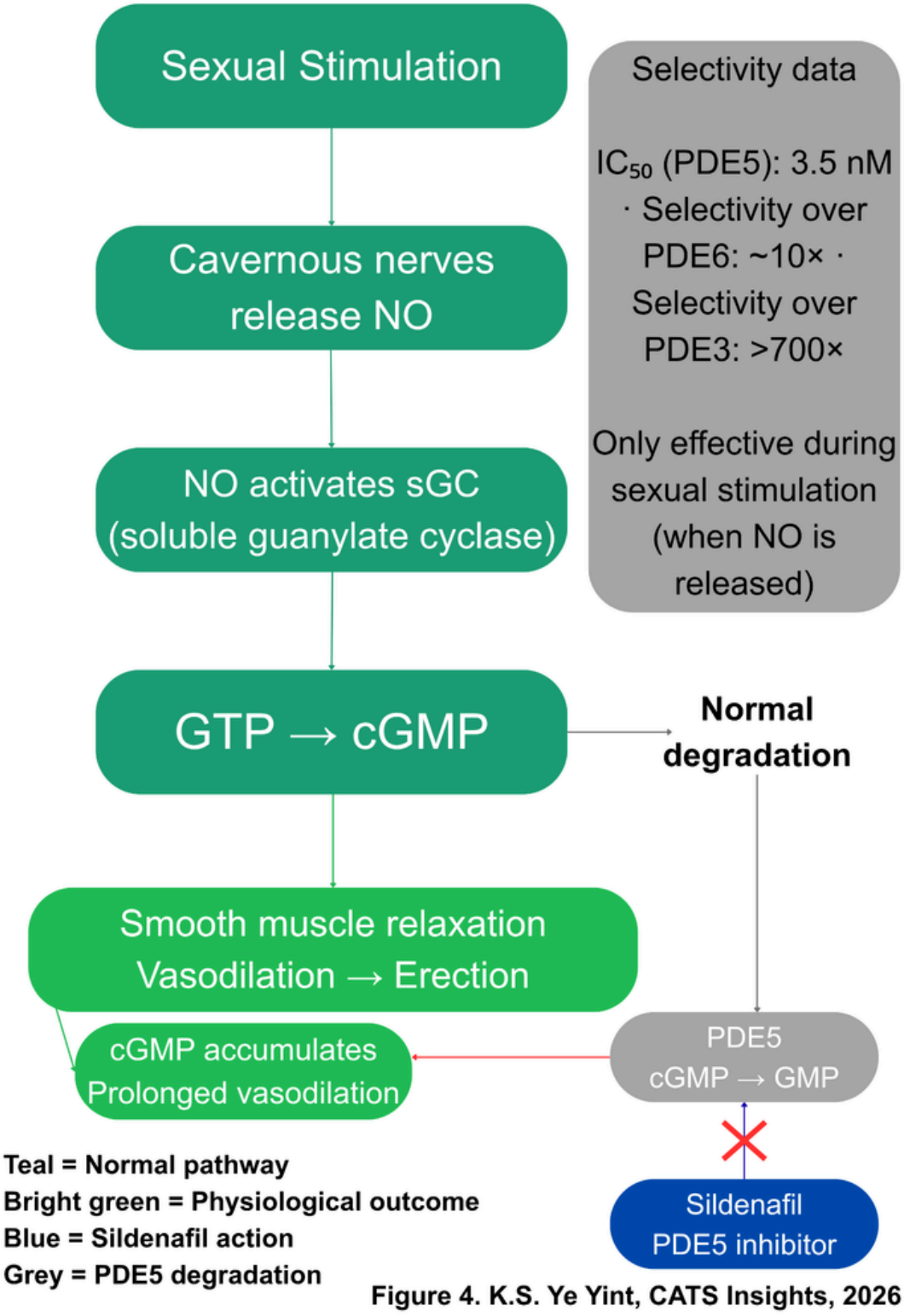

Figure 4. The nitric oxide (NO)/cGMP signalling pathway and the mechanism of action of sildenafil. During sexual stimulation, nitric oxide released from cavernous nerves activates soluble guanylate cyclase (sGC), which converts GTP to cGMP. Elevated cGMP causes smooth muscle relaxation and vasodilation in the corpus cavernosum. Phosphodiesterase type 5 (PDE5) normally hydrolyses cGMP to terminate the signal. Sildenafil selectively inhibits PDE5 (IC_{50} = 3.5 nM), prolonging cGMP accumulation and enhancing the natural erectile response. Diagram: K.S. Ye Yint, CATS Insights, 2026.

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4.3 The Clinical Pivot

Phase I clinical trials, conducted at Morriston Hospital in Swansea under the direction of Dr Ian Osterloh, produced results that were, from a cardiovascular perspective, disappointing. Sildenafil showed limited efficacy against angina. However, trial participants reported a consistent and unexpected side effect: marked penile erections.[6]

This observation coincided with emerging research demonstrating that nitric oxide is the principal neurotransmitter released from cavernous nerves during sexual stimulation and that the *corpus cavernosum* is particularly enriched with PDE5.[7] The Pfizer team recognised that a selective PDE5 inhibitor could enhance the natural erectile response by prolonging cGMP-mediated smooth muscle relaxation in penile tissue but only during sexual stimulation, when NO release is elevated. This mechanism offered a physiologically targeted approach to treating erectile dysfunction, unlike the existing invasive therapies: injectable prostaglandins, vacuum devices, and surgical implants.

Pfizer repositioned the compound, now named sildenafil. Clinical trials for erectile dysfunction commenced in 1993.[11] The drug was patented in 1996, and on 27 March 1998, the United States Food and Drug Administration approved sildenafil (Viagra®) as the first oral treatment for erectile dysfunction.[11] In the first two weeks on the American market, an estimated 150,000 prescriptions were written; within a year, annual sales surpassed one billion dollars.[12]

The story did not end there. Sildenafil was subsequently approved for the treatment of *pulmonary arterial hypertension* (marketed as Revatio®), exploiting the same PDE5 inhibitory mechanism to reduce pulmonary vascular resistance.[7] The drug's journey from angina candidate to erectile dysfunction treatment to pulmonary hypertension therapy represents a textbook example of *drug repositioning*, the process by which a compound developed for one indication is redirected to another based on observed pharmacological effects.

V. The Broader Sandwich Portfolio

Fluconazole and sildenafil are the most widely known products of the Sandwich site, but they represent only part of a broader portfolio. Other medicines discovered or significantly developed at the facility include amlodipine (Istin®/Norvasc®), a calcium channel blocker for hypertension; doxazosin (Cardura®), an alpha-adrenergic blocker; voriconazole (Vfend®), a second-generation triazole antifungal building on the fluconazole platform; and maraviroc (Celsentri®), the first orally administered CCR5 antagonist for the treatment of HIV, representing yet another 'first' from the Sandwich laboratories.[2]

The site received six Queen's Awards across its operational history: for Technological Achievement in 1979 (for oxamniquine) and 1991 (for fluconazole), Export Achievement in 1993, International Trade in 1997 and 2000, and Enterprise in 2001 (for the discovery and development of Viagra).[1] At its peak, the Sandwich facility employed over 3,500 people and constituted one of the largest pharmaceutical research operations of any American company outside the United States.[2]

VI. Discovery Park: Continuity and Reinvention

On 1 February 2011, Pfizer announced the closure of the Sandwich R&D facility as part of a global restructuring programme, placing approximately 2,400 jobs at risk.[2] The announcement represented a significant economic shock to East Kent.

Figure 6. Buildings on the former Pfizer west site, now part of Discovery Park, Sandwich, Kent. Since Pfizer's withdrawal in 2011, the campus has been redeveloped as one of the United Kingdom's leading science and innovation hubs, hosting over 180 companies and more than 3,000 people across life sciences, biotechnology, and technology. Image: David Anstiss / Geograph, CC BY-SA 2.0.

However, the site's story did not end with Pfizer's withdrawal. The campus was acquired and redeveloped as Discovery Park, which has since grown into one of the United Kingdom's leading science and innovation hubs. As of 2026, the site hosts over 180 companies and more than 3,000 people, working across life sciences, biotechnology, and technology.[13] Pfizer itself maintains a significant presence at the site, focused on mid- to late-stage pharmaceutical development.[13] Other tenants include Asymchem (a Chinese CDMO that inaugurated its first European facility at the site in 2024), Viatris, Concept Life Sciences, and Canterbury Christ Church University, which operates laboratory facilities on the campus.[13]

Discovery Park holds Enterprise Zone status and is one of only seven Life Science Opportunity Zones in the United Kingdom.[13]

Perhaps most relevant to the readership of this publication, a new science specialist sixth form, Carbon 6 Academy of Science, is due to open at the site in September 2026, designed for students aged 16–18 with a focus on science and mathematics.[13] The physical infrastructure that produced fluconazole and sildenafil is, in a meaningful sense, being repurposed to train the next generation of scientists.

VII. Conclusion: What Sandwich Teaches Us

The Pfizer Sandwich story offers several insights that extend beyond its specific institutional history. First, it demonstrates the relationship between sustained research investment and transformative outcomes: the fluconazole programme ran for over a decade from inception to clinical launch, and the sildenafil journey from cardiovascular hypothesis to erectile dysfunction approval spanned nearly as long. Breakthrough medicines are not produced quickly.

Second, the sildenafil case illustrates the importance of serendipity and scientific alertness. The compound's unexpected side effect in clinical trials could have been dismissed as a curiosity. Instead, the research team recognised its therapeutic potential, connected it to emerging basic science, and executed a successful repositioning.

This required both the intellectual flexibility to abandon the original indication and the institutional willingness to invest in an entirely new clinical programme.

Third, the Discovery Park transformation suggests that scientific infrastructure, once established, can outlast the institutions that built it. The laboratories, the talent pipeline, and the geographical advantages that attracted Pfizer to Sandwich in 1954 continue to attract innovators today.

For students in Kent studying chemistry, the legacy is not abstract. It is twenty minutes down the road.

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Written in White: The Chemistry of Dover's White Cliffs

Author: Kyal Sin Ye Yint

Figure 1. The White Cliffs of Dover, rising approximately 110 metres above the English Channel. The cliff face is composed almost entirely of Upper Cretaceous chalk, a biogenic sedimentary rock formed from the compacted remains of trillions of coccolith plates. The distinctive vertical gullies are carved by acidified rainwater dissolving calcium carbonate through the carbonate equilibria described in Section III. Image: Immanuel Giel / Wikimedia Commons, CC BY-SA 3.0.

Abstract

Stand at the edge of the Dover cliffs on a clear morning, and the facade rising 110 metres from the English Channel is visible from the French coast, approximately 32 kilometres away.15 But these are not merely a geological landmark or a national symbol. They are a chemical archive: one hundred million years of marine carbon cycling, compressed into a crystalline sedimentary record and left standing at the edge of a continent. Every square centimetre of this chalk contains the preserved remains of organisms that once dominated the Cretaceous ocean, and every winter storm that breaks against the cliff face enacts a reaction that connects atmospheric carbon dioxide to the dissolution of solid rock.

This article traces the chemistry of the White Cliffs from the cellular machinery of ancient algae to the equilibrium equations that govern their modern destruction. The cliffs, it turns out, are a chemistry lecture written in calcium carbonate, and the lesson has not yet finished.

I. Shells to Stone: Coccolith Sedimentation and Diagenesis

The chalk owes its existence to coccolithophores, unicellular marine phytoplankton belonging to the division Haptophyta. These organisms, typically 5–20 µm in diameter, construct an external armour of interlocking calcite plates called **coccoliths**, each plate measuring approximately 1–5 µm across.2 The dominant species in modern oceans is *Emiliania huxleyi*, though the Cretaceous assemblage would have been taxonomically different; genera such as *Watznaueria* and *Prediscosphaera* are the principal coccolith taxa recovered from English chalk deposits.3

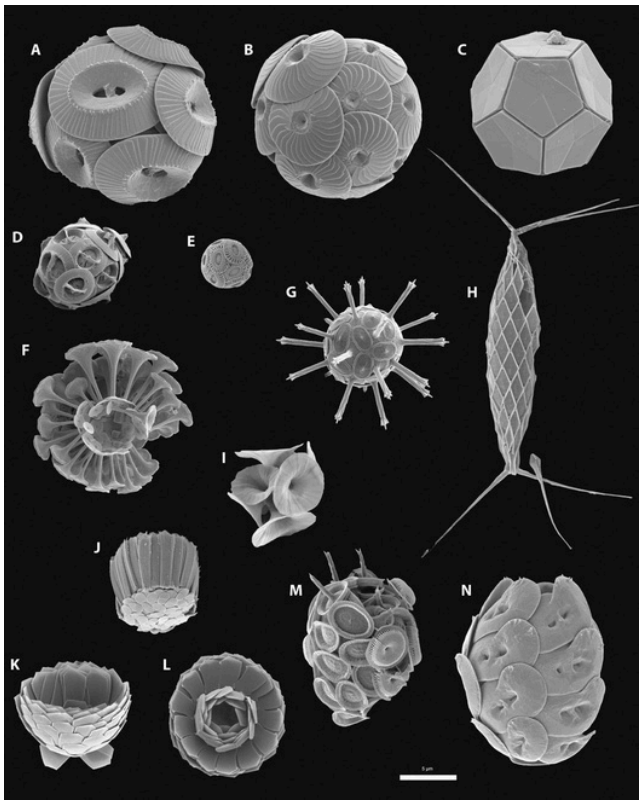


Figure 2. Scanning electron micrograph of Emiliania huxleyi (Position E), the most abundant coccolithophore species in modern oceans and 13 other species. The cell (approximately 5 µm in diameter) is encased in a coccosphere of interlocking calcite coccolith plates, each measuring 2–4 µm across. These plates are assembled inside the cell within a Golgi-derived coccolith vesicle and extruded to the cell surface upon completion. When coccolithophores die, their coccospheres disaggregate and the individual plates sink to the seabed, the process that, over 30 million years of sustained deposition, built the Dover chalk. Image: Monteiro, F.M. / Wikimedia Commons, CC BY-SA 4.0.

Coccolith formation is an entirely intracellular process. Calcite crystals are nucleated upon an organic baseplate within a specialised Golgi-derived vesicle, the coccolith vesicle, where crystal growth is directed by coccolith-associated polysaccharides (CAPs) that regulate nucleation sites and control crystal morphology.2 The net intracellular reaction is:

$Ca^{2+} + 2HCO_3^- \rightarrow CaCO_3 + CO_2 + H_2O$

This is significant since for every mole of calcite precipitated, one mole of carbon dioxide is released, meaning that coccolithophore calcification is simultaneously a carbon sink (locking carbon into solid $CaCO_3$) and a carbon source (releasing CO_2 that can re-enter the atmosphere).4 The net effect on atmospheric carbon depends on the balance between these two fluxes, a balance that remains one of the central questions in marine biogeochemistry.

During the Late Cretaceous period (approximately 100–66 million years ago), Kent's coastline lay beneath a warm, shallow epicontinental sea at a palaeolatitude of roughly 35°N.3 Conditions were ideal for sustained coccolithophore productivity: elevated atmospheric CO_2 concentrations (estimated at 600–1,200 ppm, compared to roughly 425 ppm today), warm sea-surface temperatures, and an extensive flooded continental shelf that suppressed terrigenous sediment input.5 When coccolithophores died, their coccospheres disaggregated, and the individual coccolith plates rained down through the water column, accumulating on the seabed at rates of approximately 0.5–2 millimetres per year.3,6 Over approximately 30 million years of sustained deposition, this process built chalk strata up to 500 metres thick in parts of southern England.6

The transformation from soft coccolith ooze to solid chalk rock is called diagenesis. As overlying sediment accumulated, the increasing overburden pressure expelled pore water and compacted the coccolith fragments. Small secondary calcite crystals precipitated in the intergranular spaces, cementing the compacted mass into a coherent rock with a typical porosity of 35–45% and a composition of 95–99% calcium carbonate by weight.3 Horizontal bands of black flint visible in the cliff face formed later during diagenesis, as dissolved silica from the skeletons of siliceous radiolarians and sponge spicules migrated through the chalk and precipitated as microcrystalline quartz nodules.6

II. Why Chalk Is White: Light Scattering in Microcrystalline Calcite

Pure calcite is transparent. A single crystal of Iceland spar, the optically clear variety prized since the seventeenth century for its extraordinary birefringence, transmits light from approximately 250 nm to 2,500 nm with minimal absorption.7 So why are the Dover cliffs white?

The answer lies not in the crystal structure of calcite itself, but in the microcrystalline texture of chalk. Chalk is composed of billions of calcite particles, coccolith fragments and secondary cement crystals, each measuring between 0.5 and 5 µm in diameter. When visible light (wavelengths 380–700 nm) encounters a particle whose dimensions are comparable to or larger than the wavelength, it undergoes Mie scattering, a process in which the incident radiation is redirected in all directions without wavelength-selective absorption.8 Because chalk's grain size falls squarely in this regime, all wavelengths of visible light are scattered with approximately equal efficiency. The result is diffuse, broadband reflectance: whiteness.

This is the same physics that makes snow white (ice crystals scatter light), milk white (fat globules scatter light), and clouds white (water droplets scatter light). In each case, the material's individual constituents are transparent; the opacity and whiteness emerge from the collective scattering of vast numbers of microscopic particles.8 Chalk's high porosity further enhances the effect by creating innumerable air–calcite interfaces, each of which presents a refractive index boundary (calcite: n = 1.49–1.66; air: n = 1.00) at which additional scattering occurs.

There is a practical corollary to these optics. The softness of chalk (Mohs hardness 3, inherited directly from calcite) means that the cliff face is continually being refreshed by coastal erosion. Wave undercutting at the base causes blocks of chalk to collapse, exposing fresh, unweathered surfaces. It is this constant renewal that keeps the cliffs white; a stable, unrefreshed chalk surface would gradually discolour as biological growth and atmospheric deposition accumulated.1

The erosion rate has increased to 220–320 mm per year over the last 150 years, approximately three to five times the pre-industrial rate of 20–60 mm per year, driven primarily by the loss of protective beach material and the increasing frequency of severe storms.1

III. Dissolving the Cliffs: Carbonate Equilibria and Chemical Weathering

The White Cliffs are not static. They are being dissolved, grain by grain, by a set of coupled equilibrium reactions that connect atmospheric carbon dioxide to the chemistry of every raindrop that strikes the chalk surface.

The process begins in the atmosphere. Carbon dioxide dissolves in rainwater to form carbonic acid, a weak diprotic acid:

$$CO_2(g) + H_2O(l) \rightleftharpoons H_2CO_3(aq) \quad K_1 = 3.4 \times 10^{-2} \text{ at } 25°C$$

Carbonic acid then dissociates in two stages, releasing hydrogen ions that lower the pH of the solution:

$$H_2CO_3(aq) \rightleftharpoons H^+(aq) + HCO_3^-(aq) \quad Ka_1 = 4.3 \times 10^{-7}$$
$$HCO_3^-(aq) \rightleftharpoons H^+(aq) + CO_3^{2-}(aq) \quad Ka_2 = 4.7 \times 10^{-11}$$

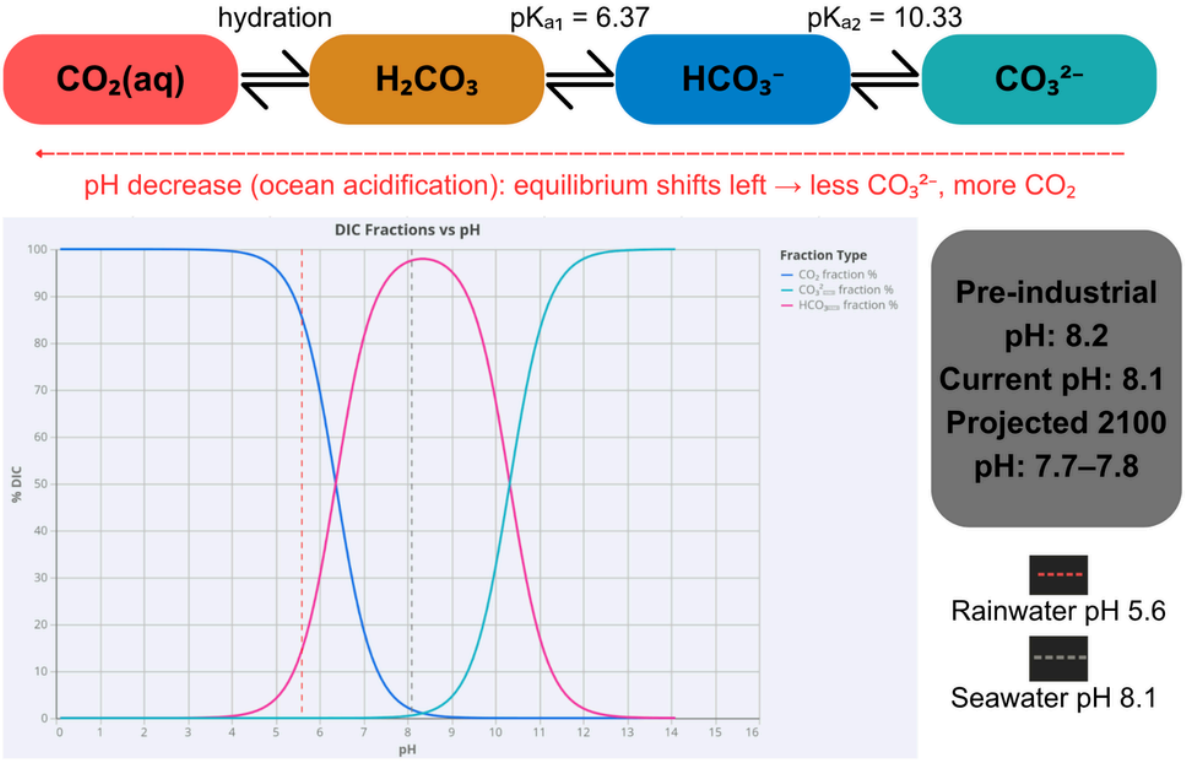


Figure 3. K.S. Ye Yint, CATS Insights, 2026

Figure 3. The marine carbonate equilibrium system. Dissolved CO_2 reacts with water to form carbonic acid, which dissociates stepwise through bicarbonate to carbonate. At typical seawater pH (8.1), approximately 90% of dissolved inorganic carbon exists as bicarbonate (HCO_3^-), 9% as carbonate (CO_3^{2-}), and less than 1% as dissolved CO_2 or H_2CO_3. As pH decreases through ocean acidification, the equilibrium shifts toward CO_2 and HCO_3^- at the expense of CO_3^{2-}, reducing the calcite saturation state (Ω) and threatening calcifying organisms. Diagram: K.S. Ye Yint, CATS Insights, 2026.

Unpolluted rainwater in equilibrium with atmospheric CO_2 at current concentrations (approximately 425 ppm) has a pH of roughly 5.6.9 When this weakly acidic solution contacts the chalk, the hydrogen ions react with calcium carbonate in a dissolution reaction:

$$CaCO_3(s) + H_2CO_3(aq) \rightleftharpoons Ca^{2+}(aq) + 2HCO_3^-(aq)$$

The overall process converts solid calcite into soluble calcium bicarbonate. This is a reversible equilibrium: the position of the reaction depends on the partial pressure of CO_2, temperature, and the pH of the solution. Le Chatelier's principle predicts, correctly, that increasing atmospheric CO_2 drives the equilibrium to the right, accelerating dissolution. Conversely, removing CO_2 (for example, when groundwater emerges at the surface and degasses) shifts the equilibrium to the left, causing calcite to precipitate, a process responsible for stalactite and stalagmite formation in chalk caves and for the calcite-rich 'hard water' characteristic of Kent's domestic water supply.9

The dissolution is visible at every scale. At the cliff face, acidified rainwater percolating through vertical fractures and bedding planes carves the distinctive gullies and undercuts that define the Dover coastline. Across geological timescales, this process has transported millions of tonnes of dissolved calcium back into the ocean, where marine organisms recycle it into new shells, completing the lithospheric component of the carbon cycle.4 The cliffs are, therefore, not merely a product of the carbon cycle, they are an active participant in it, mediating the transfer of carbon between the atmosphere, hydrosphere, and lithosphere.

IV. Fire and Powder: Thermal Decomposition and Kent's Lime Industry

Kent's chalk has been quarried and burnt for at least two thousand years. Lime burning was a technology introduced to Britain by the Romans, and there is no evidence of lime kilns in England before the Roman period.10 The chemistry is textbook thermal decomposition:

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g) \quad \Delta H = +178 \text{ kJ mol}^{-1}$$

The reaction requires temperatures above 840°C, at which point the partial pressure of CO_2 above the calcium carbonate surface exceeds one atmosphere, allowing carbon dioxide to escape irreversibly in an open kiln.11 In practice, temperatures of 900–1,000°C were used to drive the reaction to completion within a practical timeframe, though exceeding 1,200°C produced 'dead-burnt' lime, an unreactive sintered product that was useless as a building material.11

The product, quicklime (calcium oxide), is chemically unstable in the atmosphere. Mixed with water in a violently exothermic reaction called slaking, it produces calcium hydroxide:

$$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(s) \quad \Delta H = -65 \text{ kJ mol}^{-1}$$

Calcium hydroxide, or slaked lime, was the essential ingredient in lime mortar, the binding agent that built medieval Canterbury's cathedral, its city walls, and the castles at Dover, Rochester, and Leeds.10 Over months and years, the mortar hardened as calcium hydroxide reabsorbed atmospheric carbon dioxide and reconverted to calcium carbonate:

$$Ca(OH)_2(s) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$$

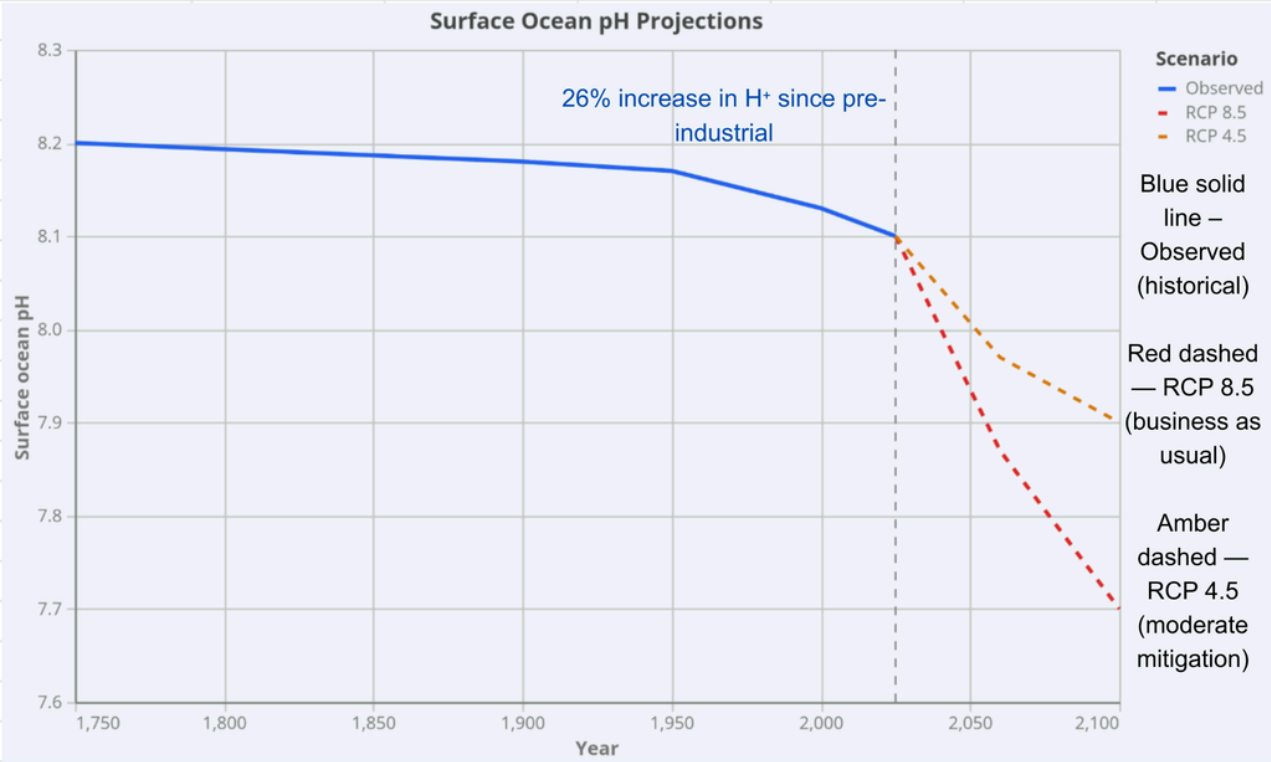
This slow carbonation means that lime mortar effectively recaptures most of the CO_2 originally released during calcination, a natural carbon loop that makes traditional lime mortar significantly less carbon-intensive over its lifecycle than modern Portland cement.11 The chemistry comes full circle: chalk is burnt to release CO_2 and produce quicklime; quicklime is slaked and used as mortar; the mortar reabsorbs CO_2 from the air and slowly reverts to calcium carbonate. The mediaeval builders of Canterbury were, without knowing it, exploiting a reversible reaction.

Beyond construction, quicklime was applied to Kent's acidic Weald Clay soils as an agricultural amendment. Calcium oxide neutralises soil acidity through the reaction:

$$CaO(s) + 2H^+(aq) \rightarrow Ca^{2+}(aq) + H_2O(l)$$

Kent's lime-burning kilns, now mostly ruined or converted into private homes, were concentrated along the chalk downlands of the North Downs. They represent the industrial infrastructure of a chemistry that sustained the county's agriculture and construction for centuries.10

V. The Acid Test: Ocean Acidification and the Carbonate Saturation State



Ω = 1 threshold: calcite undersaturation in polar surface waters

Figure 4. K.S. Ye Yint, CATS Insights, 2026 · Data: Doney et al. (2009); IPCC AR6

Figure 4. Projected decline in surface ocean pH under different CO_2 emission scenarios. Pre-industrial surface ocean pH was approximately 8.2. Current pH is approximately 8.1, representing a 26% increase in hydrogen ion concentration. Under business-as-usual projections (atmospheric CO_2 reaching 800 ppm), pH is expected to fall to approximately 7.7–7.8 by 2100. The moderate mitigation scenario projects stabilisation at approximately 7.9. Data synthesised from Doney et al. (2009) and IPCC projections. Diagram: K.S. Ye Yint, CATS Insights, 2026.

The chemistry that shaped the cliffs over geological time is now operating in reverse and at an accelerating rate. Since the beginning of the Industrial Revolution, the ocean has absorbed approximately 30% of anthropogenic CO_2 emissions.12 This absorption triggers the same carbonic acid equilibria described in Section III, but in seawater rather than rainwater, and the consequences for calcifying marine organisms are severe.

The key reactions are:

$CO_2(g) + H_2O(l) \rightleftharpoons H_2CO_3(aq) \rightleftharpoons H^+(aq) + HCO_3^-(aq)$

The excess hydrogen ions then react with dissolved carbonate ions:

$H^+(aq) + CO_3^{2-}(aq) \rightleftharpoons HCO_3^-(aq)$

This is the critical step. The hydrogen ions do not simply accumulate freely in solution; they are partially buffered by reacting with carbonate to form bicarbonate. The result is a simultaneous decrease in pH and a decrease in the concentration of dissolved carbonate ions (CO_3^{2-}).12,13 Since the pre-industrial era, average surface ocean pH has fallen from approximately 8.2 to 8.1, a 0.1-unit decline that represents a roughly 26% increase in hydrogen ion concentration, because the pH scale is logarithmic.12 The biological significance is captured by the calcite saturation state (Ω), defined as:

$\Omega = [Ca^{2+}][CO_3^{2-}] / K_{sp}$

Where Ksp is the solubility product of calcite at the relevant temperature and pressure. When $\Omega > 1$, seawater is supersaturated with respect to calcite, and shell formation is thermodynamically favourable. When $\Omega < 1$, calcite dissolution is favoured.13 Current surface ocean waters remain supersaturated ($\Omega > 1$) for calcite, but the saturation state is declining. If atmospheric CO_2 reaches 800 ppm under business-as-usual emission scenarios, surface pH is projected to fall by a further 0.3–0.4 units by 2100, representing a total increase in ocean acidity of 100–150% relative to pre-industrial conditions.12,13

For coccolithophores, the very organisms whose ancestors built the Dover chalk, this presents an existential challenge. Laboratory studies have demonstrated that calcification rates in key species decline under elevated CO_2 conditions, producing thinner coccoliths and weaker coccospheres.2 The saturation horizons, the ocean depths below which calcite dissolves, are already shoaling by 30–200 metres globally since the pre-industrial period.13 The White Cliffs, in this sense, record the conditions of a world in which marine calcification flourished; the chemistry of the modern ocean is trending away from those conditions.

VI. The Archive, Revisited

The White Cliffs of Dover compress 30 million years of marine productivity into a single geological monument. The coccolithophores that built them used intracellular calcification in Golgi-derived vesicles, and the diagenesis that hardened them illustrates pressure-driven cementation and silica remobilisation; the whiteness that defines them is an exercise in Mie scattering at the micron scale. The dissolution that reshapes them every winter is Le Chatelier's principle in real time, and the ocean acidification that threatens their living descendants is the same carbonate chemistry running under different boundary conditions.

From biomineralisation to thermal decomposition, from equilibrium thermodynamics to global biogeochemistry, these cliffs connect more branches of chemistry than any single landmark in Kent.

So the next time you see them from the Dover ferry, remember that you are looking at a scientific marvel, that, and to mind the crumbling edge.

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Brewed in the Garden: The Chemistry of Kentish Hops

Author: Kyal Sin Ye Yint

Figure 1. Traditional oast houses on the Kent landscape near Bishopsbourne, Canterbury. The conical kilns were designed to channel heated air through freshly picked hops, reducing their moisture content from approximately 80% to 6% before packing. Image: Oast House Archive / Geograph, CC BY-SA 2.0.

Abstract:

Kent has been called the Garden of England for centuries, but its most celebrated crop was never a flower. It was *Humulus lupulus* (the hop). From the first recorded English hop garden at Westbere, near Canterbury, in 15202 to the vast sea of bines that once covered over 43,000 acres of Kentish soil[1], this climbing plant shaped the county's economy, its landscape, and its cultural identity. By the late nineteenth century, Kent accounted for approximately 60% of England's total hop acreage[1], drawing over 80,000 seasonal pickers[3] from London every September in a mass migration the Victorians called 'hopping down in Kent'.[1]

But behind the heritage and the hop gardens lies a body of chemistry as rich as any in the series. The bitterness of beer, the aroma of a freshly poured pint, the shelf life of an India Pale Ale shipped across oceans, all of these are determined at the molecular level, inside the tiny lupulin glands of the female hop cone.

This article traces that chemistry from alpha-acid isomerisation to terpene volatility and from mediaeval brewing to modern antimicrobial research. The hop, it turns out, is a chemistry lesson hiding in a pint glass.

I. The Hop Garden: A Brief History of Kent's Favourite Crop

Wild hops are thought to have been introduced to Britain by the Romans.[2] Archaeobotanical evidence of *Humulus lupulus* seeds was recovered from the Graveney Boat, an Anglo-Saxon vessel dated to approximately AD 900, unearthed from the estuary mud at Seasalter in 1970.[2] However, the deliberate cultivation of hops for brewing did not begin in England until Flemish weavers and refugees introduced the practice in the fifteenth and early sixteenth centuries. The traditional English beverage was spiced ale, brewed from malt without hops, and the addition of this new bittering agent initially met considerable resistance.[2]

Figure 2. Female hop cones (Humulus lupulus). The tiny lupulin glands, visible as golden-yellow resinous deposits at the base of each bract, contain the alpha-acids responsible for beer bitterness and the terpene-rich essential oils that produce hop aroma. Image: Jean-Pol Grandmont / Wikimedia Commons, CC BY 3.0

Kent's advantages were geographical, agricultural, and economic. The county's fertile loam soils, mild maritime climate, established enclosed field system, and abundant supply of sweet chestnut coppice for poles and charcoal made it ideally suited to hop cultivation.[2] Kent farmers were also among the most prosperous of their time, able to absorb the high initial capital outlay, approximately £125 per acre in 1857, compared to £10-12 for arable crops.1 By 1655, Kent was responsible for producing one third of the national crop.[1,2] Daniel Defoe, writing in his *Tour thro' the Whole Island of Great Britain* (1724–1726), described the area near Maidstone as 'the Mother of Hop Grounds in England', noting that the region had been the first in England where hops were planted in any quantity.[13]

In 1574, Reginald Scot of Smeeth, near Ashford, published *A Perfite Platforme of a Hoppe Garden*, one of the earliest English language manuals on hop cultivation, a text that remains a landmark in English agricultural history.[1,2]

II. The Bitter Truth: Alpha-Acid Isomerisation

The bitterness of beer is not inherent in the raw hop. It is created by a chemical transformation that occurs during wort boiling, and understanding this reaction is central to understanding why hops matter at all.

The hop cone's lupulin glands contain soft resins composed primarily of alpha-acids (α-acids), a family of prenylated phloroglucinol derivatives (a group of natural chemical compounds found in plants known for antioxidant and anti-inflammatory effects). The three major analogues are humulone (35–70% of total α-acids), cohumulone (20–65%), and adhumulone (10–15%). In their native form, α-acids are poorly soluble in water and virtually tasteless. It is only when the brewer boils hops in the wort that the critical conversion occurs.[4,5]

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During wort boiling (typically at 100°C for 60–90 minutes), α-acids undergo **thermal isomerisation** to form **iso-α-acids** via an **acyloin-type ring contraction**. The mechanism, first elucidated by De Keukeleire and Verzele in 1971, proceeds in three steps: (1) deprotonation of the β-tricarbonyl system to form the required anion, (2) stereoselective ketonisation of the remaining enol function within the six-membered ring, and (3) an α-ketol rearrangement that contracts the ring from six to five members, yielding the iso-α-acid product.[5]

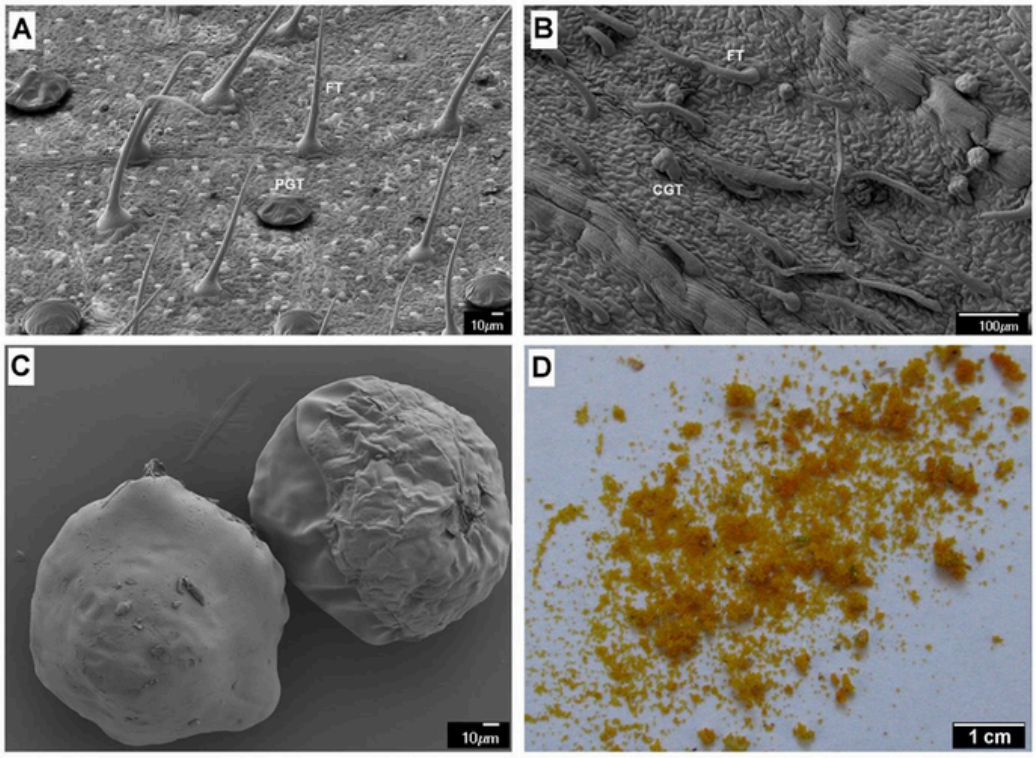


Figure 3. Lupulin glands (glandular trichomes) on the bracteoles of a female hop cone. These secretory structures biosynthesise and accumulate the alpha-acids, beta-acids, and terpenes discussed in this article. Adapted from Mishra et al. (2020), CC BY 4.0.

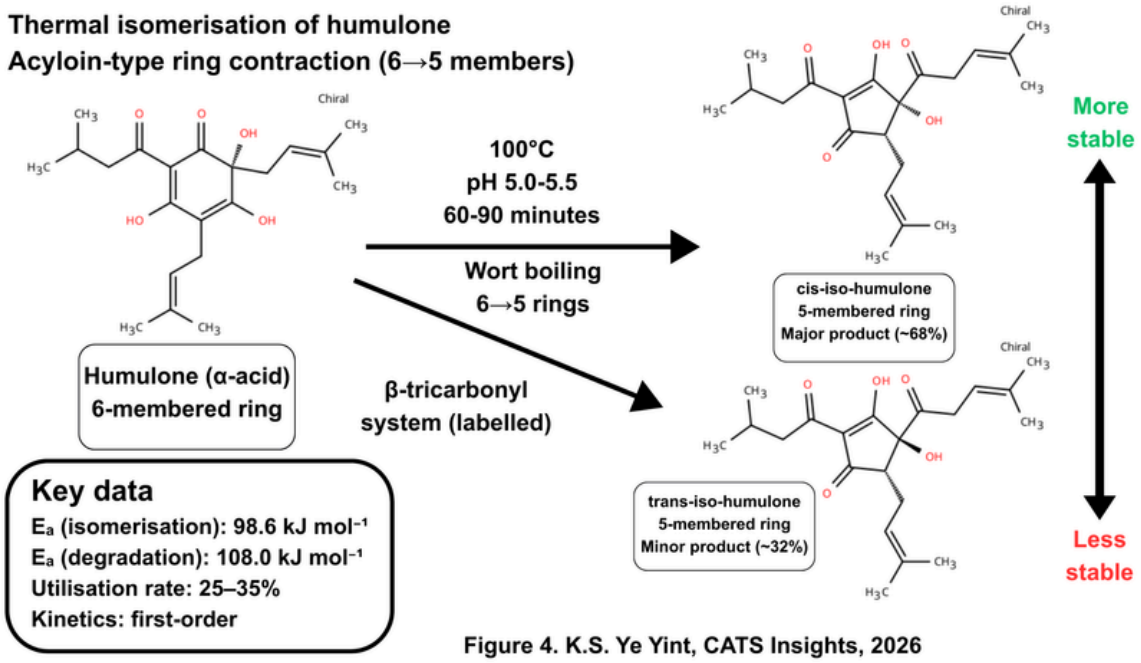


Figure 4. Thermal isomerisation of humulone to cis- and trans-iso-humulone during wort boiling. The acyloin-type ring contraction converts the six-membered ring of the native alpha-acid into the five-membered ring of the iso-alpha-acid, producing the bitter compounds responsible for beer bitterness. Under standard wort conditions (pH 5.0–5.5), the cis-isomer predominates at approximately 68:32. Diagram: K.S. Ye Yint, CATS Insights, 2026.

Each parent α-acid produces two diastereomeric iso-α-acids: a cis-isomer and a trans-isomer. Under standard wort conditions (pH 5.0–5.5), the ratio favours the cis-compounds at approximately 68:32.5 This preference arises primarily from the stereoselectivity of the ketonisation step under brewing conditions, though the cis-isomers are also thermodynamically more stable, an unusual case where kinetic and thermodynamic products coincide. The reaction follows first-order kinetics with an experimentally determined activation energy of 98.6 kJ mol^{-1} for isomerisation and 108.0 kJ mol^{-1} for the subsequent degradation of iso-α-acids to non-bitter products.6 In traditional brewing, only 25–35% of the original α-acids are converted into iso-α-acids, the rest are lost to trub formation, poor solubility, and thermal degradation.[4]

This low utilisation rate is not merely a technical footnote. It is the reason brewers have spent centuries developing new hopping technologies: pelletised hops, pre-isomerised extracts, and supercritical CO_2 extraction methods, all designed to squeeze more bitterness from each gram of precious lupulin.

III. The Shelf-Life Problem: Beta-Acids and Oxidation

Alongside the α-acids, hop resins contain a second family of bitter compounds: the **beta-acids** (β-acids). The principal β-acid analogues are **lupulone** (n-lupulone), **colupulone**, and **adlupulone**, structurally related to their α-acid counterparts but differing in the number and position of prenyl side chains. Unlike α-acids, β-acids do not isomerise significantly during wort boiling and contribute little to beer bitterness directly.[4], [7]

Their importance lies instead in **oxidative degradation**. During hop storage, β-acids undergo autoxidation through free-radical pathways, producing a cascade of transformation products, including hulupones, that can contribute bitterness and off-character notes to aged beer, though contemporary research indicates that hulupone concentrations in most beers remain below their sensory threshold of approximately 7–8 mg/L.[7]

This oxidative susceptibility explains why hop storage conditions (cold, anaerobic, and vacuum-sealed) are so critical to brewing quality and why the Kentish oast houses, those iconic conical-roofed structures visible across the county's skyline, were designed specifically to dry hops rapidly and thoroughly, minimising moisture-driven degradation before the product reached the brewer.[7]

There is, however, a useful side to this chemistry. The *trans*-iso-α-acids formed during wort boiling are significantly less stable than their cis-counterparts, meaning the bitterness profile of a beer shifts over time as the *trans*-isomers degrade preferentially. This differential stability has practical consequences: beers intended for long-distance transport, such as the heavily hopped ales traditionally understood to have been shipped from England to the colonies, retained their bitterness precisely because the *cis*-iso-α-acids survived the voyage intact, though the historical record surrounding the origins of India Pale Ale remains contested among brewing historians.[5]

IV. The Organic Chemistry of Aroma: Terpenes in Hops

Bitterness is only half the story. The aroma of hops, the floral, citrus, herbal, and resinous notes that define different beer styles, arises from a complex mixture of volatile compounds produced within the same lupulin glands. Hop essential oil is a remarkably diverse mixture: over 400 individual compounds have been identified, though three classes dominate the sensory profile.[8]

Myrcene (β-myrcene, C10H16) is the most abundant monoterpene in hop oil, typically comprising 50% or more of total oil content at harvest. It is an acyclic, unsaturated hydrocarbon biosynthesised from geranyl pyrophosphate (GPP) via the methylerythritol phosphate (MEP) pathway. Myrcene carries resinous, herbaceous, and distinctly 'hoppy' aroma notes and possesses the lowest odour threshold of all major hop oil hydrocarbons at approximately 13 parts per billion (ppb) in water14, making it among the most potent aromatic compounds in the hop cone, though its threshold is considerably higher in beer matrices owing to matrix suppression effects.[8]

However, myrcene is extremely volatile (boiling point 167°C), meaning it is almost entirely lost during traditional wort boiling.8 Late-addition hopping and dry-hopping techniques were developed specifically to preserve myrcene in the finished beer.

Linalool ($C_{10}H_{18}O$) is a terpene alcohol closely related to myrcene, contributing floral and citrus aroma notes. Although it constitutes only a small fraction of total hop oils8, linalool is considered one of the most significant aroma-active compounds in dry-hopped and late-hopped beers because it is less volatile than myrcene and more likely to survive into the finished product. The (*R*)-enantiomer, which constitutes approximately 95% of hop-derived linalool, possesses a notably low sensory threshold in beer, meaning trace quantities produce perceptible floral impact.[8,9]

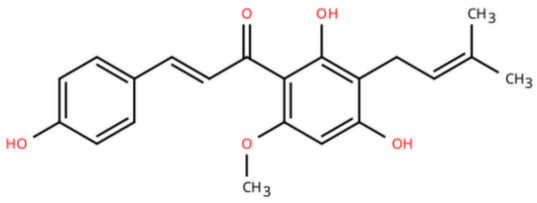
Geraniol ($C_{10}H_{18}O$) contributes rose-like and citrus aromas. Its significance extends beyond direct sensory contribution: during fermentation, yeast enzymes can convert geraniol into β-citronellol, an oxygenated terpene with woody, fresh, and citrus notes. The reduction is catalysed primarily by the NADPH oxidoreductase OYE2 (Old Yellow Enzyme 2), while β-glucosidases release glycosidically bound terpene precursors from hop material. This **biotransformation**, a chemical conversion driven by microbial metabolism, means the aroma profile of a finished beer is not simply a reflection of the hop's original terpene composition but an emergent property of the interaction between hop chemistry and yeast biochemistry.[9]

This interplay between volatile chemistry and brewing process is what gives different beer styles their characteristic aromatic identities, and it is what makes hop selection an art as much as a science. The Kent-bred variety East Kent Goldings, prized for its refined, delicate aroma, owes that character to its specific terpene profile, a profile shaped as much by Kent's terroir as by the plant's genetics.

V. Beyond the Pint Glass: Hops in Modern Pharmaceutical Research

In recent decades, researchers have turned their attention to a compound that brewers largely overlooked: **xanthohumol** (XN).

Xanthohumol is a prenylated chalcone, a type of flavonoid, found in the hard resin fraction of hop cones. Its structure was first identified in 1957, but its pharmacological properties went unrecognised until the 1990s, when a series of studies revealed a surprisingly broad spectrum of biological activity.10



Xanthohumol

Figure 5. K.S. Ye Yint, CATS Insights, 2026

Figure 5. Structure of xanthohumol, a prenylated chalcone found in hop lupulin glands. Xanthohumol has demonstrated antimicrobial activity against Clostridioides difficile with MIC values of 15–107 µg/mL, as well as anti-inflammatory and chemopreventive properties in preclinical models. Diagram: K.S. Ye Yint, CATS Insights, 2026.

Research published in the journal *APMIS* demonstrated that purified xanthohumol exhibits potent antimicrobial activity against clinically relevant anaerobic bacteria, including *Clostridium difficile*, with minimum inhibitory concentration (MIC) values of 15–107 µg/mL, values approaching those of conventional antibiotics against resistant strains.11 Subsequent *in vivo* studies using rat models confirmed that both xanthohumol and hop beta-acids significantly reduced *C. difficile* bacterial load and improved local inflammatory markers in the large intestine, with the xanthohumol effect being more pronounced.[12] Beyond antimicrobial properties, xanthohumol has demonstrated anti-inflammatory, antioxidant, and chemopreventive activity across multiple preclinical models, making it one of the most pharmacologically promising natural products currently under investigation.[10]

The broader hop bitter acids, particularly the iso-α-acids, have long been recognised for their selective antimicrobial properties against Gram-positive bacteria. This selectivity is, historically, the reason that heavily hopped beers preserved better during long sea voyages.

What modern research has done is formalise this folk knowledge into quantifiable pharmacological data, opening the door to potential applications in food preservation, wound care, and the development of novel antibiotic adjuncts at a time when antimicrobial resistance represents one of the most pressing challenges in global public health.[12]

VI. The Garden, Revisited

Kent's hop gardens have shrunk dramatically from their nineteenth-century peak. The 43,614 acres recorded in 18751 have dwindled to a fraction, and most of the county's iconic oast houses have been converted into private homes. But the chemistry of the hop, the isomerisation that creates bitterness, the terpenes that sculpt aroma, and the flavonoids that fight infection continue to drive research and industry worldwide.

The next time you see an oast house on the Kent skyline or pass through Canterbury on the road to Dover, remember that you are travelling through the landscape that gave England its brewing heritage. And the chemistry behind that heritage, from α-acid ring contractions to xanthohumol's MIC values, is as alive in the modern laboratory as it ever was in the hop garden.

That, and to try a local ale. Make it Kentish.

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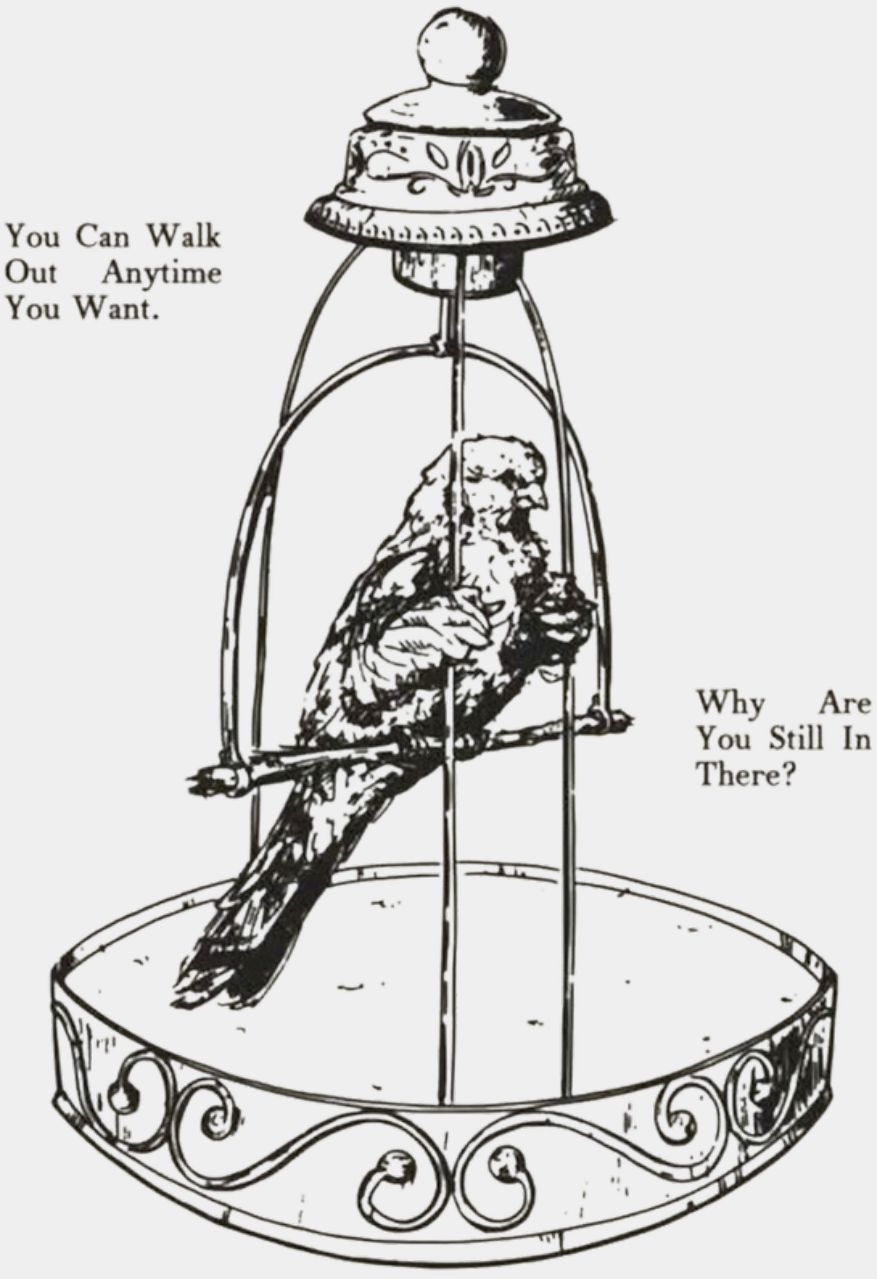
Perspectives



Empathy: Impact of genetical factors in contrast with parenting by either sex

Author: Minh Ngoc Nguyen

Abstract:

The difference in the empathetic display of male and female parents have resulted in conceived notions that mothers are more appropriate caretakers due to their higher capacity for empathy. This article explores the validity of this assumption through the insight of research providing insight into the difference in types and capacity for empathy in both sexes, along with its role in parenting and childhood development. It was suggested that the sex of the parent had little impact on the empathetic development of the child, rather the presence of responsive parenting was the determinate factor in the development of empathy in male and females.

I. Introduction

With the rise in single parent households of either sex, one may raise the question of whether one parent is more adequate for child-rearing than the other, as many continue to hold the belief that mothers are more suitable than fathers due to their supposed ability to be more empathetic. However, the validity of this assumption is debated greatly, due to it being based heavily on societal expectations and role enforcement. This leads to the question: Is empathy, although inherently learnt, influenced primarily by the sex of the parent or more dependent on innate, biological factors?

II. Concept of empathy

Empathy is the idea of sympathising with and understanding the thoughts and emotions of someone else; it is primarily conceptualised into 2 types: Cognitive empathy and emotional empathy (this essay will not discuss concepts of affective, compassionate and somatic empathy).

Cognitive empathy is understood as the comprehension of another's thoughts and feeling; however, the person themselves does not experience said emotions. While emotional empathy involves experiencing the emotions of another, their emotionality has an impact on the person's emotional state.

III. Empathy in males and females

The believed differences in empathy demonstrated through Baron-Cohen's theory (Baron-Cohen, 2003), with men being more inclined to systemisation related to problem-solving (cognitive empathy) while women tend to emotional empathise more. The former claim is supported by neuroscientific evidence provided by Barth et al (Barth et al, 2020), where brain imaging found that during social reasoning, men exhibited greater co-activation in regions involving attention and task engagement, further demonstrating a preferred utilisation of cognitive empathy. In addition, developmental findings suggest that women are more outward with their show of prosocial, empathic responses (Eisenberg et al, 1998).

Additionally, it is evidenced that the primary difference is in levels of emotional empathy, with a self-report study finding women rated themselves as having higher empathetic concern (Davis, 1983).

Furthermore, biological evidence suggests that different sexes operate using different empathetic systems when approaching situations, with women often showing stronger activation in mirror neuron-related areas, thereby, mirroring the emotions displayed by the person they are empathising towards, i.e. emotional contagion; whereas men are more reliant on regions of cognitive control (Schulte-Ruther et al, 2008). This suggests a difference in style; however, provides no conclusive differences in quantity or capacity.

In contrary, the general trend in research is that men and women possess similar empathic capacity, only having differences in display (EEG findings point to no significant differences in neural response to other's pain in either sex – Pang et al., 2023).

Furthermore, it was concluded through experimental and neuropsychological testing measures that no consistent sex effect on empathy; however, the method of assessment was suggested to have great influence on perceived differences in results, with self-reports being vulnerable to unconscious gender stereotypes (Baez et al, 2017). This is further strengthened as once expectations were manipulated, differences in self-reported studies disappeared.

Therefore, it can be concluded that males and females present their empathy differently, however, in terms of capacity, they possess the same amount. This provides biological evidence that males and females have inherently different systems of empathetic display.

IV. Developing empathy in childhood

It is suggested by modern research that empathy and its display are in part innate (temperament, brain structure) and learnt through environment (parenting, culture, and experiences).

Biologically, infants possess the basic neural capacities (systems such as the insula) for empathy, shown through acts such as crying upon seeing others cry (emotional contagion), which is suggested to aid in social bonding (shared emotionality) and survival (pack mentality), as responsible brain systems are present in primitive humans in addition to other mammals.

However, it does not encompass the entirety of empathy, consisting of stages such as affective and cognitive empathy; these must be learnt, refined, and practiced over time through social interactions.

Crucially, empathy is highly dependent on caregiving quality, as demonstrated by a longitudinal neuroscience study by Feldman (Feldman, 2015) where caregiving quality was predictive of empathy-related brain responses, with chronic adversity or unresponsive parenting producing altered neural processing of others' distress (e.g. dysregulated empathy).

It can be concluded then that empathy styles in children are not dependent on the sex or whether the empathetic display was conveyed by the maternal or paternal figure, rather it is determined by the biological brain structure of the child's sex, and their experiences in childhood were they are impacted by parenting and society.

V. Empathy in parenting

Females are often preferred as candidates for parenting children, with biases being made known in many societal procedures (e.g. courts, fostering, adoption).

Research demonstrates empathy and parental care are major factors in childhood development (Ainsworth et al, 1978); however, the idea that maternal or paternal caretaking creating different systems of empathy in children is not strongly supported; rather it is found that the quality and responsiveness of parenting is most crucial in impacting offspring empathy development.

This claim is demonstrated through a 2025 systematic review (Santana-Ferrándiz et al, 2025), where parental sensitivity had great impact on children's emotional regulation, attachment security and social ability, which are the foundation of forming emotional empathy. This suggests that children internalise and model patterns of empathic caregiving, regardless of which sex is responsible for parenting (supported by behaviourism).

Furthermore, studies were only able to conclude context-specific differences that are ungeneralisable to the majority population (Hyde, 2005), this suggests that differences in patterns are resultant of interactive styles rather than proving empathy outcomes.

Crucially, in terms of single-parent households, evidence (Amato et al, 2002) directs to the style of parenting (emotional availability, economic allowance) being a more accurate predictor for emotional development and empathy than family structure. This is further illustrated through a longitudinal study comparing single mothers and fathers, which found that children raised by single fathers were more emotionally intelligent due to their fathers' higher incomes, thereby providing stability and security and facilitating emotional growth. Subsequently, it was found that those from single mother households were more equipped to maintain peer relationships, due to their mothers' greater involvement (Sultan and Kanwal, 2013). This suggests that the combination of emotional and economic factors is determinant for the child's empathetic development.

VI. Conclusion

This conclusively demonstrates that, partially, empathy styles displayed in adulthood is determined biologically, as brain structures for empathy systems exist in infancy; with males being more likely to exhibit cognitive empathy, resorting to problem solving to display care, and females being prone to emotional empathy, displaying external emotive signals such as emotional contagion, empathy is largely learnt through parental responsiveness and quality.

Furthermore, the expression of empathy is learnt and molded by parents and other societal agents, e.g. peers, other adults, large institutions, with its development being pertinently dependent on responsive, communicative parenting, rather than the sex of the parent having impact on their child's learning and conveying of empathy later in life.

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What You Expect is What You Get - How Expectation Shapes the Senses

Author: Philip Dan Tinashe Chapanduka

Abstract:

This article explores how top-down processing (the brain's reliance on prior knowledge and expectations to interpret sensory input) can override and distort objective perception, influencing everything from everyday experiences to consumer behaviour.

I. The Thought Experiment

Consider a scenario in which someone is handed a bottle of Sprite; the lid is already open, so there is no satisfying fizz from the crack of the seal. They raise it to their lips. Their brain has already decided that it will be sweet, fizzy and sharp. They drink. But it's water. They wonder if for a fraction of a second they tasted Sprite.
** Assume they have drunk Sprite frequently enough to know/remember the taste*

This phenomenon is well-known and takes place in many forms. A frequent example is people taking extra steps on staircases.

Why does this happen?

II. Top-Down Processing

Top-down processing is a cognitive theory spearheaded by Gregory (1974). Psychologists define top-down perception as situations in which context, learning, or expectation alters a perceptual process (Gaspelin et al., 2018).

Prior knowledge changes what we perceive. It can be explained like this:

1. The brain has stored knowledge (take this to be 'expectations' or memories).
2. The brain receives a stimulus, it does not start from scratch.
3. The brain uses what it expects to 'fill in gaps'.

This helps to process stimuli quickly, but can cause misinterpretations (Cherry, n.d.).

Gregory (1974) explains that in top-down processing expectations influence our perception.

For example, if an individual reads "THE CAT" and later reads "THE CHT", most individuals will read the second word as "CAT", not "CHT".

Top-down processing enables faster decision-making; however, expectation- and knowledge-based processing can lead to mistakes (Cherry, n.d.).

As a student, a frequent phrase teachers often say is to "answer the question on the paper, not what you expect it to ask you." – This is a frequent example of how this can relate to everyday life, especially as a student.

It is important to note that cognitive theory includes bottom-up processing too. This is not relevant to this article, as it is the opposite of top-down processing and instead uses sensory input first, independent of expectation.

However, to avoid oversimplification, wherever top-down processing and bottom-up processing can have an effect has been noted.

III. Sensory Effect

The brain can't process every sensory input; it must use shortcuts to analyse information so it can process stimuli quickly. This is where top-down processing and bottom-up processing come into play. For the purpose of this article, we will focus solely on top-down processing.

The human mind only stores sensory information in the short term if it triggers a "conscious perception" (Dharani, 2015). This is due to the brain receiving an estimated 109bits/s of sensory information, while human processing capacity is approximately 10 bits/s (Zheng et al., 2024).

We use top-down processing (and bottom-up processing) to process sensory information effectively. This includes using past experiences (Cherry, n.d.) and expectations. Without these processes, it would be conceptually true that humans would have to analyse everything.

Applying this to the thought experiment, the past experience of the individual drinking Sprite from a Sprite bottle allowed their brain to have an expectation of a sweet, fizzy, and sharp taste. The fraction of a second where they may have 'tasted' Sprite was the limitation of top-down processing.

This demonstrates a clear distortion of perception in a mundane task. Top-down processing can streamline perception but also introduce bias, as interpretation is influenced by expectation rather than objective sensory input.

IV. The Placebo Palate

The earlier thought experiment has real-world applicability: expectation can directly affect taste. A study by Schmidt et al. (2017) found that the brain's valuation system mediates the price-to-taste effect.

In this study, 30 participants drank wine while lying in an MRI scanner. They were shown the prices of the wine they were tasting.

The results were measured by taste pleasantness ratings. The more expensive the wine, the higher the pleasantness rating—independent of whether participants actually paid for it. The researchers identified this as a marketing placebo effect and concluded that expectation can lead individuals to perceive a taste that is not objectively present (Schmidt et al., 2017).

This effect can be explained through top-down processing, as participants used prior knowledge and expectations to interpret sensory input.

Therefore, it can be concluded that the study provides biological evidence supporting cognitive theory. The anterior prefrontal cortex and the brain's valuation system were identified as mediating this effect.

The study provides strong evidence that top-down processing can bias sensory perception, as expectations derived from price influenced participants' experiences of taste. This supports the thesis that top-down processing can override objective sensory input, leading to distorted perceptions of reality.

V. Conclusion

The constructivist cognitive theory of top-down processing offers a powerful explanation of the malleability of perception. It can be used to draw reliable conclusions on misinterpretations of mundane tasks, demonstrated through perceptual inference errors (e.g., "CAT" vs. "CHT"), to more complex influences such as believing expensive wines taste better.

Essentially, what we perceive is not always an objective reflection of reality. Instead, it is a constructed interpretation shaped by memory, context, knowledge, and expectation (Cherry, n.d.). Top-down processing works alongside bottom-up processing to streamline sensory analysis and prevent sensory overload. However, this efficiency comes with implications that extend beyond perception, influencing everyday decisions such as consumer behaviour.

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What exactly are ideas? The concept of "idea" in Deleuzian philosophy

Author: Luca Candelot

"There's no need to fear or hope,
but only to look for new weapons."
(Gilles Deleuze)

Abstract:

Despite many attempts to understand and conceptualise the idea, this article finds in Deleuze the answer to the question put in the title. Deleuze understands the idea to be not part of essence but a multiplicity in which it can not be defined as a sole one, but rather as a complex system of problems, in which the idea itself is a problem.

I. Introduction

Since the early prolegomena of philosophy, in Plato's Theory of Forms, the concept of Idea has been widely debated. Over the course of history, many conceptions and misconceptions of what an Idea is have been elaborated. However, with the development of dialectics - more specifically the system developed by Immanuel Kant and Georg Wilhelm Hegel -, the debate towards what is the Idea gained a much more complex stance, which finds a synthesis with the French philosopher Gilles Deleuze.
This article will aim to explain his theory and foremost demonstrate the mode in which Deleuze, by his own Theory of Ideas, dialogues critically with past conceptualizations, formulating a new becoming to dialectics.

II. Background

In his book, *Difference and Repetition*[1], Deleuze synthesises his own conceptual understanding of what are Ideas. However, his work is primarily based upon previous elaborations done by Immanuel Kant throughout his transcendental dialectics.

For Kant, the Idea is an a priori form of reasoning. An idea is something that goes beyond the empirical and firms itself solely by reason.

In such a sense, it is a pure onto-epistemological object that is a *thing-by-itself*. Ideas are influenced by the externality of sensibility however it is also an understanding between different coherences of synthetical judgements, which lay early foundations for Deleuze's concept of Rhizome.

"A conception formed from notions, which transcends the possibility of experience, is an idea, or a conception of reason."
(Kant, CPR, p. 222)

So, in such a manner it is possible to argue how the concept of idea is intrinsically linked to transcendental dialectics, since the idea itself can not be synthetically thought of as a mean or an object not by itself. However, further developments will reveal the idea's rhizomatic properties.

2.1 The notion of problem

Gilles Deleuze defines the concept of the problem in a strongly unique way. Problems are universal dialectical inquiries. For example, Deleuze puts that the field of mathematics is a field of solutions, it does not deal with problems.[2]
This outlines the non-specific aspect of a problem - it may be solved through many approaches. Furthermore, when there is a solution upon a problem, the problem is not eliminated, the solution only updates one of its various possibilities. It is thought of as a dialectical multiplicity.

2.2 The concept of multiplicity

Throughout the totality of his philosophy, Deleuze will use the concept of multiplicities to explain an important part of his reasoning. The concept was developed firstly by the mathematician Bernhard Riemman and then explored in philosophy by Henri Bergson, whose work was a major inspiration for Deleuze.
The multiplicity relates to the complete opposition of the one and the many ones. Multiplicities can never be thought of as unitary manifestations or even as the relation of one with another, they are beyond that in such a sense that each relation is integrated with various multiplicities. "*In other words, the very notion of multiplicity taken as a substantive implies a displacement of all of thought: for the dialectical opposition of the one and the multiple, we substitute the typological difference between multiplicities.*"[3]

As a conclusion, multiplicities must obey the following structure:
1. Its elements must not possess a sensible form, nor conceptual meaning. Must not possess actual existence and are inseparable from a virtual potence. They must not have any previous identity. Just then, through undetermination, it becomes possible the manifestation of the free difference of the identity of the concept.
2. The elements must be defined reciprocally by relations who must not subsist their independence. Those relations form a juxtaposition of elements who create an ideal connection.[4]
3. Multiple ideal connections to the differential relationship.
Multiplicities must update themselves in space-time correlations.

III. Deleuze's conception of the Idea

For Deleuze, Kant is the utmost basis to think about the Idea. In his book *Difference and Repetition*, Deleuze dedicates an entire chapter for this discussion.5In his own elaboration, ideas are not essence, in such a way that they are formed by a cogito.[6] They are not constituted by a totality of the object but for a rather complex structure of the problem in a broader sense. In such a way, the conceptualization of the problem by itself leads to formulating the boundaries and properties of the Idea.

Hence, problems are ideas.[7] Problems are dialectical, they develop themselves by the *Aufheben*[8] of the various multiplicities that concern the problem. The same process is evident for the Idea. However, there is a teleological difference between the problem and the Idea; problems have an isomorphic relationship. To every problem, it is expected to exist a solution, independently whether it is known to man or not. Regarding the Idea, Kant would argue that ideas are "*problems to which there is no solution*"[9]. Deleuze would therefore put that true problems are ideas, since their ontological affirmation does not rely on the answer, and they will continue to exist in spite of having no solution. Due to this specific sense, they are mutually immanent and transcendental.

Upon this notion, Deleuze put forward a general conceptualization for the Idea: "*An idea is an n-dimensional, continuous, defined multiplicity*"[10]. An idea would rely upon many conditions for it to happen as a phenomenon - n-dimensions. These n-dimensions always correlate with each other in different modes of relation - continuity. Finally, the idea cannot be changed unless there is a change to its multiplicity.

Thus, the multiplicities of the idea acts as transcendental delimitators and imposed definitions for its own conceptualization.
Further on, the Idea obeys an internal temporality, split between three moments of synthesis. The moments go through the indeterminate until it finds the pseudo-concrete on the determinate. To explain this movement of the Idea, Deleuze used Kant's three moments of the *Cogito*[11], which can be represented as:
1. "I am" as undetermined
2. Time appears as the moment in which existence is determinable
3. "I think" as the determined

However, when adapting Kant's structure, Deleuze noticed that the moments of the Idea could also be thought of in the terms of differential calculus. For Deleuze, the indeterminate moment of the idea would be the differentiation in itself, notably *dx*, *dy*; while the movement is represented by *dx/dy* and finally the determinate are the values that are concluded from *dx/dy*. "*dx* is the Idea - the Platonic, Leibnizian or Kantian Idea, the 'problem' and its being."[12] Therefore, *dx* represents the idea in its first dialectical movement, which develops itself further until the concrete as a solution for dx/dy.

In simpler terms, *dx* is the infinitesimal variation of *x*, while *dx/dy* is the quotient for its variation, the instantaneous rate of change in *y* in relation to *x*. This means that *dx/dy* represents no specific quantity, it is undeterminable by itself as it only shows an infinitesimal relationship. When *dx*, as the first undetermined, becomes a relation with *dy*, they become a multiplicity in totality. Any change to dx in even the slightest intensity impacts *dy*. In such a basis, it becomes evident how ideas are determined by multiplicities, towards a specific extent where it concatenates into a determinable.

IV. Conclusion

In synthesis, Deleuze offers a new approach to the conceptualisation of the Idea. Deleuze does not - as Kant did - understand ideas only as a pure concept of reason. For him, ideas can not be formed by essence but by the *cogito* in itself.

Understanding the concept of Idea is truly crucial in order to grasp the whole of Deleuzian's philosophy. It is from the Idea that Deleuze will elaborate on the theme of rhizome and multiplicities.

By demonstrating that every idea is a multiplicity, Deleuze puts forward the notion of interdependence throughout reasoning in itself. Thus, it is foremost important that one who wishes to comprehend contemporary philosophy is able to understand what is the idea and how the idea must not be thought as a *thing-by-itself*, but rather as various possibilities of the problematic, formulating a dialectical relationship towards the aims of rhizomatic exposure of multiplicities.

V. Notes

1. It is important to note that this article retains itself from
Deleuzian philosophy solely from Difference and Repetition
and, therefore, does not consider any ideological change that
arose in Anti-Oedipus.
2. DR, p. 179
3. Deleuze, 1969
4. The concept of ideal must be thought here as the relations in a
pure epistemological mode of reasoning
5. chapter 4, 'Ideas and the Synthesis of Difference'.
6. DR, p. 168
7. "Conversely, problems are Ideas." (DR, p. 168)
8. Hegel's term for negation in the dialectical process (HEGEL,
WL. p. 58)
9. CPR, p. 317
10. DR, p. 182
11. DR, p. 180
12. DR, p. 181

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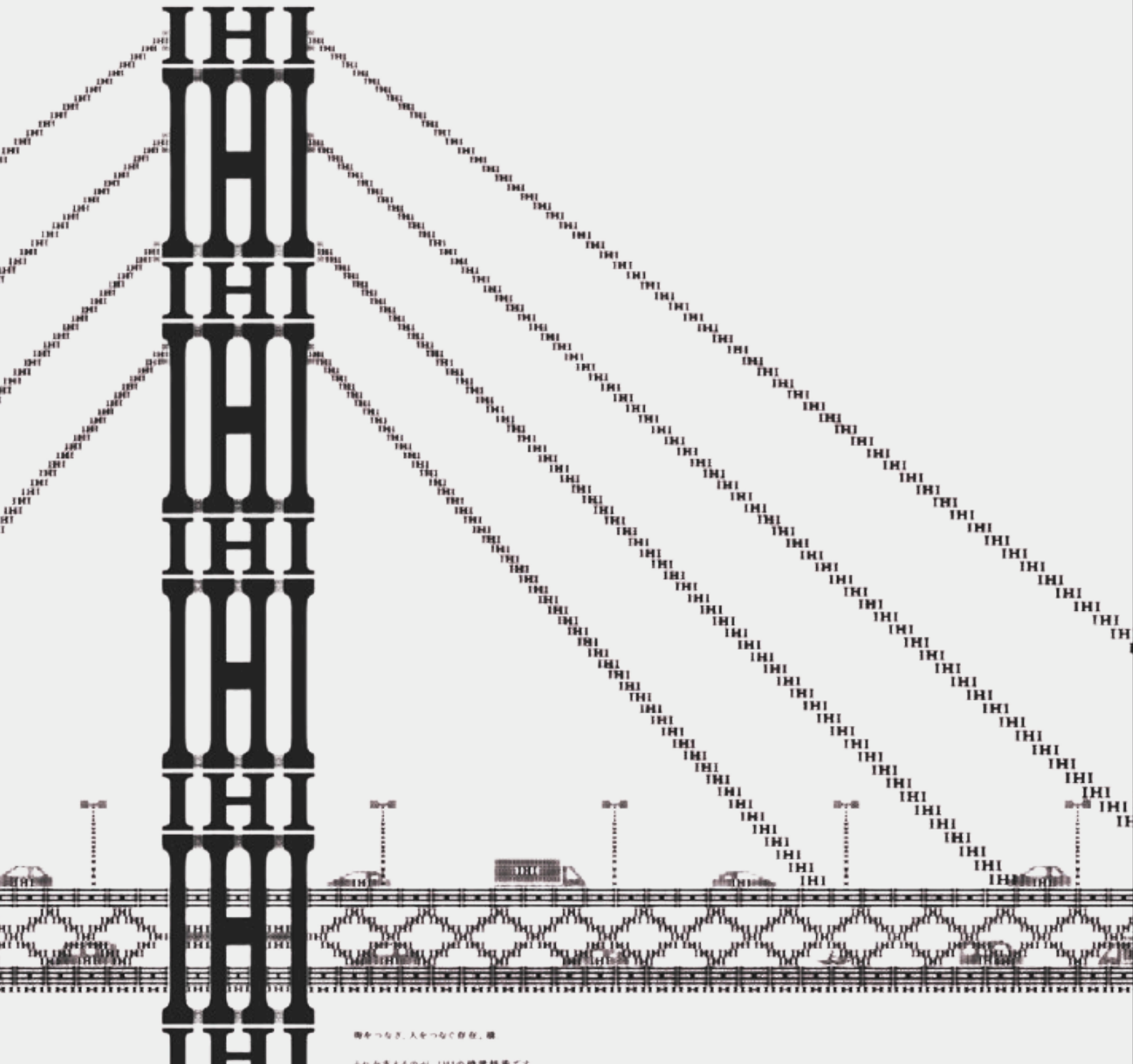
Case Studies



Failure Case Study Reflection: Podilskyi Bridge

Author: Hryhorii Suprun

Fig. 1. Night view of the Podilskyi Bridge (image source: Kashtan News .)

The Podilskyi Bridge (Figure 1) is a case of descoping and cost and schedule overrun.

I. Background information

1.1 Historical review and design

The bridge (also called the Podilsko-Voskresenskyi Bridge) was supposed to be a two-tiered structure that crosses the Dnipro and Desenka rivers in Kyiv (Figure 2). The main purpose of the bridge was to provide a metro level for the future Podilsko-Vyhurivska metro line.

The bridge was originally designed to connect the part of the historic centre (Podil) to the other part of Kyiv city—the left bank of the Dnipro River—namely Dniprovskyi district, Voskresenka and Troieshchyna, making it the main artery connecting the residential districts to the centre of the capital (Figure 2). The design dates from 1993, and construction was planned to be finished by 2002 (Kravets 2021; Suspilne Kyiv 2024); however, funding was only found in 2003 (Kravets 2021), and the road level took more than 30 years to open in stages (Interfax-Ukraine 2023; UNN 2025).

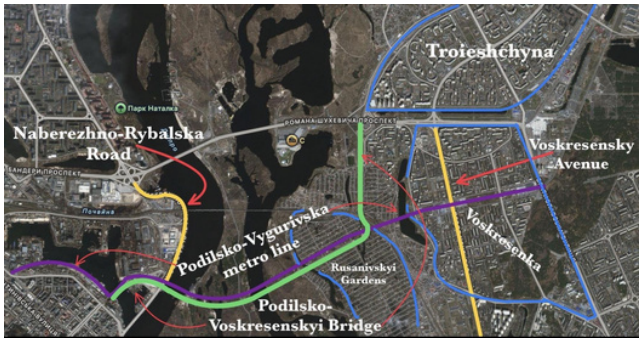


Fig. 2. Map layout showing the route of the Podilskyi Bridge across Kyiv (created by the author; base map imagery)

The bridge was planned as a 7.04 km crossing (Suspilne Kyiv 2024) that connects Naberezhno-Rybalska Road to Voskresensky Avenue, reducing the transit traffic on other bridges of Kyiv, and it is designed for about 59,000 vehicles a day (Kyiv City Guide n.d.).

1.2 Current Traffic situation

As of 2021, there were 7 road bridges across the Dnipro River in Kyiv, which formed 5 corridors for vehicles between the left and right banks of the capital (Kravets 2021). The significant importance of the Podilskyi Bridge lies in the Kyiv city plan. The Ukrainian capital originated on the right bank of the Dnipro River, and as the city developed, residential areas began to spread out to the left bank.

Although the left bank hosts major power plants, transport hubs, and manufacturing, it remains largely a residential, "sleeping" area (Kravets 2021). Therefore, many left-bank residents have to commute daily across the Dnipro River, and the bridges stand in traffic jams for at least 2 to 3 hours in the morning and evening (Kravets 2021). Furthermore, this traffic spreads jams across much of the capital's road network (Kravets 2021).

1.3 Financial situation

To alleviate this urgent issue, the Podilskyi Bridge was promised to be completed by the end of 2020 (Kravets 2021). Instead, the road level opened in stages: on 1 December 2023 for special vehicles and buses, on 1 December 2024 for cars and public transport, including the connection to Troieshchyna, and on 23 August 2025 for all lanes in both directions (Interfax-Ukraine 2023; UNN 2024; UNN 2025), while the metro level has yet to be built (Liga.net 2025). Problems with the contractor and a number of other factors delayed the work (Kravets 2021). The estimate of the funds needed has been rising for over 20 years: it grew from UAH 11.25 billion in 2016 to UAH 19.9 billion in 2021 (Ekonomichna Pravda 2021), about UAH 20 billion had been spent by 2025 (Berezhanskyi 2025), and one 2021 estimate put the total at over $950 million once exchange-rate differences over the construction period are counted (Kravets 2021). The rise was caused by the lack of constant long-term financing, contractor problems, and an annual budgeting process unsuited to a project of this scale (Kravets 2021).

However, apart from the factual rise in cost, the opportunity cost appeared to be too high. In this author's view, the city prioritised metro-based megaprojects over surface tram networks: instead of mobilising finances to construct massive rapid tram networks across the city and remote districts, the government invests in megaprojects, which just deepen and prolong the crisis.

II. Failure of the project

The project failure lies in the inability of the bridge to serve its main purpose—metro line passage. As the motorway part was under construction for the past 30 years, the metro level existed solely on paper.

Additionally, construction of the Troieshchyna part of the Podilsko-Vyhurivska metro line was delayed, with funding shortfalls a recurring problem on the project (Kravets 2021). The bridge contract has also been the subject of investigations: the Antimonopoly Committee fined the general contractor and another firm UAH 182.6 million for collusion in the 2017 tender, and prosecutors have alleged embezzlement (Berezhanskyi 2025). The delay and lack of funding likely worsened public transport problems, as residents of the left bank were left to commute by buses with uneven and sometimes delayed schedules. This probably led to the growth of the usage of private vehicles, which congested the streets even more, intensifying the transport crisis of Kyiv city.

One of the main reasons for the bridge construction delay, apart from financial issues, is the unreadiness of the construction ground (Kravets 2021). As the bridge is meant to connect Podil to the Dniprovskyi district, it is crossing a huge low-rise cottage residential area of Rusanivskyi Gardens (Figure 3), which has little infrastructure and housing. The problem, in fact, caused delays to the construction works while the area was prepared (Kravets 2021). Additionally, as the construction arose, major opposition was provoked among the residents of Rusanivskyi Gardens and ecologists who warned against the destruction of the Horbachykha natural tract (Shnaider 2023). Activists linked the protests to a failure of public consultation, as well as to the withdrawal of private land in Rusanivskyi Gardens (Shnaider 2023; Kravets 2021).

Another acute problem is the need for major construction works on the Paton Bridge, the South Bridge, and the Metro Bridge (Kravets 2021). Partial or complete blocking of these crossings might completely disrupt the flow of traffic, causing the transport collapse of the city. This could have been observed in the summer of 2022, when the South Bridge was closed to private vehicles (Slovo i Dilo 2022), leaving fewer crossings for traffic. Therefore, if these three major bridges are blocked, the finished parts of the Podilskyi Bridge are going to be exposed to a much higher amount of traffic, making completion much harder.

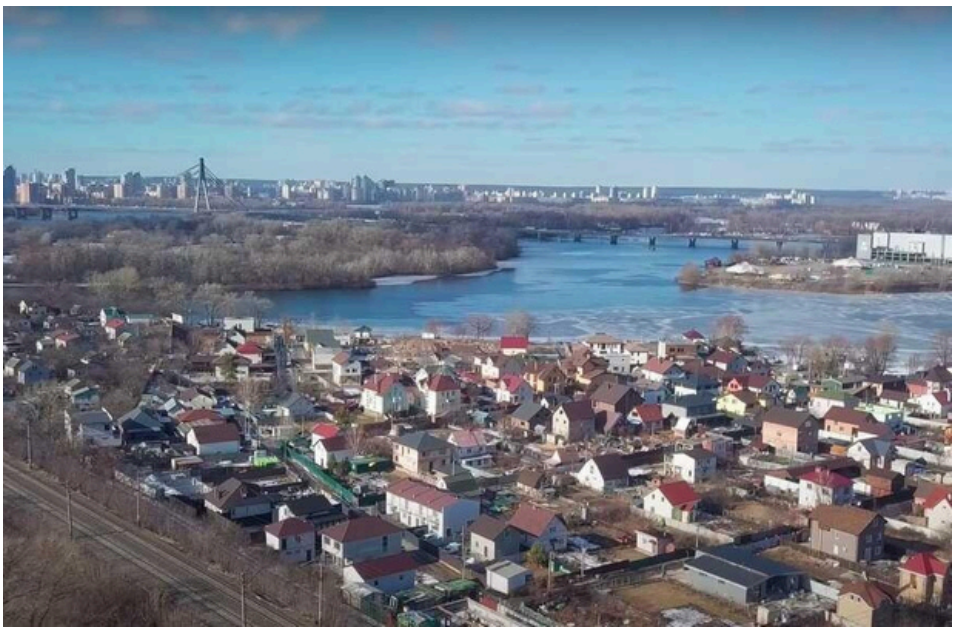


Fig. 3. Aerial view of the Rusanivskyi Gardens residential area (image source: KyivVlada n.d.)

Moreover, the effect of induced demand (Duranton and Turner 2011) is likely to cause a major traffic inflow of private vehicles to the historic centre of the capital, moving the congestion from the North Bridge to the narrow streets of Podil. The idea of the priority of car transport ahead of public transportation generally contradicts the idea of sustainable urban mobility.

As the bridge was designed to have an additional metro level for the Podilsko-Vyhurivska metro line (Figure 4), descoping arises. Authorities delivered the road level first and postponed the metro level. Buses have run on the bridge since December 2023 (Interfax-Ukraine 2023), and the city's announced next stages are an exit to Podil and only then the metro (Liga.net 2025). What's more, it also creates a delay in the construction of the line itself, as the first left-bank station, Raiduzhna, is designed to be connected to the Zatoka Desenka station, located on the bridge itself (Figure 4; Liga.net 2025).

Fig. 4. Schematic diagram of the Podilsko-Vyhurivska metro line; station labels are in Ukrainian (image source: Wikipedia 2026)

III. Conclusion

The project of the Podilskyi Bridge, designed as a two-storey bridge, whose main purpose was to serve public transport, namely the Podilsko-Vyhurivska metro line (Figure 4), partly failed. This probably led to an increase in private vehicle traffic, additionally congesting and loading other, already faulty, bridges. In this author's view, the failure of the bridge's project affected the general development of rapid transport.

As major funds were spent on this megaproject, the city was arguably unable to finance other infrastructural solutions, limiting the capability of the public transport grid and causing additional traffic jams on the roads. To conclude, large-scale infrastructure projects have to mainly focus on the needs of public transport, which is the main priority in modern-day metropolitan areas and, if ignored, may deepen the transport crisis. Additionally, it is vital to have a phased construction approach that delivers the core purpose of the project (here, the metro level) in the first phase, which will help prevent delays and further financial issues on such megaprojects. The completion of the bridge holds the potential for the Ukrainian capital, but only if it aligns with the sustainable urban mobility principles and prioritises public transport over private vehicles.

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The Lybidska–Demiivska Tunnel Failure: A Case Study

Author: Hryhorii Suprun

Figure 1. Interior of a flooded subway station in Kyiv
Source: UNN Kyiv, December 13 2023

Abstract

Heavy reliance on metro in Kyiv city has always been a major problem of the Ukrainian capital. The closure of a section of the Blue metro line clearly identified the inability of the city's public transport infrastructure to function without the metro system. The issue occurred due to both technological failure of the construction and the change in the geological situation underground, especially issues connected to the Lybid river basin. The problem has also been influenced by the urban sprawl and lack of distinct urban planning

In early December of 2023, the Kyiv City State Administration announced the failure of a section of the subway tunnel on the Obolonsko-Teremkivska line between the Lybidska and Demiivska stations (Ekonomichna Pravda 2023).

The reason for this announcement is the violation of the tunnel sealing and the flooding of the rails. Later that day, the head of the Kyiv Metro utility company, Viktor Braginskyi, claimed that new cracks appeared on the crown of the tunnel (Ekonomichna Pravda 2023). The infrastructure issue caused by this branch's failure affected more than 200,000 residents of Kyiv's Holosiivskyi district.

The Kurenivsko–Chervonoarmiyska line (former name of the Obolonsko– Teremkivska line) was planned for construction in the 1970s, aiming to connect the central bus station to the city centre (Ekonomichna Pravda 2023). The first attempts to drill the tunnel were unsuccessful due to the complex hydrogeology, the Lybid river, specifically, which flows between Lybidska and Demiivska Squares. After repeated unsuccessful attempts in the 1980s and 2004, subway builders were able to drill the first tunnel in 2006 and the second in 2008 (Ekonomichna Pravda 2023).

Later on, the tunnelling shield was damaged by unknown structures under Demiivska Square; likewise, the water flow there was very high, and the shield was progressing forward at a slow pace of around 1 metre per day. However, with the use of various waterproofing systems, from the construction of reinforced concrete structures protected by special rubber profiles to the use of external waterproofing with roll materials that are glued to the structures in several layers, the tunnel was finished

Nevertheless, in 2023, the deterioration arose due to asymmetrical soil decompaction processes, vibration around the soil mass and the load on the tunnel lining, including soil pressure, hydrostatic pressure, weight of the structure, weight of buildings and structures located in the zones of influence on the tunnel lining (Ekonomichna Pravda 2023). Furthermore, vibration-induced creep caused by passing trains led the water saturated soil to shift,creating cracks in the tunnel (Wang et al. 2021). Major pressure on the tunnel's structure increased due to many new construction sites appearing around the stations, as well as the ground swelling forces, seasonal temperature effects, and the effects of shrinkage and creep of concrete.

The tunnel's location within the Lybid River basin was a long-recognised risk factor, though this paper does not draw its own conclusion on foreseeability or fault, given that two named Kyiv Metro officials later faced criminal negligence charges over the incident.

According to the Kyiv City State Administration and Kyivmetrobud, the subway specialists, who constantly monitor the condition of the tunnels, understood that such an accident could happen sooner or later (Ekonomichna Pravda 2023). One hypothesis links additional load on the tunnel lining to heavy urban sprawl in the area: major residential complexes and shopping malls under construction above the tunnel route may have added pressure, alongside hydrostatic pressure, contributing to cracking in the lining. Additionally, due to the failure of the sealing joints, some water leaked through the sand, damaging the outer walls of the tunnel.

The technical cause of the failure remained contested for more than a year. On 24 January 2024, the Kyiv City State Administration reported preliminary findings from a research forensic-expert institution attributing the risk of collapse to poor design and poor-quality construction (Ekonomichna Pravda 2024). This was disputed six months later: on 24 July 2024, the Kyiv City Prosecutor's Office announced its own construction technical examination had found no construction defects, attributing the failure instead to inadequate maintenance between 2011 and 2023 (NV 2024; SUD.UA 2024). The next day, the Prosecutor General's Office charged two Kyiv Metro officials — the head of the track and tunnel-structures service and the chief engineer — with official negligence under Part 2, Article 367 of the Criminal Code of Ukraine (NV 2024). The question of design/construction versus maintenance responsibility has not been finally resolved by a court.

In conclusion, the Lybidska - Demiivska tunnel failure was caused by historical, geographical, and engineering factors. Due to the geology of the groundwater and the Lybid River, the tunnel was flooded and therefore needed specialised tubbing rings — prefabricated curved concrete segments bolted together to form the tunnel's circular support lining — and waterproof protection.

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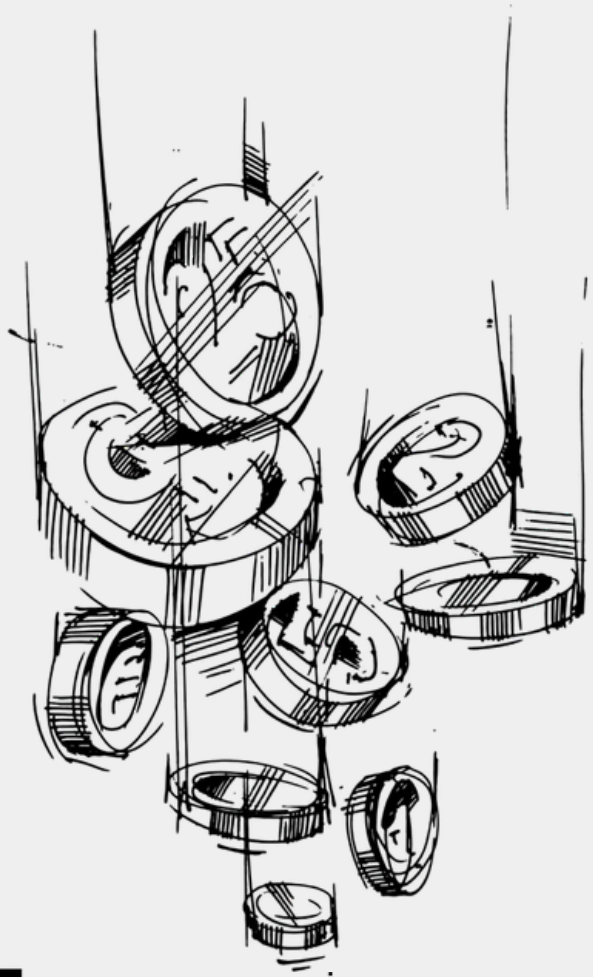
Featured Essays



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John Locke Institute Global Essay Prize 2026 Psychology Question 2: "Is mental illness over-diagnosed now, or just better recognised?"

Mental health diagnosis: Effects of the internet on modern diagnostic

Author: Minh Ngoc Nguyen

I. Introduction

Contemporary civilisation owes much of its progress to developments in the scientific field, as each successive paradigm rewires our interpretation of the world's natural order.

This is reflected in the way we conceptualise the idea of mental illness, from speculations of demonic possessions to ailments linked to the condition of the physical body to the complex interaction of internal and external factors around a person; this evolution in psychopathology has allowed humankind to better acknowledge mental disorders. In addition, the worldwide web has fastened the spread of announcements on major events and conflicts, burdening spectators with stress not witnessed during previous eras.

These refinements in diagnostic tools and criteria along with increased prevalence due to external stressors are the reasons for the rise in diagnosis, suggesting better recognition.

However, in nature with modern society, mainstream publication utilising social media has also led to the novelisation of diagnoses, and a profound phenomenon of self-diagnosis. It is plausible then that mental disorder diagnosis may be experiencing modern distortion instead.

II. The history of psychopathology

Despite mental illnesses having been previously recognised since prehistory, through the utilisation of trepanation – the drilling or cutting into the skull – to release evil spirits from those who exhibited abnormal behaviour, now commonly recognised as symptoms of mental illness, acknowledgement and explanations for mental ailments were not established until Ancient Greece, which theorised that an imbalance of humours and other physiological factors were responsible for conditions such as melancholy, madness and hysteria.

This demonstrated that the identification and categorisation of mental affliction had existed prior to contemporary science.

The late 19th century was shaped by Sigmund Freud, who reasoned that repressed memories and desires were responsible for the behavioural and emotional states, consequently serving as the basis for his psychoanalytic theory along with influencing modern learning theory.

Contradictorily, Emil Kraepelin pioneered a biological approach, which accounted for treatable (exogenous disorders caused by external factors) and incurable (endogenous factors which were products of genetical, neurochemical and structural factors) mental disorders.

Crucially, he produced a nosology – classification of disorders – for mental ailments, which was foundational to for diagnostic manuals such as the DSM-5.

It is from these fundamental pieces of research that contemporary psychopathology has developed in complexity and successfully assisted in diagnosing sufferers, providing support for the idea of increased recognition.

III. The element of social media and conflicts

The introduction of the internet and specifically social media has led to a hyper-visible world, where information, from impersonal broadcasts to individual beliefs and emotions, is made readily available to everyone through publicisation.

This has produced the influx of negative data which our brains are required to process, as news of international conflicts, health crises and social injustices are reported on by both corporations and people.

This is reported broadly by agencies, who rely on provocative wording and exclusive news 'scoops' for obtaining a monopoly in viewership. Majorly, mass-genocidal events such as the COVID-19 pandemic, Russo-Ukrainian war and conflicts within the Gaza Strip have made up a large part of news coverage in the 'age of information'; resulting in global recognition of these events, their timelines and their scope of impact (economically, physiologically and psychologically).

Consequently, this journalism produces great stress, resulting in phenomena such as vicarious traumatisation (the process of being traumatised through observing another party being scarred), along with the development of depression, anxiety and self-destructive behaviours within spectators.6

Crucially, on an individual level, citizens are unrestrained in their reporting as they are not required to comply by publishing rules, allowing for emotions and personal perspectives to take precedence over the objectivity, displaying a biased and potentially inaccurate interpretation.

Resultantly, emotional contagion (observer subconsciously imitates the behaviour and adopts the emotional state of the demonstrator) can influence the viewer's empathetic response; therefore, online messages requesting assistance are often pushed by the platform, due to the algorithm recognising their better viewership retention, with the attentiveness being caused by the activation of empathy.

Due to this, spectators' social media are further flooded with similar content displaying others in distressing situations, leading to viewers acquiring that stress 8, and developing mental ailments6.

This further suggests that mental illnesses are not being over-diagnosed, rather receiving better acknowledgment, as the increase in diagnoses may be resultant of a genuine rise in disorder presentation within the population.

IV. The phenomenon of self-diagnosis

Simultaneously, healthcare associations (NHS, APA, RSPH) have utilised the internet to raise awareness on issues of physiological and psychological wellbeing, allowing non-professionals to educate themselves.

Subsequently, disinformation and misinformation on health issues – demonstrated to receive higher engagement and spread more rapidly – have led the people, particularly adolescents, to make assumptions on their mental status, referred to as the self-diagnosis phenomenon.
This is exacerbated by their development stage, where teenagers actively seek community, approval and a sense of identity; therefore, they attach to the unverified diagnosis to fulfil their sense of belonging through identifying with a group with similar struggles10.
Whilst self-diagnosis can serve as a useful precursory acknowledgement, with research suggesting that individuals who believe they necessitated a clinical diagnosis often experience similar levels of severity (psychopathology) as those with clinical recognition of mental disorders10 (suggesting better recognition of mental recognition), professionals have expressed concern of misdiagnosis by adolescents .
Crucially, they may facilitate self-fulfilling prophecies, wherein the belief of disorder presentation results in a change in self-concept that ultimately maintains the symptoms. This provides support for the claim of excess psychiatric identification of mental disorders in modern society; in addition, suggesting that the over-diagnoses are creating more sufferers.

This possessiveness is caused by the belief that clinical diagnoses are objective truths; however, it has been argued that psychiatrics and mental disorders are subjective constructs with no concrete boundaries (anti-psychiatry).

Furthermore, within the movement, some have crtitised the de-emphasis on sufferers' awareness of their ailment, instead focusing primarily on the judgment and potential biases of the clinicians. This places scrutiny on the validity of current diagnostic criteria and methods, impacting the recognition of mental disorders and the basis of formal, psychiatric diagnosis.

V. Conclusion

It is evident that the current society, through the culmination of technological advancements and development within the field of psychiatry, has permitted the distortion of mental illness, while also allowed for improved acknowledgement of them.

Moreover, research highlights the need for humanitarian support for mentally vulnerable populations, such as adolescents, who communicate and validate their distress by using self-diagnosis in the stressful modern age.

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John Locke Institute Global Essay Prize 2026
Public Policy, Question 1: "What discount rate should be applied to long-run environmental policies? Why?"

The Fifty-Seven Pence Problem: Non-Substitutability, Irreversibility, and the Case for a Tripartite Environmental Discount Rate

Author: Kyal Sin Ye Yint

At a 5% discount rate, the complete destruction of Earth's biosphere two hundred years from now carries a present value of roughly 57 pence per person alive today.[1] Less than a chocolate bar. That single calculation which any A-Level mathematician can verify with compound interest run in reverse is why the discount rate is the most consequential number in environmental economics, and why getting it wrong is not an academic exercise but a civilisational one. I argue that no single discount rate can govern long-run environmental policy. Standard exponential discounting at market-derived rates treats ecological systems as if they behave like financial assets: continuously, reversibly, and substitutably. They do not. A biologically informed framework should instead apply declining discount rates for standard cost-benefit analysis, a separate near-zero rate for irreversible ecological harms, and an insurance-based approach for catastrophic tipping-point risks.

Nearly all modern environmental discounting begins with the Ramsey equation: $r = \delta + \eta g$.[2] The parameter δ represents pure time preference; η is the elasticity of marginal utility of consumption; g is expected growth in per-capita consumption. The UK Green Book extends this further: its Social Time Preference Rate is $STPR = \rho + \mu g$, where $\rho = \delta + L$. The government separates δ (pure time preference at 0.5%) from L (a catastrophic-risk parameter at 1.0%), then adds $\mu g = 1.0 \times 2.0\%$ to reach 3.5%.[3] The government's own formula already concedes that catastrophic risk warrants separation from pure time preference. The concession does not go far enough.

The Ramsey equation is dangerous because it compresses a philosophical disagreement into a mathematical one. Stern set δ at 0.1%, η at 1, g at 1.3%, producing a rate near 1.4% and damages of 5-20% of GDP permanently.[4] Nordhaus, using the same equation, set δ at roughly 1.5% and η at approximately 2, yielding 5–6% and conclusions dramatically less urgent.[5] The equation did not change. The ethics did.

Drupp and colleagues surveyed 197 expert economists on the appropriate long-term rate.[6] The median was 2%. But the most striking result was not the number: only 36 of 185 respondents produced a rate that matched their own Ramsey Rule parameters.[7] Extending Rawls's veil of ignorance across generations, where one does not know which generation one will inhabit, there is no rational basis for placing less weight on future welfare; the philosophical foundation for near-zero δ runs parallel to the economic arguments.[8] The Ramsey equation is an ethical claim in mathematical notation. And the framework it offers is insufficient.

Stern argued near-zero pure time preference: only extinction risk justifies discounting future welfare; any other discounting is, in Ramsey's phrase, "ethically indefensible."[9] The logic is internally coherent. But Dasgupta delivered the most technically devastating critique of Stern: accepting Stern's parameter values implies that the optimal savings rate for the current generation should be approximately 97.5% of GDP.[10] No government in history has come close. Stern's ethics produce an absurdity.

Nordhaus argued instead that the discount rate must be calibrated descriptively, to observed market returns.[11] And yet Nordhaus's own approach assigns a present value of 57 pence to the end of the world in two centuries. This is arithmetic. Weitzman broke the deadlock in 2007: Stern was "right for the wrong reasons."[12] The case for aggressive action does not depend on resolving the ethical argument about δ; it depends on recognising that climate change presents fat-tailed catastrophic risks whose expected disutility may be unbounded under standard utility assumptions.[13] The danger is that we have no idea how bad the worst outcomes could be, and standard models cannot price that ignorance.

Weitzman's intervention was the most productive move in the debate. But it shared with Stern and Nordhaus a premise so deeply embedded that none of them questioned it: that ecological damages are continuous and at least partially reversible through future wealth and technology. Studying both mathematics and biology has made one thing clear to me. This assumption is not merely optimistic. It is biologically false.

Population dynamics exhibit carrying capacities and what ecologists call regime shifts: states that do not smoothly return to equilibrium once disrupted beyond a threshold. Haemoglobin's oxygen-binding curve is sigmoidal, binding barely proceeds below a threshold partial pressure; above it, cooperative conformational change drives saturation. Ecological systems operate by the same logic, not as continuous functions of temperature or CO_2 concentration, but as threshold-dependent systems capable of abrupt state changes.

Lenton and colleagues identified nine policy-relevant tipping elements in 2008.[14] The Greenland Ice Sheet may pass its threshold at roughly 1–2°C of global warming, with potential sea-level rise of 2–7 metres; the Atlantic thermohaline circulation and the Amazon rainforest each carry their own thresholds and timescales. These transitions are not smooth curves on a damage function. They are discontinuities. Dietz and colleagues showed that incorporating eight tipping points into climate-economy models increases the social cost of carbon by more than 25%.[15]

Cai and colleagues found the optimal carbon tax should be 50–200% higher when tipping points are endogenised in stochastic integrated assessment models.[16] Standard models built on smooth polynomial damage functions underestimate the cost of inaction.[17]

The biological problem runs deeper than tipping points alone. Standard discounting assumes that future generations can substitute wealth for environmental quality, that a richer society can always compensate itself for ecological loss. Sterner and Persson exposed the flaw.[18] When the elasticity of substitution σ between environmental and consumption goods approaches zero, meaning no amount of GDP growth can replace a functioning ecosystem, the relative price of environmental goods rises without bound as they become scarcer; the growth in relative prices exactly offsets the consumption discount rate, and the effective rate for ecological damages converges mathematically toward zero. The near-zero rate is not an ethical assertion. It is an algebraic consequence of $\sigma \to 0$.

In an amended DICE model that accounts for rising relative scarcity, Sterner and Persson obtained Stern-like abatement results using Nordhaus's high discount rate. The rate barely mattered once substitutability was correctly specified. What changed my understanding was that specific result. The entire Stern-Nordhaus debate assumes a single composite consumption good. But environmental goods are not consumption goods with green labels. Coral reef collapse has no price. Species extinction is not a depreciating asset. Ice sheet disintegration cannot be reversed by a wealthier civilisation writing a larger cheque. When the damage function is a step function rather than a smooth curve, standard discounting is not merely imprecise. It is structurally wrong.

The answer is not a single number. It is a framework that matches the discount rate to the category of harm. The first component is a declining discount rate for standard environmental cost-benefit analysis. Three independent theoretical routes converge on this conclusion. Weitzman demonstrated in 1998 that if the discount rate is uncertain, the certainty-equivalent rate declines over time via Jensen's inequality.[19]

Gollier showed in 2002 that persistent shocks to consumption growth produce the same declining structure through an entirely different mechanism.[20] Schernberg's 2025 result adds a third: social welfare functions exhibiting aversion to persistent consumption risk also generate declining rates, with the decline steepening as the planner's risk aversion and horizon increase.[21] Three independent derivations reach the same destination.

And the evidence is not only theoretical. Fesselmeyer, Liu, and Salvo documented declining discount rates empirically in Singapore's private apartment market.[22] The UK Green Book implements a declining schedule: 3.5% (years 0–30), 3.0% (31–75), and 2.5% beyond 75 years; sensitivity analysis applies further reductions.[23] France and Denmark have adopted similar structures. In June 2025, HM Treasury's Green Book Review acknowledged stakeholder concerns that current rates "may unfairly undermine the appraisal of long-term transformational schemes that produce benefits well into the future." The Treasury commissioned an independent STPR review under Professors Ben Groom and Mark Freeman, with final findings due before end of June 2026.[24] Whether declining rates should govern long-run policy is no longer a theoretical question. The UK is already answering it. Arrow and colleagues, an expert panel sympathetic to both sides, concluded that "theory provides compelling arguments for a declining certainty-equivalent discount rate."[25] What struck me most is that Nordhaus himself co-authored it. The strongest critic of low rates endorsed declining rates.

The second component is a near-zero rate for irreversible ecological harms. When no technological replacement exists for a functioning Amazon rainforest or a stable thermohaline circulation, the appropriate rate for those specific damages converges toward zero. The economics already supplies the mechanism. What it lacks is the institutional willingness to apply separate rates to separate categories of harm.

The framework's final element addresses a category of harm that lies outside cost-benefit analysis entirely. Weitzman's Dismal Theorem holds that under fat-tailed distributions, expected disutility can become unbounded.[26] Rate-of-return analysis ceases to function; expected welfare loss diverges, and no finite abatement cost can be rejected on cost-benefit grounds. Standard CBA was not designed for risks of this magnitude.[27] Insurance is the policy instrument designed for exactly this class of problem: paying a finite premium to avoid losses whose expected value cannot be calculated. Nordhaus responded in 2012 that "no loaded gun of strong tail dominance has been uncovered to date."[28] But this misses the structural point. Policy itself can thin the tails; mitigation converts strong tail dominance to weak.[29] The Green Book has already conceded that catastrophic risk warrants its own parameter, L set at 1.0%; the framework here extends that concession to risks L alone cannot capture.

The strongest objection is the market-rates argument: observed returns on capital reflect genuine social preferences, and the social planner should respect them. Goulder and Williams showed that the social welfare discount rate and the financial rate of return serve different functions and should not be conflated.30 Environmental policy requires the consumption rate of discount, not the opportunity cost of capital. Conflating them is a category error.

A second institutional objection comes from HM Treasury itself. The 2021 Environmental Discount Rate Review concluded that lowering the discount rate for environmental impacts was "an imprecise way of accounting for effects such as lack of substitutability and increasing scarcity in the environment," favouring improved valuation methods over a separate rate.[31] This objection has force for marginal environmental goods whose scarcity can be monetised through relative-price adjustments. It does not apply to the class of harms the second pillar targets: goods for which no valuation function exists because no substitute exists at any price. One cannot improve the valuation of a collapsed thermohaline circulation if the valuation requires a market that does not and cannot exist.

A third objection holds that the framework is too complex to implement. But the UK already implements declining rates. France already implements declining rates. The question is not whether governments can manage variable discount structures (they demonstrably can) but whether those structures capture the distinct categories of environmental harm. I acknowledge a difficult operational question: where precisely should the boundary fall between the three categories? This is a limitation, but the difficulty of drawing boundaries does not invalidate the distinction. The difference between a reversible productivity loss and the irreversible collapse of the Atlantic thermohaline circulation is not a matter of degree. It is a difference in kind.

Return to the opening calculation. At a declining rate starting at 3.5% and converging toward 1%, the present value of preventing planetary destruction two hundred years out is no longer 57 pence. It is orders of magnitude higher. Under the near-zero ecological rate, for damages that admit no substitution and no recovery, the present value approaches the undiscounted cost itself.

The answer is not a number. It is a structure: declining rates for standard cost-benefit analysis, grounded in three independent derivations and operational in UK fiscal policy; a near-zero rate for irreversible ecological harms where no substitution function exists; and an insurance-based approach for catastrophic tipping-point risks that standard models cannot price. Ramsey called the practice of discounting future utility "ethically indefensible" and attributed it to "the weakness of the imagination."[9] The framework here attempts to correct that weakness, not through ethical prescription alone, but through biological and mathematical precision.

Endnote:

1. At a 5% annual discount rate, the present value factor for 200 years is $(1.05)^{-200} \approx 5.78 \times 10^{-5}$. Applied to global GDP of approximately $100 trillion divided by 8 billion people (≈ $12,500 per person), total destruction has a present value of roughly $0.72 per person, or approximately 57 pence at prevailing exchange rates. This calculation uses per-capita annual GDP as a lower-bound proxy for the annualised value of a functioning biosphere; a stock-based valuation would produce a higher figure, reinforcing the argument. The precise figure varies with assumptions; the order of magnitude is robust.

2. Ramsey, F. P. (1928). "A Mathematical Theory of Saving." The Economic Journal, 38(152), 543–559.

3. HM Treasury (2026). Green Book Supplementary Guidance: Discounting. February 2026. STPR = ρ + μg with ρ = δ + L, δ = 0.5%, L = 1.0%, μ = 1.0, g = 2.0%, yielding 3.5%. Main Table 3.A applies the declining schedule 3.5% (years 0–30), 3.0% (31–75), and 2.5% (76–125), with NPV capped at 125 years since 2018. Table 3.B Reduced STPR (excluding δ) is used for sensitivity analysis on long-horizon proposals, declining through 2.14%, 1.71%, 1.29%, and 0.86% beyond 300 years.

4. Stern, N. (2007). The Economics of Climate Change: The Stern Review. Cambridge University Press. The Stern Review sets δ = 0.1%, η = 1, g = 1.3%, yielding r ≈ 1.4%.

5. Nordhaus, W. D. (2007). "A Review of the Stern Review on the Economics of Climate Change." Journal of Economic Literature, 45(3), 686–702.

6. Drupp, M. A., Freeman, M. C., Groom, B., & Nesje, F. (2018). "Discounting Disentangled." American Economic Journal: Economic Policy, 10(4), 109–134.

7. Drupp et al. (2018), p. 123. The SDR coincided with the imputed simple Ramsey Rule for only 36 of 185 quantitative respondents.

8. Rawls, J. (1971). A Theory of Justice. Harvard University Press. The original veil is intra-generational; the intergenerational extension follows the just-savings argument in Chapter V.

9. Ramsey (1928), p. 543. The full sentence: "it is assumed that we do not discount later enjoyments in comparison with earlier ones, a practice which is ethically indefensible and arises merely from the weakness of the imagination."

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10. Dasgupta, P. (2008). "Discounting Climate Change." Journal of Risk and Uncertainty, 37, 141–169. The 97.5% savings-rate implication is derived on pp. 156–158.
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26. Weitzman (2009), op. cit. Under fat-tailed distributions with constant relative risk aversion, the integral defining expected disutility fails to converge — the structural sense in which "rate-of-return analysis ceases to function."
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29. This argument follows Weitzman (2009)'s acknowledgement that willingness-to-pay for tail-slimming is itself a policy variable: the Dismal Theorem's force is mitigated precisely by the precautionary investments it motivates.
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Kyal Sin Ye Yint
Founder and President
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Canterbury, September 2026

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CATS INSIGHTS

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